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5.73 Quantum Mechanics I
Fall 2018

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Eigenvalues, Eigenvectors, and the Discrete Variable Representation (DVR)

should have read CDTL pages 94-144

Last time:

a "*bra*" $\quad {}_{\phi}\langle\ \ | = \begin{pmatrix} a_1^* & \dots & a_N^* \end{pmatrix}_{\phi}$

a"*ket*" $\quad |\ \ \rangle_{\phi} = \begin{pmatrix} a_1 \\ \vdots \\ a_N \end{pmatrix}_{\phi}$

$|\ \ \rangle\langle\ \ |$ $\quad$ an $N \times N$ matrix

$\langle\ \ | \ \ \rangle$ $\quad$ a $\left(\text{complex}\right)\#$

ψ in $\{\phi\}$ basis set

$$|\psi_i\rangle = \begin{pmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{pmatrix}_{\psi} = \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ \vdots \\ a_N \end{pmatrix}_{\phi}$$

$$a_j = {}_{\phi}\langle \phi_j | \psi_i \rangle_{\phi}$$

$$\mathbb{1} = \begin{pmatrix} 1 & & & \\ & 1 & & 0 \\ & & 1 & \\ & & & \\ & 0 & & \ddots \end{pmatrix} = \sum_k |k\rangle\langle k| \qquad \text{The "unit" matrix}$$

at end of lecture #10 we saw

$$\langle \phi_i | \mathbf{AB} | \phi_j \rangle = \sum_k \langle \phi_i | \mathbf{A} \underbrace{|\phi_k\rangle\langle\phi_k|}_{\mathbb{1}} \mathbf{B} | \phi_j \rangle$$

$$= \sum_k A_{ik} B_{kj} = \left(\mathbf{AB}\right)_{ij}$$

What is the connection between the Schrödinger (wavefunction) and Heisenberg (matrix) representations?

$$\psi_i(x) = \langle x | \psi_i \rangle$$

$$|x_0\rangle = \delta(x, x_0)$$ eigenfunction of **x** with eigenvalue x_0

Using this formulation for $\psi_i(x)$, you can go freely (and rigorously) between the Schrödinger and Heisenberg representations.

Today: **eigenvalues** of a matrix – what are they? how do we get them? (secular equation). Why do we need them?

eigenvectors – how do we get them?

Arbitrary V(x) in Harmonic Oscillator Basis Set (Discrete Variable Representation)

The Schrödinger Equation is an eigenvalue equation

$$\hat{A}\psi = a\psi$$ (a: an eigenvalue)

$$\mathbf{A}|\psi_i\rangle = a_i|\psi_i\rangle$$

In matrix language:

$$A^{\phi} = \begin{pmatrix} a_1 & 0 & 0 & 0 \\ 0 & a_2 & 0 & 0 \\ 0 & 0 & \ddots & 0 \\ 0 & 0 & 0 & a_N \end{pmatrix}_{\psi}$$

$$|1\rangle_{\psi} = \begin{pmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix} \qquad |2\rangle_{\psi} = \begin{pmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$$

etc…

$$|n\rangle_{\psi} = \begin{pmatrix} \vdots \\ \vdots \\ 1 \\ \vdots \\ 0 \end{pmatrix}$$ ← n'th position

The ψ representation is special. We want to derive it from a computationally explicit starting point. We compute A^{ϕ} for a complete ortho-normal basis set $\{\phi\}$.

$$A^{\phi} = \begin{pmatrix} A_{11} & A_{12} & \cdots & A_{1N} \\ \vdots & A_{22} & & \vdots \\ \vdots & \cdots & \ddots & \vdots \\ A_{N1} & \cdots & \cdots & A_{NN} \end{pmatrix}_{\phi}$$

a non-diagonal matrix

The basis set that we choose can be the eigenfunctions for a simplified problem or they can be computationally convenient functions.

How do we find the eigenvalues and eigenfunctions of a non-diagonal $\mathbf{A}^{\phi}$?

I am using unconventional notation, ϕ or ψ subscripts or superscripts, which make the meaning of all symbols explicit. (They are like training wheels on a bicycle, discarded as soon as you learn to ride.

Some notation

$$\left|i\right\rangle_{\phi} = \begin{pmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ j \end{pmatrix}_{\phi}$$

← i'th position

$$\mathbf{T}\left|i\right\rangle_{\phi} = \begin{pmatrix} T_{1i} \\ \vdots \\ T_{Ni} \end{pmatrix}_{\phi}$$

the i'th column of $\mathbf{T}$

Suppose we have a diagonal matrix, $\mathbf{A}^{\psi}$, an eigenvector of $\mathbf{A}^{\psi}$ is given by

$$\mathbf{A}^{\psi}\left|i\right\rangle_{\psi} = \mathbf{A}^{\psi}\begin{pmatrix}0\\ \vdots \\ 1 \\ \vdots \\ 0\end{pmatrix} = \begin{pmatrix}0\\ 1 \\ a_i \\ \vdots \\ 0\end{pmatrix} = a_i \begin{pmatrix}0\\ \vdots \\ 1 \\ \vdots \\ 0\end{pmatrix}_i$$

The following **Non-Lecture** section is a derivation and discussion of the most important and useful equation in matrix mechanics.

$$\mathbf{S}^{\dagger}\mathbf{A}^{\phi}\mathbf{S}\left(\mathbf{S}^{\phi}\left|\ \right\rangle_{\phi}\right) = \mathbf{A}^{\psi}\left|\ \right\rangle_{\psi}$$

The eigenvectors are given by

$$\left(\mathbf{S}^{\dagger}\left|\ \right\rangle_{\phi}\right)$$

The i-th eigenvector is the i-th column or $\mathbf{S}^{\dagger}$

Sometimes, as for a time-dependent wavepacket, you want to know how a non-eigenvector is expressed as a linear combination of eigenkets. These are given by the columns of $\mathbf{S}$.

$$\mathbf{S}\left|\ \right\rangle_{\psi} = \left|\ \right\rangle_{\phi}$$

Another very useful trick will be to evaluate the elements of $\mathbf{S}$ or $\mathbf{S}^{\dagger}$ using perturbation theory rather than diagonalization of the full matrix by a computer.

For the exact **H**, we use perturbation theory, discussed in Lectures #14 – 17.

$$\mathbf{H} = \mathbf{H}^{(0)} + \mathbf{H}^{(1)}$$

$\mathbf{H}^{(0)}$: exactly solved (diagonal)

$\mathbf{H}^{(1)}$: bad stuff (not diagonal)

$$\left(\mathbf{S}^{\dagger}\mathbf{H}\mathbf{S}\right)_{ii} = \mathbf{H}_{i}^{(0)} + \sum_{j\neq i} \frac{\left|\mathbf{H}_{ij}^{(1)}\right|^{2}}{E_{i}^{(0)} - E_{j}^{(0)}}$$

$$\mathbf{S}_{ij}^{\dagger} = \frac{\mathbf{H}_{ji}^{(1)}}{E_{j}^{(0)} - E_{i}^{(0)}}$$

which is the amount of $\left|i\right\rangle_{(0)}$ mixed into $\left|j\right\rangle$

where $E_{j}^{(0)}$ and $E_{i}^{(0)}$ are two eigen-energies of $\mathbf{H}^{(0)}$

The equation

$$\mathbf{S}^\dagger \mathbf{A}^\phi \mathbf{S}\mathbf{S}^\dagger \Big| \; \Big\rangle_\phi = \mathbf{A}^\psi \Big| \; \Big\rangle_\psi$$

is the most important equation in matrix mechanics

$\Big| \; \Big\rangle_\phi$ a complete, ortho-normal set of basis-kets

$\Big| \; \Big\rangle_\psi$ the complete set of eigen-kets of Hermitian operator $\mathbf{A}$

$\mathbf{A}^\phi$ is the $N \times N$ (N can, in principle, be ∞) matrix representation of $\mathbf{A}$ where the elements of $\mathbf{A}$ are

$$A_{ij}^\phi = {}_\phi\left\langle i \middle| \mathbf{A} \middle| j \right\rangle_\phi = \begin{pmatrix} A_{11} & A_{12} & \cdots & A_{1N} \\ A_{21} & A_{22} & \cdots & A_{2N} \\ \vdots & \vdots & \vdots & \vdots \\ A_{N1} & \cdots & \cdots & A_{NN} \end{pmatrix}_\phi$$

$\mathbf{A}^\psi$ is the diagonal matrix representation of $\mathbf{A}$ in the eigen-basis.

$$A^\psi = \begin{pmatrix} a_1 & 0 & 0 & 0 \\ 0 & a_2 & 0 & 0 \\ 0 & 0 & \ddots & 0 \\ 0 & 0 & 0 & a_N \end{pmatrix}_\psi$$

$\mathbf{A}$ is Hermitian, which means that $\mathbf{A}^\dagger = \mathbf{A}$ where $A_{ij}^\dagger = A_{ji}^*$ and all the $\{a_i\}$ are real.

$\mathbf{S}$	is a unitary matrix
$\mathbf{S}^\dagger = \mathbf{S}^{-1}$	is the definition of a unitary matrix
$\mathbf{S}^\dagger\mathbf{S} = \mathbb{1}$	unit matrix
$\mathbf{S}^\dagger\mathbf{A}\mathbf{S}$	is a unitary transformation of $\mathbf{A}$

We are interested in the unitary transformation that "diagonalizes" $\mathbf{A}$

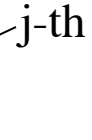
$$\mathbf{S}^\dagger \mathbf{A}^\phi \mathbf{S} = \mathbf{A}^\psi$$

and gives us the eigen-kets of $\mathbf{A}$,

$$\mathbf{S}^\dagger \left| \; \right\rangle_\phi = \left| \; \right\rangle_\psi$$

This means that the j-th column of $\mathbf{S}^\dagger$ is the j-th eigenvector of **A**.

$$\sum_i \mathbf{S}^\dagger_{ij} \left| i \right\rangle_\phi = \begin{pmatrix} S^\dagger_{1j} \\ \vdots \\ S^\dagger_{Nj} \end{pmatrix}_\phi = \begin{pmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{pmatrix}_\psi = \left| j \right\rangle_\psi$$



If $|\;\rangle_\phi$ is orthonormal, the unitary transformation preserves the orthonormality of $|\;\rangle_\psi$.

Efficient computer programs exist that diagonalize any Hermitian matrix. The diagonalization is done in sequence of 2 × 2 transformations $\{S_i^\dagger\}$, and the final diagonalizing matrices, **S**, $\mathbf{S}^\dagger$, are expressed as a product of partial diagonalization steps

$$\mathbf{S}^\dagger = \prod_{i=1}^{N_{\max}} S_i^\dagger$$

which means that the computer gives us both the eigenvalues and the eigenvectors of **A**.

Now where does this come from? Linear Algebra.

This is discussed in elegant detail in Merzbacher, Chapters 10.1 – 10.2, pages 207-212.

Eigenvalue Equation

$$\psi_j = \sum_i c_i^j \phi_i$$

completeness, an eigenstate is obtained from a linear combination of basis functions.

$$\mathbf{A}\sum_i c_i^j \phi_i = a_j \sum c_i^j \phi_i$$

left multiply by ϕ_k^* and integrate

$$\int \phi_k^* \mathbf{A} \phi_i \, d\tau = A_{ki}^{\phi}$$

$$\sum_i c_i^j A_{ki}^{\phi} - a_j c_k^j = 0$$

a linear homogeneous equation in unknown coefficients $\left\{c_k^j\right\}$

rearrange:

$$\sum_i \left[c_i^j A_{ki} - a_j \delta_{ki} c_i^j \right] = 0$$

$$\sum_i c_i^j \left(A_{ki} - a_j \delta_{ki} \right) = 0$$

left multiply by $\phi_{k'}^*$, and integrate, get another homoeneous linear equation.

The determinant of the unknown coefficients of the set of $\left\{c_i^j\right\}$ must be zero to yield a non-trivial solution: non-zero values of the c_i^j.

$$\left|\mathbf{A}^{\phi} - a\delta_{ki}\right| = 0$$

The vertical lines denote a determinant.

If we diagonalize **A**, then we have an equation that satisfies the determinant of |**A**| = 0 requirement.

$$\prod_{i=1}^{N}\left[A_{ii}^{\psi} - a\right] = 0 \quad \text{for each eigenvalue in the set } \left\{a_j\right\}$$

For each member of the set of eigenvalues $\{a_j\}$, we get one factor of the N-term product that is zero, which ensures that $|A^{\psi} - a| = 0$.

Two Useful Properties of Hermitian Matrices

Determinant Invariance

$$\left|\mathbf{ABC}\right| = \left|\mathbf{A}\right|\left|\mathbf{B}\right|\left|\mathbf{C}\right|$$

This means $\left|\mathbf{S}^{\dagger}\mathbf{AS}\right| = \left|\mathbf{S}^{\dagger}\mathbf{S}\right|\left|\mathbf{A}\right| = \left|\mathbf{A}\right|$

$$\text{so } \left|A^{\phi}\right| = \left|A^{\psi}\right| = \prod_{i=1}^{N} a_i$$

Invariance of the <u>product</u> of eigenvalues

Trace Invariance

$$\sum_{i=1}^{N} A_{ii} = \sum_{i=1}^{N} a_i$$

Representation of the <u>sum</u> of eigenvalues

End of Non-Lecture

Can now solve many difficult appearing problems!

Start with a **matrix representation** of *any operator* that is expressable as a function of a matrix.

e.g. $e^{-i\mathbf{H}(t-t_0)/\hbar}$ propagator , $f(\mathbf{x})$ potential curve

prescription example

$$f(\mathbf{x}) = \mathbf{T} f\underbrace{\left(\mathbf{T}^\dagger \mathbf{x} \mathbf{T}\right)}_{\text{diagonalize } \mathbf{x} \text{ – so } f(\,) \text{ is applied to each diagonal element}} \mathbf{T}^\dagger$$

$$\mathbf{T}^\dagger \mathbf{x} \mathbf{T} = \begin{pmatrix} x_1 & & 0 \\ & x_2 & \\ & & \ddots \\ 0 & & x_N \end{pmatrix}$$

$$f\left(\mathbf{T}^\dagger \mathbf{x} \mathbf{T}\right) = \begin{pmatrix} f(x_1) & & 0 \\ & f(x_2) & \\ & & \ddots \\ 0 & & f(x_N) \end{pmatrix}$$

Then perform the inverse transformation, $\mathbf{T}\, f(\mathbf{T}^\dagger \mathbf{x} \mathbf{T})\mathbf{T}^\dagger$ – undiagonalizes matrix, which gives matrix representation of the desired function of a matrix.

Show that this actually is valid for simple example

$$f(x) = \mathbf{x}^N$$

$$\underline{\underline{f\left(\mathbf{x}^N\right)}} = \mathbf{T}\left[\left(\mathbf{T}^\dagger \mathbf{x} \mathbf{T}\right)\left(\mathbf{T}^\dagger \mathbf{x} \mathbf{T}\right)\cdots\left(\mathbf{T}^\dagger \mathbf{x} \mathbf{T}\right)\right]\mathbf{T}^\dagger \quad \text{apply prescription}$$

(1) (2) (N)

$$= \mathbf{T}\left[\mathbf{T}^\dagger \mathbf{x}^N \mathbf{T}\right]\mathbf{T}^\dagger = \mathbf{x}^N \quad \text{get expected result}$$

general proof for arbitrary $f(\mathbf{x}) \rightarrow$ expand in power series. Use previous result for each integer power.

John Light: Discrete Variable Representation (DVR)
General $V(x)$ evaluated in Harmonic Oscillator Basis Set.
we did not do H-O yet, but the general formula for all of the nonzero matrix elements of **x** in the harmonic oscillator basis set is:

$$\langle n|\mathbf{x}|n+1\rangle = \left[\frac{\hbar}{2\omega\mu}\right]^{1/2}(n+1)^{1/2} \qquad \omega = (k/\mu)^{1/2}$$

(infinite dimension matrix)

$$\mathbf{x} = \left[\frac{\hbar}{2\omega\mu}\right]^{1/2}\begin{pmatrix} 0 & \sqrt{1} & 0 & \cdots & \cdots \\ \sqrt{1} & 0 & \sqrt{2} & \cdots & \cdots \\ 0 & \sqrt{2} & 0 & \sqrt{3} & 0 \\ \vdots & \vdots & \sqrt{3} & 0 & \sqrt{4} \\ \vdots & \vdots & \vdots & \sqrt{4} & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

[CARTOON]

$$\mathbf{x}^2 = \left[\frac{\hbar}{2\omega\mu}\right]\begin{array}{c} n = \\ \\ \\ \\ \\ \\ \\ \\ \\ \end{array}\begin{array}{c} \begin{array}{ccccccccc} 0 & 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \end{array} \\ \begin{pmatrix} 1 & 0 & \sqrt{2} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 3 & 0 & \sqrt{6} & 0 & 0 & 0 & 0 & 0 \\ \sqrt{2} & 0 & 5 & 0 & \sqrt{12} & 0 & 0 & 0 & 0 \\ 0 & \sqrt{6} & 0 & 7 & 0 & & 0 & 0 & 0 \\ 0 & 0 & \sqrt{12} & 0 & 9 & 0 & & 0 & 0 \\ 0 & 0 & 0 & & 0 & 11 & 0 & & 0 \\ 0 & 0 & 0 & 0 & & 0 & 13 & 0 & \\ 0 & 0 & 0 & 0 & 0 & & 0 & 15 & \ddots \\ 0 & 0 & 0 & 0 & 0 & 0 & & \ddots & \ddots \\ & & & & & & & & \end{pmatrix} \end{array}\begin{array}{l} 0 \\ 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \\ \\ \\ \end{array}$$

-two off-main diagonal $[n(n+1)]^{1/2}$

-main diagonal $|2n+1\rangle$

etc. matrix multiplication

to get matrix for f(**x**) diagonalize e.g., 1000 × 1000 (truncated) **x** matrix that was expressed in harmonic oscillator basis set.

$$\mathbf{T}^{\dagger}\mathbf{x}\mathbf{T} = \begin{pmatrix} x_1 & 0 & 0 & 0 \\ 0 & x_2 & 0 & 0 \\ 0 & 0 & \ddots & 0 \\ 0 & 0 & 0 & x_{1000} \end{pmatrix}_x$$

diagonalized-**x** basis $\{x_i\}$ are eigenvalues. They have no obvious physical significance.

$$\mathbf{V}(x)_x = \begin{pmatrix} V(x_1) & 0 & 0 & 0 \\ 0 & V(x_2) & 0 & 0 \\ 0 & 0 & \ddots & 0 \\ 0 & 0 & 0 & V(x_{1000}) \end{pmatrix}_x$$

next transform back from x-basis to H-O basis set

$$\mathbf{V}(x)_{H-O} = \mathbf{T}\mathbf{V}(x)_x \mathbf{T}^{\dagger} = \begin{pmatrix} \text{full} \\ \text{complicated} \\ \text{matrix} \\ \\ \\ \end{pmatrix}_{\substack{1000 \times 1000 \\ H-O}}$$

T was determined when **x** was diagonalized

$$\mathbf{H} = \frac{\mathbf{p}^2}{2\mu} + V(x)$$

need matrix for $\mathbf{p}^2$, get it from **p** (the general formula for all non-zero matrix elements of **p**)

$$\langle n|p|n+1\rangle = -i\left[\frac{\hbar(\omega\mu)}{2}\right]^{1/2}(n+1)^{1/2}$$

$$\mathbf{p} = -i\left[\frac{\hbar(\omega\mu)}{2}\right]^{1/2}\begin{pmatrix} 0 & \sqrt{1} & 0 & 0 & 0 \\ -\sqrt{1} & 0 & \sqrt{2} & 0 & 0 \\ 0 & -\sqrt{2} & 0 & 0 & 0 \\ 0 & 0 & \ddots & \ddots & \ddots \\ 0 & 0 & \ddots & \ddots & \ddots \end{pmatrix} \quad \text{same structure as } \mathbf{x}$$

$$\mathbf{p^2} = -\left[\frac{\hbar(\omega\mu)}{2}\right]\begin{pmatrix} -1 & 0 & \sqrt{2} & 0 & 0 \\ 0 & -3 & 0 & \ddots & 0 \\ \sqrt{2} & 0 & -5 & 0 & \ddots \\ 0 & \ddots & \ddots & \ddots & \ddots \\ 0 & \ddots & \ddots & \ddots & \end{pmatrix}$$

if $\quad H=\frac{p^2}{2\mu}+\frac{1}{2}kx^2 \qquad \left(\frac{1}{2}k=\frac{1}{2}\omega^2\mu\right)$

$$H = \frac{\hbar\omega}{4}\begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 6 & 0 & 0 \\ 0 & 0 & 10 & 0 \\ 0 & 0 & 0 & 14 \end{pmatrix} = \hbar\omega\begin{pmatrix} 1/2 & 0 & 0 & 0 \\ 0 & 3/2 & 0 & 0 \\ 0 & 0 & 5/2 & 0 \\ 0 & 0 & 0 & \ddots \end{pmatrix}$$

but for arbitrary V(x), express **H** in HO basis set,

$$\mathbf{H}_{HO} = \frac{\mathbf{p}^2_{HO}}{2\mu} + \underbrace{\mathbf{V(x)}_{HO}}_{\mathbf{TV(x)}_x\mathbf{T}^\dagger}$$

eigenvalues obtained by $\mathbf{S}^\dagger\mathbf{H}_{HO}\mathbf{S} = \begin{pmatrix} E_1 & 0 & 0 \\ 0 & \ddots & 0 \\ 0 & 0 & E_N \end{pmatrix}$

columns of $\mathbf{S}^\dagger$ are eigenvectors in HO basis set!

1. Express matrix of $\mathbf{x}$ in H-O basis (automatic; easy to program a computer to do this), get $\mathbf{x}_{HO}$.
2. Diagonalize $\mathbf{x}_{HO}$. Get $\mathbf{x}_x$ and $\mathbf{T}$.
3. Trivial to write $V(x)_x$ as $V(x_i) = \mathbf{V}(\mathbf{x})_x$ in $\mathbf{x}$ basis
4. Transform $\mathbf{V}(\mathbf{x})_x$ back to $\mathbf{V}(x)_{HO}$
5. Diagonalize $\mathbf{H}_{HO}$.

Solve many V(x) problems in this basis set.

1000×1000 $\mathbf{T}$ matrix diagonalizes $\mathbf{x} \Rightarrow 1000$ x_i's

Save the $\mathbf{T}$ and the $\{x_i\}$ for future use on *all* V(x) problems.

To verify convergence, repeat for new $\mathbf{x}$ matrix of dimension 1100×1100. There will be no obvious resemblance between

$$\{\mathbf{x}_i\}_{1000} \text{ and } \{\mathbf{x}_i\}_{1100}.$$

If the lowest energy eigenvalues of $\mathbf{H}$ (i.e. the ones you care about) do not change (by measurement accuracy), converged!

Next: Matrix Solution of Harmonic Oscillator (completely without wavefunctions, starting from the [$\mathbf{x}$,$\mathbf{p}$] commutation rule)

Then (at last) Perturbation Theory

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Matrix Solution of Harmonic Oscillator

Last time:

* $\mathbf{T}^\dagger \mathbf{A}^\phi \mathbf{T} = \begin{pmatrix} a_1 & 0 & 0 \\ 0 & \ddots & 0 \\ 0 & 0 & a_N \end{pmatrix}_\psi$ transformation to the diagonal form of **A**

* eigenbasis $|i\rangle = \begin{pmatrix} T^\dagger_{1i} \\ \vdots \\ T^\dagger_{Ni} \end{pmatrix}_\phi$ $i-th$ column of $\mathbf{T}^\dagger$ eigenbasis for **A**

* matrix representation of a function of a matrix is given by $\mathbf{T} f\left(\mathbf{T}^\dagger \mathbf{x} \mathbf{T}\right) \mathbf{T}^\dagger$

e.g. $f(x) = \mathbf{x}^N = \mathbf{T}\overbrace{(\mathbf{T}^\dagger \mathbf{x} \mathbf{T})(\mathbf{T}^\dagger \mathbf{x} \mathbf{T})\ldots}^{\text{N diagonal terms}}\mathbf{T}^\dagger = \mathbf{T}\tilde{\mathbf{x}}^N \mathbf{T}^\dagger = \mathbf{x}^N$

* Discrete Variable Representation: Matrix representation for *any* 1-D problem

Matrix version of numerical integration — works even for repulsive V(x) via addition of an infinite well

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TODAY: Harmonic Oscillator: Derive all matrix elements of **x**, **p**, **H** from the [**x**,**p**] commutation rule and the definition of **H**.

Example of how one can get matrix results entirely from commutation rule definitions (e.g. we will soon see this for an angular momentum: $\mathbf{J}^2$, $\mathbf{J}_x$, $\mathbf{J}_y$, $\mathbf{J}_z$, and the Wigner-Eckart Theorem)

NO WAVEFUNCTIONS, NO INTEGRALS, ALL MAGIC!

Outline of steps:

1. Assumptions
   * $\mathbf{H} = \frac{\mathbf{p}^2}{2m} + \frac{k\mathbf{x}^2}{2}$ Specific Model
   * eigen-basis exists for **H**
   * $\left[\hat{x},\hat{p}\right] = i\hbar$ **Central postulate of QM**
   * $\hat{x}$ and $\hat{p}$ are Hermitian (they have real expectation values)
2. x_{nm} and p_{nm} in terms of $(E_n - E_m)$
3. x_{nm} in terms of p_{nm}
4. Block Diagonalize **x, p, H** matrices (this is the most difficult step to understand)
5. Lowest quantum number must exist (call it 0) → explicit values for $|x_{01}|^2$ and $|p_{01}|^2$
6. Recursion relationship for $x_{nn\pm1}$ and $p_{nn\pm1}$ (x_{nn+1} from $x_{n-1n}\ldots$)
7. Magnitudes and phases for $x_{nn\pm1}$ and $p_{nn\pm1}$
8. Possibility of noncommunicating blocks along diagonal of **H, x, p**?

See CTDL pages 488-500 for similar treatment. IN MORE ELEGANT NOTATION
You will never use this methodology - only the results!

Enjoy this!

1. recall assumptions
2. **x** and **p** matrix elements are derived here from Commutation Rules: x_{nm} and p_{nm} in terms of $E_n - E_m$

$$[\mathbf{x},\mathbf{H}]=\left[\mathbf{x},\frac{\mathbf{p}^2}{2m}+\frac{1}{2}k\mathbf{x}^2\right]=\frac{1}{2m}\left[\mathbf{x},\mathbf{p}^2\right]=\frac{1}{2m}\left(\mathbf{p}\underbrace{[\mathbf{x},\mathbf{p}]}_{i\hbar}+\underbrace{[\mathbf{x},\mathbf{p}]}_{i\hbar}\mathbf{p}\right)=\frac{2i\hbar}{2m}\mathbf{p}$$

** $[\mathbf{x},\mathbf{p}]=i\hbar$ $\quad\rightarrow[\mathbf{x},\mathrm{H}]=\dfrac{\mathbf{p}}{2\mathrm{m}}2i\hbar=\dfrac{i\hbar}{\mathrm{m}}\mathbf{p}$

$$\mathbf{p}=\left(\frac{\mathrm{m}}{i\hbar}\right)[\mathbf{x},\mathbf{H}]$$

Take *n*,*m* matrix elements of both sides, insert completeness operator, $\sum_{\ell}|\ell\rangle\langle\ell|$, between **x** and **H**.

$$p_{nm}=\left(\frac{m}{i\hbar}\right)\sum_{\ell}\left(x_{n\ell}H_{\ell m}-H_{n\ell}x_{\ell m}\right).$$

Similarly, starting from $[\mathbf{p},\mathbf{H}]=\left[\mathbf{p},\tfrac{1}{2}k\mathbf{x}^2\right]=-i\hbar k\mathbf{x}$

rearrange and solve for x_{nm}

$$x_{nm}=\frac{i}{k\hbar}\sum_{\ell}\left(p_{n\ell}H_{\ell m}-H_{n\ell}p_{\ell m}\right).$$

But we know that some basis set of functions (the "harmonic oscillator eigenbasis") must exist in which **H** is diagonal. Use it implicitly: ∴ replace $H_{\ell m}$ by $E_m\delta_{m\ell}$. This kills the sum over ℓ.

$$p_{nm}=\left(\frac{m}{ih}\right)\left(x_{nm}E_m-E_nx_{nm}\right)$$

$$p_{nm}=\left(\frac{m}{i\hbar}\right)x_{nm}\left(E_m-E_n\right)$$ so for the special case of $n = m$, $p_{nn} = 0$

(but, in addition, $p_{nn} = 0$ if **H** for the harmonic oscillator has a degenerate eigenvalue, then $p_{nm} = 0$ if $E_n = E_m$)

similarly for the x_{nm} equation

$$x_{nm}=\frac{i}{\hbar k}p_{nm}\left(E_m-E_n\right)$$

∴ $x_{nn} = 0$ (and $x_{nm} = 0$ if $E_n = E_m$)

3. solve for the non-zero x_{nm} in terms of the non-zero p_{nm}

multiply the x_{nm} equation by p_{nm}
multiply the p_{nm} equation by x_{nm}

The LHSs of both resulting equations are equal

equate RHSs: $\frac{\mathrm{m}}{\mathrm{i}\hbar} x_{nm}^2 (E_m - E_n) = \frac{i}{\hbar k} p_{nm}^2 (E_m - E_n)$

* If $E_n = E_m$ (degeneracy) – then we already know that $x_{nm} = 0$, $p_{nm} = 0$

* If $E_n \neq E_m$ then we can divide by $(E_m - E_n)$ and rearrange

$$x_{nm}^2 = -\frac{1}{km} p_{nm}^2$$

$$x_{nm} = \pm i (km)^{-1/2} p_{nm}$$

THERE IS A PHASE AMBIGUITY HERE!

Need to find out what phase choice is consistent with other requirements

Earlier we derived $p_{nm} = \frac{m}{i\hbar} x_{nm} (E_m - E_n)$

plug in new result for x_{nm} $p_{nm} = \frac{m}{i\hbar} \left(\pm i (km)^{-1/2} \right) p_{nm} (E_m - E_n)$

Thus either:

* $p_{nm} \neq 0$ AND $E_m - E_n = \pm \hbar (k/m)^{1/2} \equiv \pm \hbar\omega$!!

(Why is it OK to divide thru by p_{nm}?)

OR

when $E_m \neq E_n \pm \hbar\omega$

* $p_{nm} = 0 \Rightarrow x_{nm} = 0$

This is the most difficult but most crucial point in the logic.

The *only* non-zero off-diagonal matrix elements of **x** and **p** involve eigenfunctions of **H** that have energies differing by exactly $\hbar\omega$!
A "selection rule"! The only nonzero matrix elements of **x** and **p** are those where the indices for each non-zero x_{nm} and p_{nm} differ by ±1.

Updated 8/13/20 8:23 AM

4. **x, p, H** are block "diagonalized"

within this set one finds all of the non-zero elements of **x** and **p**.

In what sense? There is a set of eigenstates of **H** that have energies that fall onto a comb of evenly spaced $E_n^{(1)}$ values:

$$E_n^{(1)} = n(\hbar\omega) + \varepsilon_1$$

But there could be another set:

$$E_n^{(2)} = n(\hbar\omega) + \varepsilon_2 \qquad \text{where } \varepsilon_2 - \varepsilon_1 \neq n\hbar\omega \text{ for all n}$$

Set II $E_0^{(2)}$ is lowest

$E_1^{(2)} = E_0^{(2)} + \hbar\omega$

But within each set, there must be a lowest energy level

Set I $E_0^{(1)}$ is lowest

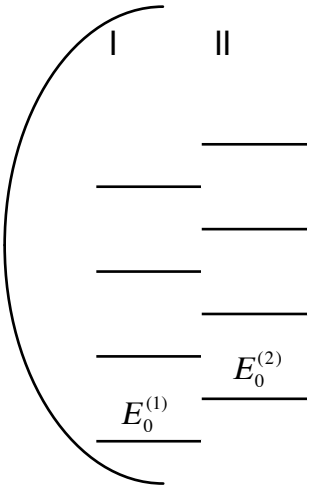
$E_1^{(1)} = E_0^{(1)} + \hbar\omega$

etc.

Set I $E_0^{(2)}$ is lowest

$E_1^{(2)} = E_0^{(2)} + \hbar\omega$

etc.



possibly interleaved but are they non-communicating?

Since **x** and **p** have nonzero elements only within communicating sets for **H**, thus **x**, **p**, **H** are block diagonalized into sets I, II, etc.

$$\mathbf{H,x,p} = \begin{pmatrix} \mathrm{I} & 0 & 0 & 0 \\ 0 & \mathrm{II} & 0 & 0 \\ 0 & 0 & \mathrm{III} & 0 \\ 0 & 0 & 0 & \ddots \end{pmatrix}$$

We will eventually show that all of these blocks along the diagonal are identical (and that each energy level is nondegenerate). If **x, p** are "block diagonal", then $\mathbf{x^2}$, $\mathbf{p}^2$ are similarly block diagonal.

5. <u>A lowest index must exist within each block. Call it 0</u>.

$[\mathbf{x},\mathbf{p}] = i\hbar$ is a diagonal matrix: $i\hbar\mathbb{1}$

$$\sum_{\ell}\left(x_{n\ell}p_{\ell m} - p_{n\ell}x_{\ell m}\right) = i\hbar\delta_{nm} \quad \text{because } [\mathbf{x},\mathbf{p}] = i\hbar\mathbb{1}$$

$(\mathbf{xp})_{nm}$ $(\mathbf{px})_{nm}$

$$i\hbar = \left(x_{nn+1}p_{n+1n} - p_{nn+1}x_{n+1n}\right) + \left(x_{nn-1}p_{n-1n} - p_{nn-1}x_{n-1n}\right)$$

indices must be equal

These are the only surviving nonzero terms in the sum over ℓ. because we showed that the ***x,p*** matrix elements are non-zero only between states with E different by $\pm\hbar\omega$. This is the basis for the indexing of the basis states within a block.

There *must* be a lowest E_i because, classically,

$E = T + V$ and $T \geq 0$, $E \geq V_{min}$.

Let $n = 0$ be the lowest allowed index (this is an arbitrary choice of labels).

$p_{0,-1} = x_{0,-1} = 0$ (–1 is not an allowed index.)

$$\boxed{x_{01}p_{10} - p_{01}x_{10} = i\hbar}$$

used Hermiticity here

$\mathbf{x},\mathbf{p}$ are Hermitian $\left(\mathbf{A} = \mathbf{A}^{\dagger}\right)$ thus $x_{01}p_{01}^{*} - p_{01}x_{01}^{*} = i\hbar$

we are going to insert this into the above $x_{01}p_{10}$ equation

previously $x_{nm} = \pm i(km)^{-1/2}p_{nm}$ (note that the same symbol is used for mass and one of the basis state indices)

$x_{01} = \pm i(km)^{-1/2}p_{01}$

We must make phase choices so that both $\mathbf{x}$ and $\mathbf{p}$ are Hermitian.

Phase ambiguity: we are free to specify the absolute phase of **x** or **p** BUT NOT BOTH because that would affect the value of [**x**,**p**]

BY CONVENTION:

all non-zero matrix elements of **x** are REAL
all non-zero matrix elements of **p** are IMAGINARY

try $x_{01} = +i(km)^{-1/2} p_{01}$ and eliminate p_{01} by plugging this value of x_{01} into $x_{01}p_{01}^* - p_{01}x_{01}^* = i\hbar$: we get $i(km)^{-1/2}\left|p_{01}\right|^2 + i(km)^{-1/2}\left|p_{01}\right|^2 = i\hbar$

thus

$$\left|x_{01}\right|^2 = \frac{\hbar}{2}(km)^{-1/2}$$

$$\left|p_{01}\right|^2 = \frac{\hbar}{2}(km)^{+1/2} \text{ !!!}$$

[If we had chosen $x_{01} = -i(km)^{-1/2} p_{01}$ we would have obtained $\underbrace{\left|x_{01}\right|^2}_{\geq 0} = -\frac{\hbar}{2}(km)^{1/2}$ which is impossible!]

There are two things that must be checked for self-consistency of seemingly arbitrary phase choices at every opportunity:

* Hermiticity
* $\left|\quad\right|^2 \geq \mathbf{0}$.

6. Recursion Relation for $\left|x_{n,n+1}\right|^2$

Start again with the general equation derived in part #3 above using the phase choice that worked in part #5 above

$$\underbrace{x_{nn+1}}_{\text{index increasing by 1}} = i(km)^{-1/2} \underbrace{p_{nn+1}}_{\text{index increasing by 1}}$$

Hermiticity of **x** and **p**: $x_{n+1n}^* = i(km)^{-1/2} p_{n+1n}^*$

c.c. of both sides

$$\underbrace{x_{n+1n}}_{\text{index decreasing by 1}} = -i(km)^{-1/2} \underbrace{p_{n+1n}}_{\text{index decreasing by 1}}$$

$\therefore \quad x_{nn\pm1} = \pm i(km)^{-1/2}\, p_{nn\pm1}$

these 3 choices of ± go together

Now the arbitrary part of the phase ambiguity in the relationship between **x** and **p** has been eliminated

Apply this to the general term in [**x**,**p**] ⇒ requires a lot of algebra

---

NONLECTURE : evaluation of the four non-zero terms in [**x**,**p**] = i$\hbar$, given here:

$$x_{nn+1}p_{n+1n} = x_{nn+1}p^{*}_{nn+1} = x_{nn+1}\left(-\frac{(km)^{1/2}}{i}x^{*}_{nn+1}\right)$$
$$= \left|x_{nn+1}\right|^2\left(+i(km)^{1/2}\right)$$

$$-p_{nn+1}x_{n+1n} = -\left(\frac{(km)^{1/2}}{i}x_{nn+1}\right)\left(x^{*}_{nn+1}\right) = \left|x_{nn+1}\right|^2\left(+i(km)^{1/2}\right)$$

$$x_{nn-1}p_{n-1n} = x_{nn-1}p^{*}_{nn-1} = x_{nn-1}\left(+\frac{(km)^{1/2}}{i}x^{*}_{nn-1}\right)$$
$$= \left|x_{nn-1}\right|^2\left(-i(km)^{1/2}\right)$$

$$-p_{nn-1}x_{n-1n} = -\left(-\frac{(km)^{1/2}}{i}x_{nn-1}\right)\left(x^{*}_{nn-1}\right) = \left|x_{nn-1}\right|^2\left(-i(km)^{1/2}\right)$$

Combine the 4 terms in [**x**,**p**] = i$\hbar$ to get

$$i\hbar = 2i(km)^{1/2}\left[\left|x_{nn+1}\right|^2 - \left|x_{nn-1}\right|^2\right]$$

$$\left|x_{nn+1}\right|^2 = \frac{\hbar(km)^{-1/2}}{2} + \left|x_{nn-1}\right|^2 \qquad \text{recursion relation}$$

but $\left|x_{01}\right|^2 = \left|x_{10}\right|^2 = \frac{\hbar}{2}(km)^{-1/2}$

OK to reverse indices in $\left|\ \right|^2$

$$\left|x_{n+1n}\right|^2 \leftarrow \left|x_{nn+1}\right|^2$$

Each step-up in quantum number produces another additive term: $\frac{\hbar}{2}(km)^{-1/2}$

thus

$$\left|x_{nn+1}\right|^2 = (n+1)\frac{\hbar}{2}(km)^{-1/2}$$
$$\left|p_{nn+1}\right|^2 = (n+1)\frac{\hbar}{2}(km)^{+1/2}$$

general result

7. Magnitudes and Phases for $x_{nn\pm1}$ and $p_{nn\pm1}$

Verify phase consistency and Hermiticity for **x** and **p.**

In part #3 we derived $x_{nn\pm1} = \pm i(km)^{-1/2} p_{nn\pm1}$

one self-<u>consistent set is</u>

x real and positive

$$\begin{cases} x_{nn+1} = +(n+1)^{1/2}\left(\dfrac{\hbar}{2(km)^{1/2}}\right)^{1/2} = +x_{n+1n} \\ x_{nn-1} = +(n)^{1/2}\left(\dfrac{\hbar}{2(km)^{1/2}}\right)^{1/2} = +x_{nn-1} \end{cases}$$

AND

p imaginary with sign flip for up vs. down

$$\begin{cases} p_{nn+1} = -i(n+1)^{1/2}\left(\dfrac{\hbar(km)^{1/2}}{2}\right)^{1/2} = -p_{n+1n} \\ p_{nn-1} = +i(n)^{1/2}\left(\dfrac{\hbar(km)^{1/2}}{2}\right)^{1/2} = -p_{n-1n} \end{cases}$$

Note that nonzero matrix elements of x and p are always ∝ to the SQRT of the larger quantum number.
This is the usual phase convention: $x_{nn\pm1}$ real and positive, $p_{nn+1} = -p_{n+1n}$ imaginary.
Must be careful about phase choices because in the matrix form of QM one never really looks at wavefunctions, operators, or integrals

---

8. <u>Possible existence of noncommunicating blocks along the diagonal of **H, x, p**</u>

You show that $H_{nm} = \left(n+1/2\right)\hbar\left(\dfrac{k}{m}\right)^{1/2}\delta_{nm}$ from definition of **H**

$$\left(\begin{array}{l} \text{Note that } \mathbf{x}^2 \text{ and } \mathbf{p}^2 \text{ have nonzero } \Delta n = \pm 2 \text{ elements but} \\ \dfrac{1}{2}k\mathbf{x}^2 + \dfrac{\mathbf{p}^2}{2m} \text{ has cancelling contributions in the } \Delta n = +2 \text{ and } \Delta n = -2 \text{ locations} \end{array}\right)$$

This result implies

* all of the possibly independent blocks in **x**, **p**, **H** are identical because they all start from the same lowest energy
* $E^{(i)}_n = n\hbar\omega + \varepsilon_i$ where $\varepsilon_i = (1/2)\hbar\omega$ for all i
* degeneracy of all E_n? All E_n must have the same degeneracy, but I can't prove that this degeneracy is 1. Are there parallel non-communicating universes?

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Fall 2018

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End of Matrix Solution of H-O, and Feel the Power of the a and a† Operators

1. starting from $\mathbf{H} = \frac{\mathbf{p}^2}{2m} + \frac{1}{2}k\mathbf{x}^2$ *and* $[\mathbf{x},\mathbf{p}] = i\hbar$

2. we showed $p_{nm} = \frac{m}{i\hbar} x_{nm}\left(E_m - E_n\right)$

$$x_{nm} = \frac{i}{\hbar k} p_{nm}\left(E_m - E_n\right)$$

$$\therefore x_{nn} = 0, p_{nn} = 0 \text{ and } \left\{ \begin{array}{c} x_{nm} \\ p_{nm} \end{array} \right\} = 0 \text{ if } E_n = E_m$$

3. $x_{nm}^2 = -\frac{1}{km} p_{nm}^2$

$E_m - E_n = \pm\hbar\omega \qquad \omega = \left(k/m\right)^{1/2}$

$\therefore$ the only non-zero **x** and **p** elements are between states whose E' s differ by $\pm\hbar\omega$

4. combs of connected states, "block diagonalization" of **H**, **x**, **p**, $\mathbf{x}^2$, $\mathbf{p}^2$: $E_n^{(i)} = \hbar\omega n + \varepsilon_i$ ← constant offset for block i

5. lowest index must exist because lowest E must exist. Call this index 0

$$|x_{01}|^2 = \frac{\hbar}{2}(km)^{-1/2}$$

$$|p_{01}|^2 = \frac{\hbar}{2}(km)^{+1/2}$$

from arbitrary (but almost universally chosen) phase choice

$$x_{01} = +i(km)^{-1/2} p_{01}$$

Today

6. Recursion Relationship: $|x_{nn+1}|^2$ in terms of $|x_{nn-1}|^2$

⇩

general matrix elements $|x_{nn+1}|^2, |p_{nn+1}|^2$

7. general **x** and **p** elements

8. the only blocks of **H** correspond to $\varepsilon_i = \frac{1}{2}\hbar\omega$

We are ready to derive a powerful, compact, and intuitive algebra:

Dimensionless $\underset{\sim}{\mathbf{x}}$, $\underset{\sim}{\mathbf{p}}$, $\underset{\sim}{\mathbf{H}}$ and **a** (annihilation) and $\mathbf{a}^\dagger$ (creation) operators.

Partial repetition from Lecture #12

phase ambiguity: we can specify absolute phase of **x** or **p** BUT NOT BOTH because that would affect the value of [**x**,**p**]

BY CONVENTION:

matrix elements of **x** are REAL
p are IMAGINARY

try $x_{01} = +i(km)^{-1/2} p_{01}$ and plug this into

$x_{01}p_{01}^{*} - p_{01}x_{01}^{*} = i\hbar$

get

$$|x_{01}|^2 = \frac{\hbar}{2}(km)^{-1/2}$$

$$|p_{01}|^2 = \frac{\hbar}{2}(km)^{+1/2}$$

[If we had chosen $x_{01} = -i(km)^{-1/2} p_{01}$ we would have obtained $|x_{01}|^2 = -\frac{\hbar}{2}(km)^{1/2}$ which is impossible!]

check for self-consistency of seemingly arbitrary phase choices at every opportunity: * Hermiticity $\left(\mathbf{A}^{\dagger} = \mathbf{A}, \quad A_{ij}^{*} = A_{ji}\right)$

* $\left|\quad\right|^2 \geq 0$ for any A_{ij}

6. Recursion Relation for $|x_{ii+1}|^2$

start again with general equation derived in part #3 of Lecture #12 using the phase choice that worked

$x_{nn+1} = i(km)^{-1/2} p_{nn+1}$

index increasing

Hermiticity $x_{n+1n}^{*} = i(km)^{-1/2} p_{n+1n}^{*}$

c.c. of both sides $x_{n+1n} = -i(km)^{-1/2} p_{n+1n}$

index decreasing

$$\therefore \quad x_{nn\pm1} = \pm i\left(km\right)^{-1/2} p_{nn\pm1}$$

now the arbitrary part of the phase ambiguity in the relationship between **x** and **p** is eliminated

Apply this to the general term in [**x**,**p**] ⇒ algebra

NONLECTURE : from four terms in [**x**,**p**] = $i\hbar$

$$x_{nn+1}p_{n+1n} = x_{nn+1}p^{*}_{nn+1} = x_{nn+1}\left(-\frac{(km)^{1/2}}{i}x^{*}_{nn+1}\right)$$
$$= \left|x_{nn+1}\right|^2\left(+i(km)^{1/2}\right)$$

$$-p_{nn+1}x_{n+1n} = -\left(\frac{(km)^{1/2}}{i}x_{nn+1}\right)\left(x^{*}_{nn+1}\right) = \left|x_{nn+1}\right|^2\left(+i(km)^{1/2}\right)$$

$$x_{nn-1}p_{n-1n} = x_{nn-1}p^{*}_{nn-1} = x_{nn-1}\left(+\frac{(km)^{1/2}}{i}x^{*}_{nn-1}\right)$$
$$= \left|x_{nn-1}\right|^2\left(-i(km)^{1/2}\right)$$

$$-p_{nn-1}x_{n-1n} = -\left(-\frac{(km)^{1/2}}{i}x_{nn-1}\right)\left(x^{*}_{nn-1}\right) = \left|x_{nn-1}\right|^2\left(-i(km)^{1/2}\right)$$

$$\therefore i\hbar = 2i\left(km\right)^{1/2}\left[\left|x_{nn+1}\right|^2 - \left|x_{nn-1}\right|^2\right]$$

$$\left|x_{nn+1}\right|^2 = \frac{\hbar(km)^{-1/2}}{2} + \left|x_{nn-1}\right|^2$$ recursion relation

but $\left|x_{01}\right|^2 = \left|x_{10}\right|^2 = \frac{\hbar}{2}\left(km\right)^{-1/2}$

thus

$$\left|x_{nn+1}\right|^2 = (n+1)\frac{\hbar}{2}(km)^{-1/2}$$
$$\left|p_{nn+1}\right|^2 = (n+1)\frac{\hbar}{2}(km)^{+1/2}$$

general result

7. Magnitudes and Phases for $x_{nn\pm1}$ and $p_{nn\pm1}$

verify phase consistency and Hermiticity for **x** and **p**

in #3 we derived $x_{nn\pm1} = \pm i(km)^{-1/2} p_{nn\pm1}$

one self-consistent set is

matrix elements of **x** real and positive

$$x_{nn+1} = +(n+1)^{1/2}\left(\frac{\hbar}{2(km)^{1/2}}\right)^{1/2} = +x_{n+1n}$$

$$x_{nn-1} = +(n)^{1/2}\left(\frac{\hbar}{2(km)^{1/2}}\right)^{1/2} = +x_{nn-1} \qquad (km)^{1/2} = m\omega$$

AND

matrix elements of **p** imaginary with sign flip for up vs. down

$$p_{nn+1} = -i(n+1)^{1/2}\left(\frac{\hbar(km)^{1/2}}{2}\right)^{1/2} = -p_{n+1n}$$

$$p_{nn-1} = +i(n)^{1/2}\left(\frac{\hbar(km)^{1/2}}{2}\right)^{1/2} = -p_{n-1n}$$

This is the usual phase convention

Phase is a recurrent problem in matrix mechanics because we never look at wavefunctions or evaluate integrals explicitly.

---

8. Possible existence of noncommunicating blocks along diagonal of **H, x, p**

you show that $H_{nm} = (n+1/2)\hbar\left(\frac{k}{m}\right)^{1/2}\delta_{nm}$

(note that $\mathbf{x}^2$ and $\mathbf{p}^2$ have non-zero $\Delta n = \pm 2$ elements but $\frac{1}{2}k\mathbf{x}^2 + \frac{\mathbf{p}^2}{2m}$ has cancelling contributions in $\Delta n = \pm 2$ locations)

This result implies

* all of the possibly independent blocks in **x**, **p**, **H** are identical
* $\varepsilon_i = (1/2)\hbar\omega$ for all i
* degeneracy of all E_n? all are the same, but can't prove that this universal degeneracy is 1.

End of repetition from Lecture #12

updated 8/13/20 8:23 AM

Creation and Annihilation Operators (CTDL pages 488-508)

* Dimensionless operators
* simple operator algebra rather than complicated real algebra
* matrices arranged according to "selection rules"
* matrix elements calculated by extremely simple rules
* automatic generation of any basis function by repeated operations on lowest (nodeless) basis state

---

Get rid of system-specific factors k, μ, ω and also $\hbar$.

$$\omega = (k/m)^{1/2}$$

$$\underset{\sim}{\mathbf{x}} \equiv \left(\frac{m\omega}{\hbar}\right)^{1/2} \mathbf{x} \qquad \mathbf{x}^2 = \left(\frac{\hbar}{m\omega}\right)\underset{\sim}{\mathbf{x}}^2$$

dimensionless ↑ ; regular ↑

$$\underset{\sim}{\mathbf{p}} \equiv (\hbar m\omega)^{-1/2}\mathbf{p} \qquad \mathbf{p}^2 = \hbar m\omega\,\underset{\sim}{\mathbf{p}}^2$$

We choose these factors to make everything come out dimensionless.

$$\underset{\sim}{\mathbf{H}} = \frac{1}{\hbar\omega}\mathbf{H} = \frac{1}{2}\left(\underset{\sim}{\mathbf{x}}^2 + \underset{\sim}{\mathbf{p}}^2\right) \qquad \mathbf{H} = \frac{1}{2}k\mathbf{x}^2 + \frac{\mathbf{p}^2}{2m} = \frac{1}{2}\hbar\omega\left(\underset{\sim}{\mathbf{x}}^2 + \underset{\sim}{\mathbf{p}}^2\right)$$

$$\frac{1}{2}k\left(\frac{\hbar}{m\omega}\right) = \frac{1}{2}\frac{k}{m}\frac{\hbar}{\omega} = \frac{\omega^2}{2}\frac{\hbar}{\omega} = \frac{\hbar\omega}{2}$$

$$\frac{1}{2m}(\hbar m\omega) = \frac{1}{2}\hbar\omega$$

$$\left[\underset{\sim}{\mathbf{x}}, \underset{\sim}{\mathbf{p}}\right] = \left(\frac{m\omega}{\hbar}\frac{1}{\hbar m\omega}\right)^{1/2}\left[\mathbf{x},\mathbf{p}\right] = \frac{1}{\hbar}(i\hbar) = i$$

dimensionless

from results for **x, p, H**

$$\underset{\sim}{x}_{mn} = 2^{-1/2}\left[(n+1)^{1/2}\delta_{mn+1} + n^{1/2}\delta_{mn-1}\right]$$

$$\underset{\sim}{p}_{mn} = 2^{-1/2}i\left[(n+1)^{1/2}\delta_{mn+1} - n^{1/2}\delta_{mn-1}\right]$$

$$\underset{\sim}{H}_{mn} = (n+1/2)\delta_{mn} \qquad \text{diagonal}$$

square root of larger quantum number

note the negative sign

Kronecker − δ's specify selection rules for all nonzero matrix elements

Now define something new: use **a**, $\mathbf{a}^\dagger$ to clean things up even more!

$$\mathbf{a} = 2^{-1/2}\left(\underset{\sim}{\mathbf{x}} + i\,\underset{\sim}{\mathbf{p}}\right) \qquad \mathbf{a}^\dagger = 2^{-1/2}\left(\underset{\sim}{\mathbf{x}} - i\,\underset{\sim}{\mathbf{p}}\right)$$

$$\underset{\sim}{\mathbf{x}} = 2^{-1/2}\left(\mathbf{a} + \mathbf{a}^\dagger\right) \qquad \underset{\sim}{\mathbf{p}} = 2^{-1/2} i\left(\mathbf{a}^\dagger - \mathbf{a}\right)$$

Let's examine the matrix elements of $\mathbf{a}$ and $\mathbf{a}^\dagger$

now plug in m,n matrix elements of $\underset{\sim}{\mathbf{x}}$ and $\underset{\sim}{\mathbf{p}}$ from previous page

$$a_{mn} = \left[2^{-1/2}\,\underset{\sim}{x}_{mn} + 2^{-1/2}\,i\,\underset{\sim}{p}_{mn}\right]$$

$$= \left[\frac{1}{2}(n+1)^{1/2}\delta_{mn+1} - \frac{1}{2}(n+1)^{1/2}\delta_{mn+1} + \frac{1}{2}n^{1/2}\delta_{mn-1} + \frac{1}{2}n^{1/2}\delta_{mn-1}\right]$$

group terms according to "selection rule"

$\underset{\sim}{\mathbf{x}}$ $i\underset{\sim}{\mathbf{p}}$ — two terms cancel

$\underset{\sim}{\mathbf{x}}$ $i\underset{\sim}{\mathbf{p}}$ — two terms add

$$\boxed{a_{mn} = n^{1/2}\delta_{mn-1}}$$

the first (left) index is one smaller than the second (right)

column

$$a_{mn} = \langle m|\mathbf{a}|n\rangle = n^{1/2}$$

row

$$n^{1/2}\,|\,n-1\rangle$$

$\mathbf{a}$ is lowering or "annihilation" operator

similarly

$$\boxed{a^\dagger_{mn} = (n+1)^{1/2}\,\delta_{mn+1}}$$

$$a^\dagger_{mn} = \left\langle m\left|\mathbf{a}^\dagger\right|n\right\rangle = (n+1)^{1/2}$$

the first index is one larger than the second

$\mathbf{a}^\dagger$ is a "creation" operator

$$\mathbf{a}^\dagger = \begin{array}{c} \\ 0 \\ 1 \\ 2 \\ 3 \\ 4 \end{array} \begin{array}{c} \text{columns} \\ \begin{array}{ccccc} 0 & 1 & 2 & 3 & 4 \end{array} \\ \begin{pmatrix} 0 & 0 & 0 & 0 & 0 \\ 1^{1/2} & 0 & 0 & 0 & 0 \\ 0 & 2^{1/2} & 0 & 0 & 0 \\ 0 & 0 & \boxed{3^{1/2}} & 0 & 0 \\ 0 & 0 & 0 & 4^{1/2} & 0 \end{pmatrix} \end{array} \quad \text{rows}$$

Square root of integers always only one step below main diagonal. $\mathbf{a}$, $\mathbf{a}^\dagger$ are obviously not Hermitian!

$$\mathbf{a} = \begin{pmatrix} 0 & 1^{1/2} & 0 & 0 & 0 \\ 0 & 0 & 2^{1/2} & 0 & 0 \\ 0 & 0 & 0 & 3^{1/2} & 0 \\ 0 & 0 & 0 & 0 & \boxed{4^{1/2}} \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Square root of integers always only one step above main diagonal.

e.g. $\langle 3|\mathbf{a}^\dagger|2\rangle = 3^{1/2}$

row — $\mathbf{a}^\dagger$ raises — column

e.g. $\langle 3|\mathbf{a}|4\rangle = 4^{1/2}$

$\mathbf{a}$ lowers

What is so great about $\mathbf{a}$, $\mathbf{a}^\dagger$?

$\mathbf{a}|n\rangle = n^{1/2}|n-1\rangle$ annihilates 1 quantum

$\mathbf{a}^\dagger|n\rangle = (n+1)^{1/2}|n+1\rangle$ creates 1 quantum

$|n\rangle = [n!]^{-1/2}(\mathbf{a}^\dagger)^n|0\rangle$ generate any state, $|n\rangle$, from the lowest state, $|0\rangle$

needed to normalize. Each application of $\mathbf{a}^\dagger$ gives the next larger integer. Do it n times on $|0\rangle$, get $n!\,|n\rangle$.

More tricks: look at $\mathbf{aa}^\dagger$ and $\mathbf{a}^\dagger\mathbf{a}$

is $\mathbf{aa}^\dagger$ Hermitian?

$[(\mathbf{AB})^\dagger = \mathbf{B}^\dagger\mathbf{A}^\dagger]$ definition of Hermitian

$$(\mathbf{aa}^\dagger)^\dagger = \mathbf{a}^{\dagger\dagger}\mathbf{a}^\dagger = \mathbf{aa}^\dagger$$

$\therefore$ $\mathbf{aa}^\dagger$ and $\mathbf{a}^\dagger\mathbf{a}$ are Hermitian — to what "observable" quantity do they correspond? We will see that one of these is called the "number operator."

$$\mathbf{aa}^\dagger = \frac{1}{2}\left(\underset{\sim}{\mathbf{x}} + i\underset{\sim}{\mathbf{p}}\right)\left(\underset{\sim}{\mathbf{x}} - i\underset{\sim}{\mathbf{p}}\right) = \frac{1}{2}\left(\underset{\sim}{\mathbf{x}}^2 + i\underset{\sim}{\mathbf{p}}\underset{\sim}{\mathbf{x}} - i\underset{\sim}{\mathbf{x}}\underset{\sim}{\mathbf{p}} + \underset{\sim}{\mathbf{p}}^2\right)$$

$$= \frac{1}{2}\left(\underset{\sim}{\mathbf{x}}^2 + \underset{\sim}{\mathbf{p}}^2 - i\underbrace{[\underset{\sim}{\mathbf{x}}, \underset{\sim}{\mathbf{p}}]}_{i}\right) = \frac{1}{2}\left(\underset{\sim}{\mathbf{x}}^2 + \underset{\sim}{\mathbf{p}}^2 + 1\right)$$

similarly $\mathbf{a}^\dagger\mathbf{a} = \frac{1}{2}\left(\underset{\sim}{\mathbf{x}}^2 + \underset{\sim}{\mathbf{p}}^2 - 1\right)$

$$\left.\begin{aligned} \therefore \underset{\sim}{\mathbf{H}} &= \frac{1}{2}\left(\mathbf{a}^\dagger\mathbf{a} + \mathbf{aa}^\dagger\right) \quad \text{and} \quad \left[\mathbf{a}, \mathbf{a}^\dagger\right] = 1 \\ \underset{\sim}{\mathbf{H}} &= \mathbf{a}^\dagger\mathbf{a} + 1/2 \end{aligned}\right\} \text{ simple form for } \underset{\sim}{\mathbf{H}}$$

$\underset{\sim}{\mathbf{H}}$ is the **number operator** + 1/2

$$\mathbf{H} = \hbar\omega\underset{\sim}{\mathbf{H}} = \hbar\omega\left(\mathbf{a}^\dagger\mathbf{a} + 1/2\right)$$

↑ # of quanta

$\mathbf{a}^\dagger\mathbf{a}|n\rangle = n|n\rangle$ $\mathbf{a}^\dagger\mathbf{a}$ is "number operator"

$\left[\mathbf{aa}^\dagger|n\rangle = (n+1)|n\rangle\right]$ not as useful

What have we done? We have exposed all of the "symmetry" and universality of the H–O basis set. We can now trivially work out what the matrix for any $\mathbf{x}^n\mathbf{p}^m$ operator looks like and organize it according to selection rules.

What about $\mathbf{x}^3$?

$$\mathbf{x}^3 = \left(\frac{m\omega}{\hbar}\right)^{-3/2} \underset{\sim}{\mathbf{x}}^3$$

When you multiply this out, preserve the order of $\mathbf{a}$ and $\mathbf{a}^\dagger$ factors.

$$\underset{\sim}{\mathbf{x}}^3 = \left(2^{-3/2}\right)\left(\mathbf{a}+\mathbf{a}^\dagger\right)^3$$

$$= \left(2^{-3/2}\right)\left[\mathbf{a}^3 + \left(\mathbf{a}^\dagger\mathbf{aa} + \mathbf{aa}^\dagger\mathbf{a} + \mathbf{aaa}^\dagger\right) + \left(\mathbf{aa}^\dagger\mathbf{a}^\dagger + \mathbf{a}^\dagger\mathbf{aa}^\dagger + \mathbf{a}^\dagger\mathbf{a}^\dagger\mathbf{a}\right) + \mathbf{a}^{\dagger 3}\right]$$

$\Delta n =$ $\quad -3, \quad -1, \quad +1, \quad +3$ (# of † minus # of non-†)

Simplify each group using commutation properties so that it has form

$$\mathbf{a}\left[\mathbf{a}^\dagger\mathbf{a}\right]|n\rangle \quad or \quad \mathbf{a}^\dagger\left[\mathbf{a}^\dagger\mathbf{a}\right]|n\rangle$$

$$\Downarrow \qquad\qquad \Downarrow$$

$$n^{1/2}n|n-1\rangle \qquad (n+1)^{1/2}n|n+1\rangle$$

NONLECTURE: Simplify the $\Delta n = -1$ terms.

$$\mathbf{a}^\dagger\mathbf{aa} = \mathbf{aa}^\dagger\mathbf{a}\underbrace{-\mathbf{aa}^\dagger\mathbf{a} + \mathbf{a}^\dagger\mathbf{aa}}_{[\mathbf{a}^\dagger,\mathbf{a}]\mathbf{a} = -\mathbf{a}} = \mathbf{aa}^\dagger\mathbf{a} - \mathbf{a}$$

add and subtract the term needed to reverse order

$$\mathbf{aaa}^\dagger = \mathbf{aa}^\dagger\mathbf{a}\underbrace{-\mathbf{aa}^\dagger\mathbf{a} + \mathbf{aaa}^\dagger}_{\mathbf{a}[\mathbf{a},\mathbf{a}^\dagger]=\mathbf{a}} = \mathbf{aa}^\dagger\mathbf{a} + \mathbf{a}$$

$$\left[\mathbf{a}^\dagger\mathbf{aa} + \mathbf{aa}^\dagger\mathbf{a} + \mathbf{aaa}^\dagger\right] = 3\mathbf{aa}^\dagger\mathbf{a}$$

try to put everthing into $\mathbf{aa}^\dagger\mathbf{a}$ order

---

NONLECTURE: Simplify the $\Delta n = +1$ terms.

$$\mathbf{aa}^\dagger\mathbf{a}^\dagger = \mathbf{a}^\dagger\mathbf{aa}^\dagger\underbrace{-\mathbf{a}^\dagger\mathbf{aa}^\dagger + \mathbf{aa}^\dagger\mathbf{a}^\dagger}_{\left[\mathbf{a},\mathbf{a}^\dagger\right]\mathbf{a}^\dagger=\mathbf{a}^\dagger} = \mathbf{a}^\dagger\mathbf{aa}^\dagger + \mathbf{a}^\dagger$$

$$\mathrm{a}^\dagger\mathbf{aa}^\dagger = \mathbf{a}^\dagger\mathbf{a}^\dagger\mathbf{a}\underbrace{-\mathbf{a}^\dagger\mathbf{a}^\dagger\mathbf{a} + \mathbf{a}^\dagger\mathbf{aa}^\dagger}_{\mathbf{a}^\dagger\left[\mathbf{a},\mathbf{a}^\dagger\right]=\mathbf{a}^\dagger} = \mathbf{a}^\dagger\mathbf{a}^\dagger\mathbf{a} + \mathbf{a}^\dagger$$

$$\left[\mathbf{aa}^\dagger\mathbf{a}^\dagger + \mathbf{a}^\dagger\mathbf{aa}^\dagger + \mathbf{a}^\dagger\mathbf{a}^\dagger\mathbf{a}\right] = 3\mathbf{a}^\dagger\mathbf{a}^\dagger\mathbf{a} + 3\mathbf{a}^\dagger$$

$\underline{\Delta n = \pm 3}$

$$\mathbf{a}^3|n\rangle = \left[n(n-1)(n-2)\right]^{1/2}|n-3\rangle$$

$$\mathbf{a}^{\dagger 3}|n\rangle = \left[(n+1)(n+2)(n+3)\right]^{1/2}|n+3\rangle$$

$\underline{\Delta n = \pm 1}$

$$\left[\mathbf{a}^\dagger\,\mathbf{aa} + \mathbf{aa}^\dagger\mathbf{a} + \mathbf{aaa}^\dagger\right]|n\rangle = 3\left(n^{3/2}\right)|n-1\rangle$$

$$\left[\mathbf{aa}^\dagger\mathbf{a}^\dagger + \mathbf{a}^\dagger\mathbf{aa}^\dagger + \mathbf{a}^\dagger\mathbf{a}^\dagger\mathbf{a}\right]|n\rangle = 3\left[(n+1)^{1/2}\,n|n+1\rangle + (n+1)^{1/2}|n+1\rangle\right]$$

$$= 3(n+1)^{1/2}(n+1)|n+1\rangle$$

$$= 3(n+1)^{3/2}|n+1\rangle$$

---

no need to do matrix multiplication. Just play with $\mathbf{a}$, $\mathbf{a}^\dagger$ and the $[\mathbf{a}, \mathbf{a}^\dagger]$ commutation rule and the $\mathbf{a}^\dagger\mathbf{a}$ number operator

"Second Quantization"

$$\mathbf{x}^3_{mn} = \left(\frac{\hbar}{2m\omega}\right)^{3/2}\Big[\delta_{mn+3}\left((n+1)(n+2)(n+3)\right)^{1/2}$$

(δ_{mn+3}: same as $|n+3\rangle\langle n|$)

$$+\,\delta_{mn+1}\,3(n+1)^{3/2} \quad \leftarrow |n+1\rangle\langle n|$$

$$+\,\delta_{mn-1}\,3n^{3/2} \quad \leftarrow |n\rangle\langle n-1|$$

$$+\,\delta_{mn-3}\left(n(n-1)(n-2)\right)^{1/2}\Big]$$

(δ_{mn-3}: $|n\rangle\langle n-3|$, same as $|n\rangle\langle n-3|$)

simple! $\mathbf{x}^3$ is arranged into four separate terms, each with its own explicit selection rule.

---

* $V(x) = \frac{1}{2}k\mathbf{x}^2 + \underbrace{a\mathbf{x}^3 + b\mathbf{x}^4}_{\text{anharmonic terms} \rightarrow \text{perturbation theory}}$

* IR transition intensities $\propto \left|\langle n|\mathbf{x}|n+1\rangle\right|^2$

* Survival and transfer probabilities of initially prepared pure harmonic oscillator non-eigenstate in an anharmonic potential.

* Expectation values of any function of $\mathbf{x}$ and $\mathbf{p}$.

Universality: all k,m (system-specific) constants are removed until we put them back in at the end of the calculation.

e.g., What is $\langle \Delta x \rangle^2 = \left[\langle x^2 \rangle - \langle x \rangle^2 \right]$

$$\mathbf{x}^2 = \frac{\hbar}{m\omega} \underset{\sim}{x} = \frac{\hbar}{m\omega} \left[\frac{1}{2} \left(\mathbf{a} + \mathbf{a}^\dagger \right)^2 \right] \leftarrow$$

pure numbers in [ ]

$$\langle \Delta x \rangle^2 = \frac{\hbar}{2m\omega} \left[\left\langle \left(\mathbf{a} + \mathbf{a}^\dagger \right)^2 \right\rangle - \left\langle \mathbf{a} + \mathbf{a}^\dagger \right\rangle^2 \right] ?$$

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5.73 Quantum Mechanics I
Fall 2018

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5.73 Lecture #14

Perturbation Theory I

(See CTDL 1095-1107, 1110-1119)

Last time: derivation of all matrix elements for Harmonic-Oscillator: **x, p, H**

"selection rules" $x_{ij}^n \quad |i-j| \le n$ in steps of 2 $\left(\text{e.g. } x^3 : \Delta n = \pm 3, \pm 1\right)$

"quantum number scaling" $x_{ii}^n \propto i^{n/2}$

dimensionless quantities

$$\underset{\sim}{\mathbf{x}} = \left(\frac{m\omega}{\hbar}\right)^{1/2} \mathbf{x}$$

$$\underset{\sim}{\mathbf{p}} = (\hbar m\omega)^{-1/2} \mathbf{p}$$

$$\underset{\sim}{\mathbf{H}} = \frac{1}{\hbar\omega}\mathbf{H}$$

$$\underset{\sim}{\mathbf{x}} = 2^{-1/2}(\mathbf{a} + \mathbf{a}^\dagger)$$

$$\underset{\sim}{\mathbf{p}} = 2^{-1/2} i(\mathbf{a}^\dagger - \mathbf{a})$$

"annihilation"	$\mathbf{a} = 2^{-1.2}\left(\underset{\sim}{\mathbf{x}} + i\mathbf{p}\right)$	$\mathbf{a}\left\|n\right\rangle = n^{1/2}\left\|n-1\right\rangle$
"creation"	$\mathbf{a}^\dagger = 2^{-1/2}(\mathbf{x} - i\mathbf{p})$	$\mathbf{a}^\dagger\left\|n\right\rangle = (n+1)^{1/2}\left\|n+1\right\rangle$
"number"	$\mathbf{a}^\dagger\mathbf{a}$ (not $\mathbf{a}\mathbf{a}^\dagger$)	$\mathbf{a}^\dagger\mathbf{a}\left\|n\right\rangle = n\left\|n\right\rangle$
"commutator"	$[\mathbf{a},\mathbf{a}^\dagger] = +\mathbf{1}$	

a little more:

$$a_{01} = 1^{1/2}$$

$$\mathbf{a} = \begin{array}{c} \begin{array}{cccc} 0 & 1 & 2 & 3 \end{array} \\ \left(\begin{array}{ccccc} 0 & \sqrt{1} & 0 & 0 & 0 \\ 0 & 0 & \sqrt{2} & 0 & 0 \\ 0 & 0 & 0 & \sqrt{3} & 0 \\ 0 & 0 & 0 & 0 & \ddots \\ 0 & 0 & 0 & 0 & 0 \end{array}\right) \end{array} \begin{array}{l} 0 \\ 1 \\ 2 \\ 3 \end{array} \text{ row}$$

column

(one step to right of main diagonal)

convenience of having only one and not two diagonals filled

$$\mathbf{a}^n = \begin{pmatrix} 0 & \cdots & (n!)^{1/2} & 0 & 0 \\ 0 & 0 & \cdots & \left(\frac{(n+1)!}{1!}\right)^{1/2} & 0 \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ 0 & \cdots & \cdots & \cdots & \left(\frac{(n+q)!}{q!}\right)^{1/2} \end{pmatrix} \begin{array}{l} 0 \\ 1 \\ \\ q \end{array}$$

(n steps to right)

selection rule for a_{ij}^{n} $\quad j - i = n$

selection rule for $a_{ij}^{\dagger n}$ $\quad j - i = -n$

$$|n\rangle = [n!]^{-1/2} \left(\mathbf{a}^{\dagger}\right)^n |0\rangle$$

operate on right with $\mathbf{a}^n$ first

operate on left with $(\mathbf{a}^{\dagger})^m$ second

$$\left[\left(\mathbf{a}^{\dagger}\right)^m (\mathbf{a})^n\right]_{jk} = \underbrace{\delta_{j,k-n+m}}_{\text{selection rule}} \left[\frac{(k!)}{(k-n)!} \frac{(j!)}{(j-m)!}\right]^{1/2}$$

Selection rules are obtained simply by counting the numbers of $\mathbf{a}^{\dagger}$ and $\mathbf{a}$ operators and taking the difference.

The actual value of the matrix element depends on the order in which the individual $\mathbf{a}^{\dagger}$ and $\mathbf{a}$ factors are arranged, but the selection rule does not.

Lots of nice tricks and shortcuts using $\mathbf{a}$, $\mathbf{a}^{\dagger}$, and $\mathbf{a}^{\dagger}\mathbf{a}$

This makes writing computer programs trivially easy and transparent.

One of the places where these tricks come in handy is perturbation theory.

We already have:

1. WKB: local solution, local k(x), locate and exploit stationary phase (Lecture #3)
2. Numerov–Cooley: exact solution - no restrictions (Lecture #9)
3. Discrete Variable Representation: exact solution, ψ as linear combination of H-O eigenstates (Lecture #11)

Why perturbation theory?

- replace exact **H**, which is usually of ∞ dimension, by $\mathbf{H}^{eff}$, which is of finite dimension. Truncate infinite matrix so that any eigenvalue and eigenfunction can be computed with error < some preset tolerance.
  "Fit model" that is physical (because it makes localization and coupling mechanisms explicit) yet parametrically parsimonious

- derive explicit ***functional*** relationship between an n-dependent observable (e.g. E_n) and n
  e.g. $$\frac{E_n}{hc} = \omega_e\left(n+1/2\right) - \omega_e x_e\left(n+1/2\right)^2 + \omega_e y_e\left(n+1/2\right)^3$$

- establish relationship between a molecular constant (ω_e, $\omega_e x_e$, …) and the parameters that define V(x) e.g. $\omega_e x_e \leftrightarrow a x^3$

There are 2 kinds of garden variety perturbation theory:

1. Nondegenerate (Rayleigh-Schrödinger) P.T. → simple formulas.
   This breaks down when the interacting basis states are "near degenerate."

2. Quasi-Degenerate P.T. → matrix $\mathbf{H}^{eff}$
   Finite $\mathbf{H}^{eff}$ is corrected for "out-of-block" perturbers by "van Vleck" or "contact" transformation

~4 Lectures

Derive Perturbation Theory Formulas

* correct E_n and ψ_n directly for effects of "neglected" terms in exact **H**
* correct all other observables indirectly through corrected ψ

Perturbation Theory I. Begin Cubic Anharmonic Perturbation

Formal treatment

$E_n = \lambda^0 E_n^{(0)} + \lambda E_n^{(1)} + \lambda E_n^{(2)}$ usually stops at λ^2

$\psi_n = \lambda^0 \psi_n^{(0)} + \lambda^1 \psi_n^{(1)}$ usually stops at λ^1 (because all observables involve $\psi \times \psi'$, hence orders go up to λ^2).

$\mathbf{H} = \lambda^0 \mathbf{H}^{(0)} + \lambda^1 \mathbf{H}^{(1)}$ order sorting is MURKY

λ is an order-sorting parameter with no physical significance. Set $\lambda = 1$ after all is done. $\lambda = 0 \rightarrow 1$ is like turning on the effect of $\mathbf{H}^{(1)}$. Equations must be valid for the entire range of λ. $0 \leq \lambda \leq 1$.

Plug 3 equations into Schrödinger Equation, $\mathbf{H}\psi_n = E_n\psi_n$, and collect terms into separate equations according to the order of λ.

λ^0 terms:

$\mathbf{H}^{(0)} \left| \psi_n^{(0)} \right\rangle = E_n^{(0)} \left| \psi_n^{(0)} \right\rangle$

left multiply by $\left\langle \psi_m^{(0)} \right|$

$$\boxed{H_{mn}^{(0)} = E_n^{(0)} \delta_{mn}}$$

requires that $\mathbf{H}^{(0)}$ be diagonal in $\left\{ \psi_n^{(0)} \right\}$

know the eigenvalues $\left\{ E_n^{(0)} \right\}$ and eigenfunctions $\left\{ \psi_n^{(0)} \right\}$ of $\mathbf{H}^{(0)}$

CALLED BASIS FUNCTIONS

CALLED "ZERO–ORDER" MODEL

So we choose $\mathbf{H}^{(0)}$ to be the part of $\mathbf{H}$ for which:

* it is easy to write and exploit a *complete* set of eigenfunctions and eigenvalues
* it is easy to evaluate matrix elements of commonly occurring "perturbation" terms in this basis set
* sometimes the choice of basis set is based on convenience rather than "goodness" — doesn't matter as long as the basis set is complete.

} easier to think "nature" intended a simpler reality

examples:

Harmonic Oscillator	$V(x) = \frac{1}{2}kx^2$
Morse Oscillator	$V(x) = D[1 - e^{-ax}]^2$ $[\text{D, a, and } R_e]$
Quartic Oscillator	$V(x) = bx^4$
n-fold hindered rotor	$V_n(\phi) = (V_n^0/2)(1 - \cos n\phi)$

Now return to the Schrödinger Equation and examine the λ^1 and λ^2 terms.

λ^1 terms

$$\mathbf{H}^{(1)}\left|\psi_n^{(0)}\right\rangle + \mathbf{H}^{(0)}\left|\psi_n^{(1)}\right\rangle = E_n^{(1)}\left|\psi_n^{(0)}\right\rangle + E_n^{(0)}\left|\psi_n^{(1)}\right\rangle$$

multiply by $\left\langle\psi_n^{(0)}\right|$

from $\mathbf{H}^{(0)}$ operating to left

$$\underbrace{H_{nn}^{(1)} + E_n^{(0)}\left\langle\psi_n^{(0)}\middle|\psi_n^{(1)}\right\rangle}_{\text{same}} = \underbrace{E_n^{(1)} + E_n^{(0)}\left\langle\psi_n^{(0)}\middle|\psi_n^{(1)}\right\rangle}_{\text{get rid of them}}$$

$\left(\text{could also require } \left\langle\psi_n^{(0)}\middle|\psi_n^{(1)}\right\rangle = 0\right)$

we do require this later

$H_{nn}^{(1)} = E_n^{(1)}$ 1st-order correction to E is just the expectation value of the perturbation term in **H**: $\mathbf{H}^{(1)}$.

Return to λ^1 equation and this time multiply by $\left\langle \psi_m^{(0)} \right|$

$$H_{mn}^{(1)} + E_m^{(0)} \left\langle \psi_m^{(0)} \middle| \psi_n^{(1)} \right\rangle = 0 + E_n^{(0)} \left\langle \psi_m^{(0)} \middle| \psi_n^{(1)} \right\rangle$$

$$H_{mn}^{(1)} = \left\langle \psi_m^{(0)} \middle| \psi_n^{(1)} \right\rangle \left(E_n^{(0)} - E_m^{(0)} \right)$$

$$\left\langle \psi_m^{(0)} \middle| \psi_n^{(1)} \right\rangle = \frac{H_{mn}^{(1)}}{E_n^{(0)} - E_m^{(0)}}$$

Now, to get $\left| \psi_{\mathrm{n}}^{(1)} \right\rangle$, we use the completeness of $\left\{ \psi^{(0)} \right\}$: $\sum_k \left| \psi_k^{(0)} \right\rangle \left\langle \psi_k^{(0)} \right|$

$$\psi_n^{(1)} = \sum_k \left| \psi_k^{(0)} \right\rangle \underbrace{\left\langle \psi_k^{(0)} \middle| \psi_n^{(1)} \right\rangle}_{\text{but we already know this}}$$

$$\psi_n^{(1)} = \sum_k \left| \psi_k^{(0)} \right\rangle \frac{H_{kn}^{(1)}}{E_n^{(0)} - E_k^{(0)}}$$

indices in opposite order from naïve expectation

* index of $\psi_n^{(1)}$ matches 1st index of $E_n^{(0)}$ in denominator
* $n = k$ is problematic. Insist on Σ_k' which means exclude the $k = n$ term
* we cold have demanded $\left\langle \psi_n^{(0)} \middle| \psi_n^{(1)} \right\rangle = 0$
* counter - intuitive order of indices $\mathrm{H}_{kn}^{(1)}$

λ^2 terms

most important in real problems although mindlessly excluded from most textbooks.

$$\mathbf{H}^{(1)}\left|\psi_n^{(1)}\right\rangle = E_n^{(1)}\left|\psi_n^{(1)}\right\rangle + E_n^{(2)}\left|\psi_n^{(0)}\right\rangle$$

multiply by $\left\langle\psi_n^{(0)}\right|$ $\qquad \left\langle\psi_n^{(0)}\middle|\psi_n^{(1)}\right\rangle = 0$

$$\left\langle\psi_n^{(0)}\middle|\mathbf{H}^{(1)}\middle|\psi_n^{(1)}\right\rangle = 0 + E_n^{(2)}$$

↑ completeness

$$\sum_k \underbrace{\left\langle\psi_n^{(0)}\middle|\mathbf{H}^{(1)}\middle|\psi_k^{(0)}\right\rangle}_{\mathrm{H}_{n,k}^{(1)}}\underbrace{\left\langle\psi_k^{(0)}\middle|\psi_n^{(1)}\right\rangle}_{\sum_k' \frac{\mathrm{H}_{n,k}^{(1)}}{E_n^{(0)}-E_k^{(0)}}} = E_n^{(2)}$$

$$E_n^{(2)} = \sum_k{}' \frac{\left|H_{k,n}^{(1)}\right|^2}{E_n^{(0)} - E_k^{(0)}}$$

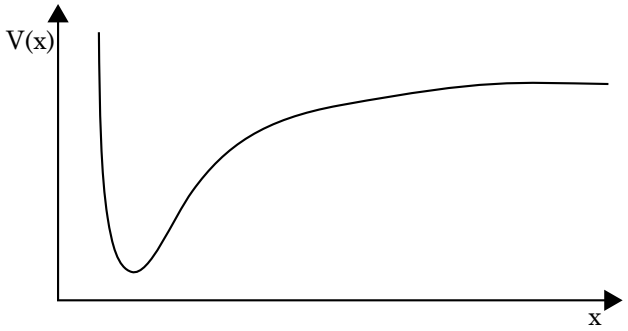
↑ always first

"matrix element squared" over energy difference in "energy denominator"

We have derived all needed formulas $E_n^{(0)}, E_n^{(1)}, E_n^{(2)}; \psi_n^{(0)}, \psi_n^{(1)}$!

Examples

$V(x) = \frac{1}{2}kx^2 + ax^3$ $\qquad$ (a < 0)



$\mathbf{H}^{(0)} = \frac{1}{2}k\mathbf{x}^2 + \frac{\mathbf{p}^2}{2m}$

$\mathbf{H}^{(1)} = a\mathbf{x}^3$

(actually the ax^3 term with a < 0 makes all potentials unbound. How can we pretend that this catastrophe does not affect the results from perturbation theory?)

modified 8/13/20 1:07 PM

We need matrix elements of $\mathbf{x}^3$

Two ways to get them:

* matrix multiplication $x^3_{i\ell} = \sum_{j,k} x_{ij} x_{jk} x_{k\ell}$

* $\mathbf{a}, \mathbf{a}^\dagger$ tricks

$$\mathbf{x}^3 = \left(\frac{\hbar}{m\omega}\right)^{3/2} \underset{\sim}{\mathbf{x}}^3 = \left(\frac{\hbar}{m\omega}\right)^{3/2} \left[2^{-1/2}\left(\mathbf{a}+\mathbf{a}^\dagger\right)\right]^3$$

$$= \left(\frac{\hbar}{2m\omega}\right)^{3/2} \left[\mathbf{a}^3 + \left(\mathbf{a}^\dagger\mathbf{a}\mathbf{a} + \mathbf{a}\mathbf{a}^\dagger\mathbf{a} + \mathbf{a}\mathbf{a}\mathbf{a}^\dagger\right) + \left(\mathbf{a}\mathbf{a}^\dagger\mathbf{a}^\dagger + \mathbf{a}^\dagger\mathbf{a}\mathbf{a}^\dagger + \mathbf{a}^\dagger\mathbf{a}^\dagger\mathbf{a}\right) + \mathbf{a}^{\dagger 3}\right]$$

Each group in ( ) has its own Δv selection rule (see lecture #13 notes). Simplify using $[\mathbf{a}, \mathbf{a}^\dagger] = 1$.

Goal is to manipulate each mixed $\mathbf{a}, \mathbf{a}^\dagger$ term so that "the number operator", $\mathbf{a}^\dagger\mathbf{a}$, appears at the far right and then exploit $\mathbf{a}^\dagger\mathbf{a}|n\rangle = n|n\rangle$

All of the nonzero elements:

square root of larger q.n.

$$\Delta n = \pm 3$$

$$\mathbf{a}^3_{n-3n} = \left[n(n-1)(n-2)\right]^{1/2}$$

$$\mathbf{a}^{\dagger 3}_{n+3n} = \left[(n+3)(n+2)(n+1)\right]^{1/2}$$

$\Delta n = -1$: $\left(\mathbf{a}^\dagger\mathbf{a}\mathbf{a} + \mathbf{a}\mathbf{a}^\dagger\mathbf{a} + \mathbf{a}\mathbf{a}\mathbf{a}^\dagger\right) = 3\mathbf{a}\mathbf{a}^\dagger\mathbf{a}$ $\boxed{-1}$

$$\text{because } \mathbf{a}^\dagger\mathbf{a}\mathbf{a} = \mathbf{a}\mathbf{a}^\dagger\mathbf{a} + \underbrace{\left[\mathbf{a}^\dagger, \mathbf{a}\right]}_{\boxed{-1}}\mathbf{a} = \mathbf{a}\mathbf{a}^\dagger\mathbf{a} - \mathbf{a}$$

$$\mathbf{a}\mathbf{a}\mathbf{a}^\dagger = \mathbf{a}\mathbf{a}^\dagger\mathbf{a} + \mathbf{a}\underbrace{\left[\mathbf{a}, \mathbf{a}^\dagger\right]}_{\boxed{+1}} = \mathbf{a}\mathbf{a}^\dagger\mathbf{a} + \mathbf{a}$$

$$\left(\mathbf{a}\mathbf{a}^\dagger\mathbf{a}\right)_{n-1n} = n^{3/2}$$

$\Delta n = +1$: $\left(\mathbf{a}\mathbf{a}^\dagger\mathbf{a}^\dagger + \mathbf{a}^\dagger\mathbf{a}\mathbf{a}^\dagger + \mathbf{a}^\dagger\mathbf{a}^\dagger\mathbf{a}\right) = 3\mathbf{a}^\dagger\mathbf{a}^\dagger\mathbf{a} + 3\mathbf{a}^\dagger$

$$\Big[\underbrace{3\underbrace{\mathbf{a}^\dagger\mathbf{a}}_{n}\mathbf{a}^\dagger\mathbf{a} + 3\mathbf{a}^\dagger}_{(n+1)^{1/2}}\Big]_{n+1\,n} = 3n\left(n+1\right)^{1/2} + 3\left(n+1\right)^{1/2} = 3\left(n+1\right)^{3/2}$$

(This is neither typo nor approximation.)

So we have worked out all $\mathbf{x}^3$ matrix elements — leave the rest to P.S. #5.

Properties other than E_n? Use $\psi_n = \psi_n^{(0)} + \psi_n^{(1)}$

e.g. transition probability (electric dipole allowed vibrational transitions)

$P_{nn'} \propto \left|x_{nn'}\right|^2$

for H-O

$$\left|x_{nn'}\right|^2 = \left(\frac{\hbar}{2\cancel{(km)^{1/2}}}\right) n_> \delta_{n_>, n_<+1} \qquad \frac{}{m\omega}$$

($n_>$ here means the larger of $n_>$ and $n_<$)

(selection rule: only $\Delta n = \pm 1$ transitions).

For a perturbed H–O, e.g. $\mathbf{H}^{(1)} = a\mathbf{x}^3$

$$\left|\psi_n\right\rangle = \left|\psi_n^{(0)}\right\rangle + \Sigma_k' \frac{H_{kn}^{(1)}}{E_n^{(0)} - E_k^{(0)}} \left|\psi_k^{(0)}\right\rangle$$

$$\left|\psi_n\right\rangle = \left|\psi_n^{(0)}\right\rangle + \underbrace{\frac{H_{nn+3}^{(1)}}{-3\hbar\omega}\left|\psi_{n+3}^{(0)}\right\rangle + \frac{H_{nn+1}^{(1)}}{-\hbar\omega}\left|\psi_{n+1}^{(0)}\right\rangle + \frac{H_{nn-1}^{(1)}}{+\hbar\omega}\left|\psi_{n-1}^{(0)}\right\rangle + \frac{H_{nn-3}^{(1)}}{+3\hbar\omega}\left|\psi_{n-3}^{(0)}\right\rangle}$$

Note the pairwise simplicity in the denominators.

The $\mathbf{H}_{nn'}^{(1)}$ terms are matrix elements of $\mathbf{x}^3$.

1st index (initial state)		Allowed 2nd indices (final state)
$n+3$		$n+4, n+2$
$n+1$		$n+2, n$
n	**x**	$n+1, n-1$
$n-1$		$n, n-2$
$n-3$		$n-2, n-4$

1st index reflects anharmonic mixing of initial state due to a$\mathbf{x}^3$ term

2nd indices refer to final state reached via electric dipole allowed transition, controlled by matrix elements of **x**.

Cubic anharmonicity of V(x) can give rise to $\Delta n = \pm 7, \pm 5, \pm 4, \pm 3, \pm 2, \pm 1, 0$ transitions.

$$\left\langle n\left|\mathbf{x}\right|n+7\right\rangle = \left(\frac{\hbar}{2\omega m}\right)^{7/2} \frac{a^2}{\left(-3\hbar\omega\right)^2}\left[\underbrace{\frac{(n+7)!}{n!}}_{\approx n^{7/2}}\right]^{1/2}$$

$$\left|x_{nn+7}\right|^2 \approx \frac{a^4}{m^7\omega^{11}} n^7$$

Other less extreme Δn transition strengths are given by smaller powers of $\frac{1}{\omega}$ and n.

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Perturbation Theory II

(See CTDL 1095-1104, 1110-1119)

Last time:

$$\mathbf{H}^{(0)}\psi_n^{(0)} = E_n^{(0)}\psi_n^{(0)}$$

$\mathbf{H}^{(0)}$ is diagonal

$\left\{\psi_n^{(0)}\right\}, \left\{E_n^{(0)}\right\}$ are basis functions and zero-order energies

$$E_n^{(1)} = H_{nn}^{(1)}$$

expectation value for $\psi_n^{(0)}$ of the perturbation operator

$$E_n^{(2)} = \Sigma_k' \frac{\left|H_{nk}^{(1)}\right|^2}{E_n^{(0)} - E_k^{(0)}}$$

sum excludes k = n

matrix element vs. energy denominator

↑ 1st index

$$E_n = E_n^{(0)} + E_n^{(1)} + E_n^{(2)}$$

$$\psi_n^{(1)} = \Sigma_k' \frac{H_{nk}^{(1)}}{E_n^{(0)} - E_k^{(0)}} \psi_k^{(0)}$$

sum excludes k = n

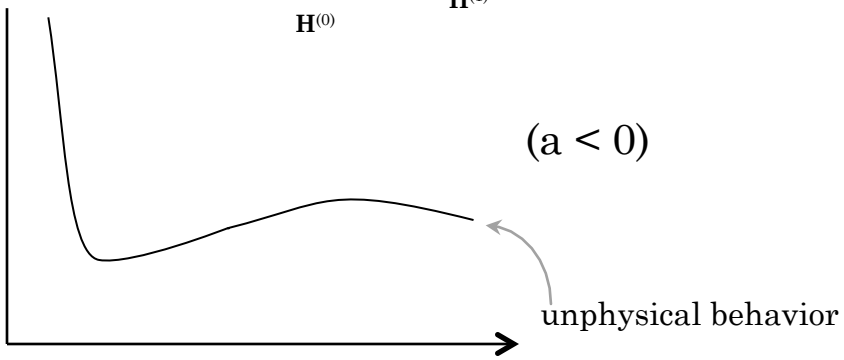
mixing coefficient, order-sorting parameter, convergence criterion

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Today:

1. cubic anharmonic perturbation
   $\mathbf{x}^3$ vs. $\mathbf{a}, \mathbf{a}^\dagger$ matrix elements
   $a\mathbf{x}^3$ contributions to ωx and Y_{00}
2. nonlecture Morse oscillator ↔ pert. theory for $a\mathbf{x}^3$
3. transition probabilities — orders and convergence of perturbation theory
   Mechanical and electronic anharmonicities.

Example 1. $\mathbf{H} = \underbrace{\frac{\mathbf{p}^2}{2m} + \frac{1}{2}k\mathbf{x}^2}_{\mathbf{H}^{(0)}} + \underbrace{a\mathbf{x}^3}_{\mathbf{H}^{(1)}}$



$$\left[(i+1)(i+2)(i+3)\right]^{1/2}$$

Need matrix elements of $\mathbf{x}^3$

one (longer) way $x^3_{i\ell} = \sum_{j,k} x_{ij} x_{jk} x_{k\ell}$ matrix multiplication

4 different selection rules: $\ell - i = 3, 1, -1, -3$

$\ell - i = 3$ $\quad i \longrightarrow i+1,\ i+1 \longrightarrow i+2,\ i+2 \longrightarrow i+3$ one path for $\ell - i = 3$

$$[(i+1)(i+2)(i+3)]^{1/2}$$

$\ell - i = 1$ $\quad i \longrightarrow i+1,\ i+1 \longrightarrow i+2,\ i+2 \longrightarrow i+1$ three paths for $\ell - i = 1$

$i \longrightarrow i-1,\ i-1 \longrightarrow i,\ i \longrightarrow i+1$

$i \longrightarrow i+1,\ i+1 \longrightarrow i,\ i \longrightarrow i+1$

There are three 3-step paths from i to $i+1$. Add them.

$$\left[(i+1)(i+2)(i+2)\right]^{1/2} + \left[(i)(i)(i+1)\right]^{1/2} + \left[(i+1)(i+1)(i+1)\right]^{1/2}$$

algebraically complicated (but only apparently!)

an other (much shorter) alternative method: using $\mathbf{a}$, $\mathbf{a}^\dagger$, and $\mathbf{a}^\dagger\mathbf{a}$
[operator algebra rather than ordinary algebra]

$$\mathrm{x}^3 = \left(\frac{\hbar}{m\omega}\right)^{3/2} \underset{\sim}{\mathrm{x}}^3 = \left(\frac{\hbar}{m\omega}\right)^{3/2} \left[2^{-1/2}\left(\mathrm{a} + \mathrm{a}^\dagger\right)\right]^3$$

$$= \left(\frac{\hbar}{2m\omega}\right)^{3/2} \left(\mathrm{a} + \mathrm{a}^\dagger\right)^3$$

$$\left(\mathbf{a} + \mathbf{a}^\dagger\right)^3 = \mathbf{a}^3 + \left[\mathbf{a}^\dagger\mathbf{a}\mathbf{a} + \mathbf{a}\mathbf{a}^\dagger\mathbf{a} + \mathbf{a}\mathbf{a}\mathbf{a}^\dagger\right] + \left[\mathbf{a}\mathbf{a}^\dagger\mathbf{a}^\dagger + \mathbf{a}^\dagger\mathbf{a}\mathbf{a}^\dagger + \mathbf{a}^\dagger\mathbf{a}^\dagger\mathbf{a}\right] + \mathbf{a}^{\dagger 3}$$

four additive terms, four different selection rules.

Use simple $\mathbf{a}$,$\mathbf{a}^\dagger$ algebra to work out all matrix elements and selection rules by inspection.

recall: $\mathbf{a}^\dagger|n\rangle = (n+1)^{1/2}|n+1\rangle$, $\mathbf{a}|n\rangle = n^{1/2}|n-1\rangle$, $\mathbf{a}^\dagger\mathbf{a}|n\rangle = n|n\rangle$

$[\mathbf{a},\mathbf{a}^\dagger] = 1$ $\quad\therefore\quad$ $\boxed{\mathbf{aa}^\dagger = 1 + \mathbf{a}^\dagger\mathbf{a}}$ prescription for permuting $\mathbf{a}$ through $\mathbf{a}^\dagger$

$\Delta n = -3$ $\quad \mathbf{a}^3_{n-3,n} = \left[(n-2)(n-1)(n)\right]^{1/2}$

$\Delta n = +3$ $\quad \mathbf{a}^{\dagger 3}_{n+3,n} = \left[(n+3)(n+2)(n+1)\right]^{1/2}$

$\Delta n = -1$ $\quad \left[\mathbf{a}^\dagger\mathbf{aa} + \mathbf{aa}^\dagger\mathbf{a} + \mathbf{aaa}^\dagger\right]_{n-1,n}$

goal is to rearrange each product so that it has the number operator at the far right

$\mathbf{a}^\dagger\mathbf{aa} = \mathbf{aa}^\dagger\mathbf{a} - \mathbf{a}$

$\mathbf{aaa}^\dagger = \mathbf{aa}^\dagger\mathbf{a} + \mathbf{a}$

$\mathbf{aa}^\dagger\mathbf{a} = \mathbf{aa}^\dagger\mathbf{a}$

$3\mathbf{aa}^\dagger\mathbf{a} + 0$ 3 operators combined into only one!

$\mathbf{a}^\dagger\mathbf{aa} = \underbrace{[\mathbf{a}^\dagger,\mathbf{a}]}_{-1}\mathbf{a} + \mathbf{aa}^\dagger\mathbf{a}$

$\Delta n = -1$ $\quad [\quad]_{n-1,n} = 3\left(\mathbf{aa}^\dagger\mathbf{a}\right)_{n-1,n} = \langle n-1|3\mathbf{a}\left(\mathbf{a}^\dagger\mathbf{a}\right)|n\rangle = 3n^{3/2}$

$\Delta n = +1$ $\quad \left[\mathbf{aa}^\dagger\mathbf{a}^\dagger + \mathbf{a}^\dagger\mathbf{aa}^\dagger + \mathbf{a}^\dagger\mathbf{a}^\dagger\mathbf{a}\right]$ simplify as below

$\mathbf{aa}^\dagger\mathbf{a}^\dagger = \mathbf{a}^\dagger\mathbf{aa}^\dagger + \mathbf{a}^\dagger = \mathbf{a}^\dagger\mathbf{a}^\dagger\mathbf{a} + 2\mathbf{a}^\dagger$

$\mathbf{a}^\dagger\mathbf{aa}^\dagger = \mathbf{a}^\dagger\mathbf{a}^\dagger\mathbf{a} + \mathbf{a}^\dagger$

$\mathbf{a}^\dagger\mathbf{a}^\dagger\mathbf{a} = \mathbf{a}^\dagger\mathbf{a}^\dagger\mathbf{a}$

$3\mathbf{a}^\dagger\mathbf{a}^\dagger\mathbf{a} + 3\mathbf{a}^\dagger$

$$3\langle n+1|\left(\mathbf{a}^\dagger\mathbf{a}^\dagger\mathbf{a} + \mathbf{a}^\dagger\right)|n\rangle = 3\left(n(n+1)^{1/2} + (n+1)^{1/2}\right) = 3\left[(n+1)(n+1)^{1/2}\right] = 3(n+1)^{3/2}$$

All done — not necessary to massage the algebra as would have been necessary for $\mathbf{x}^3$ by direct $\mathbf{x}$ multiplication!

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Now do the perturbation theory:

$$E_n = E_n^{(0)} + E_n^{(1)} + E_n^{(2)} = \hbar\omega\left(n+1/2\right) + 0 + \Sigma_k' \frac{\left|H_{nk}^{(1)}\right|^2}{E_n^{(0)} - E_k^{(0)}}$$

$\uparrow$ $\boxed{x_{nn}^3 = 0}$

	$\left\|H_{nk}^{(1)}\right\|^2$	$E_n^{(0)} - E_k^{(0)}$
$k = n-3$	$a^2\left(\frac{\hbar}{2m\omega}\right)^3 (n-2)(n-1)(n)$	$+3\hbar\omega$
$k = n-1$	$a^2\left(\frac{\hbar}{2m\omega}\right)^3 9n^3$	$+1\hbar\omega$
$k = n+1$	$a^2\left(\frac{\hbar}{2m\omega}\right)^3 9(n+1)^3$	$-1\hbar\omega$
$k = n+3$	$a^2\left(\frac{\hbar}{2m\omega}\right)^3 (n+3)(n+2)(n+1)$	$-3\hbar\omega$

$$E_n^{(2)} = \underbrace{\frac{a^2\left(\frac{\hbar}{2m\omega}\right)^3}{\hbar\omega}}_{\text{all of the constants}} \left[\underbrace{\frac{(n-2)(n-1)(n)}{3} - \frac{(n+3)(n+2)(n+1)}{3}}_{\text{2 nearly-cancelling pairs}} + \underbrace{\frac{9n^3}{1} - \frac{9(n+1)^3}{1}} \right]$$

Simplest path is to combine the pairs of $\Delta n = 3$ and -3, $\Delta n = 1$ and -1 terms

$$E_n^{(2)} = \frac{a^2\hbar^2}{8m^3\omega^4}\left[-30\left(n+1/2\right)^2 - 3.5\right] \qquad \text{algebra}$$

$$E_n^{(2)} = -\frac{a^2\hbar^2}{m^3\omega^4}\left[\frac{15}{4}\left(n+1/2\right)^2 + \frac{7}{16}\right] \qquad \left(m^3\omega^4 = mk^2\right)$$

$\uparrow$ *all levels are shifted down*, regardless of sign of a. Can't measure the sign of the cubic anharmonicity constant, a, from vibrational structure alone!

$$E_n = \hbar\omega\left(n+1/2\right) - \underbrace{\hbar\frac{15}{4}\left(\frac{a^2\hbar}{m^3\omega^4}\right)}_{\hbar\omega_e x_e}\left(v+1/2\right)^2 - \underbrace{\hbar\frac{7}{16}\left(\frac{a^2\hbar}{m^3\omega^4}\right)}_{\hbar Y_{00}}$$

$$E_n = \hbar\left[Y_{00} + \omega_e\left(v+1/2\right) - \omega_e x_e\left(v+1/2\right)^2 + \omega_e y_e\left(v+1/2\right)^3 \ldots\right]$$

a$\mathbf{x}^3$ makes contributions exclusively to Y_{00} and $\omega_e x_e$.

NON-LECTURE

Relationship between Morse Oscillator and Perturbation Theory Treatment of Cubic Plus Quartic Anharmonic Oscillator

Morse oscillator

$$V_{\text{Morse}}(x) = D_e\left[1 - e^{-\alpha x}\right]^2 \qquad (D_e \text{ is the dissociation energy})$$

Cubic Plus Quartic Oscillator

$$V_{3,4}(x) = \frac{1}{2}kx^2 + ax^3 + bx^4$$

The exact energy levels of V_{Morse} (obtained via WKB or DVR) have the simple form

$$E_n = \hbar\left[(n+1/2)\omega - (n+1/2)^2\omega x\right].$$

First we determine the relationship between D_e,α and ω, ωx for the Morse oscillator.

At the dissociation limit, $n \equiv n_{\text{D}}$

$$\frac{dE}{dn} = 0$$

$$\frac{dE}{dn} = 0 = \hbar\omega - \hbar\omega x\left(2n_D + 1\right)$$

$$\boxed{n_D = \frac{\omega}{2\omega x} - \frac{1}{2}}$$

$$E\left(n_D\right) = D_e$$

$$E(n_D) = \hbar\omega\left(\frac{\omega}{2\omega x}\right) - \hbar\omega x\left(\frac{\omega}{2\omega x}\right)^2$$

$$= \hbar\frac{\omega^2}{4\omega x}$$

$$\boxed{D_e = \hbar\frac{\omega^2}{4\omega x}}$$

This is neat because we have related two easily measured molecular constants, ω and ωx, to one less easily measured molecular constant, D_e.

Now, in preparation for the perturbation theoretic comparison of V_{Morse} to $V_{3,4}$, we compute the derivatives of V_{Morse} at $x = 0$.

$$V(0) = 0$$

$$\frac{dV}{dx} = V'(x) = \frac{\hbar\omega^2}{4\omega x}\left[2\alpha e^{-\alpha x} - 2\alpha e^{-2\alpha x}\right]$$

As expected, $V(x)$ is a minimum at $x = 0$,

$$V'(0) = 0$$

$$\frac{d^2V}{dx^2} = V''(x) = \frac{\hbar\omega^2}{4\omega x}\left[-2\alpha^2 e^{-\alpha x} + 4\alpha^2 e^{-2\alpha x}\right]$$

$$V''(0) = \frac{\hbar\omega^2}{4\omega x}2\alpha^2 = k = m\omega^2 \quad \left(\omega^2 = k/m\right)$$

$$\boxed{\alpha = \left[\frac{2m\omega x}{\hbar}\right]^{1/2}}$$

Thus we know <u>both</u> D_e and α for V_{Morse} in terms of ω and ωx for an anharmonic oscillator.

$$V'''(x) = \frac{\hbar\omega^2}{4\omega x}\left[2\alpha^3 e^{-\alpha x} - 8\alpha^3 e^{-2\alpha x}\right]$$

$$V'''(0) = -\frac{3}{2}\frac{\hbar\omega^2\alpha^3}{\omega x} = -\frac{3}{2}\frac{\hbar\omega^2}{\omega x}\left[\frac{2m\omega x}{\hbar}\right]^{3/2}$$

$$V''''(x) = \frac{\hbar\omega^2}{4\omega x}\left[-2\alpha^4 e^{-\alpha x} + 16\alpha^4 e^{-2\alpha x}\right]$$

$$V''''(0) = \frac{\hbar\omega^2}{4\omega x}\left[14\alpha^4\right] = \frac{7}{2}\frac{\hbar\omega^2}{\omega x}\left[\frac{2m\omega x}{\hbar}\right]^2$$

$$= 14\frac{(\omega x)\omega^2}{\hbar}$$

Now we look at the same set of derivatives for $V_{3,4}$

$$V_{3,4}(x) = \frac{1}{2}kx^2 + ax^3 + bx^4$$

$$V''_{3,4}(0) = k$$

$$V'''_{3,4}(0) = 6a$$

$$V''''_{3,4}(0) = 24b$$

$$V''''_{\text{Morse}}(0) = V''''_{3,4}(0)$$

$$-3\left(\frac{2\omega x}{\hbar}\right)^{1/2} = 6a$$

$$\boxed{\omega x = \frac{2a^2\hbar}{\omega^4 m^3}}$$

$$V''''_{\text{Morse}}(0) = V''''_{3,4}(0)$$

$$\boxed{14\frac{(\omega x)\omega^2}{\hbar} = 24b}$$

Applying perturbation theory to $V_{3,4}$(x), we saw on page 15-4 that

$$\omega x = \frac{15}{4}\frac{a^2\hbar}{m^3\omega^4}$$

but the algebraic approach to V_{Morse} led to

$$\omega x = 2\frac{a^2\hbar}{m^3\omega^4}$$

This difference is due to neglect of the first order contribution from the $\mathbf{x}^4$ term in the power series expansion of $V_{\text{Morse}}(x)$.

$$E_n^{(1)} = V''''(0)\,\mathbf{x}^4/4! = \left[7/2\,\frac{\hbar\omega^2\alpha^4}{\omega x}\right]\mathbf{x}^4/24$$

$$\langle n|\mathbf{x}^4|n\rangle = \left(\frac{\hbar}{2m\omega}\right)^2\left[4(n+1/2)^2+2\right]$$

$$E_n^{(1)} = \frac{7}{12}\omega x\left(n+1/2\right)^2 + \frac{7}{24}\omega x$$

It turns out that input of the algebraic relationships between k, a, b for the $V_{3,4}$ potential and D_e, α for V_{Morse} into perturbation theory gives correct results if the $a\mathbf{x}^3$ term is treated through second-order of perturbation theory but the $b\mathbf{x}^4$ term is treated only through first order of perturbation theory.

END OF NON-LECTURE

One reason that the result from second-order perturbation theory applied directly to V(x) = k$\mathbf{x}^2$/2 + a$\mathbf{x}^3$ and the term-by-term comparison of the power series expansion of the Morse oscillator are not identical is that contributions to the $(n + 1/2)^2$ term have been neglected from higher derivatives of the Morse potential in the energy level expression. In particular

$$E_n^{(1)} = V''''(0)\mathbf{x}^4/4! = \left[7/2\frac{\hbar\omega^2\alpha^4}{\omega\mathbf{x}}\right]x^4/24$$

$$\left\langle n\left|\mathbf{x}^4\right|n\right\rangle = \left(\frac{\hbar}{2m\omega}\right)^2\left[4\left(n+1/2\right)^2+2\right]$$

contributes in first order of perturbation theory to the $(n + 1/2)^2$ term in E_n.

$$E_n^{(1)} = \frac{7}{12}\omega x\left(n+1/2\right)^2 + \frac{7}{24}\omega x$$

Example 2 Use perturbation theory to compute some property other than Energy.

To do this we need $\psi_n = \psi_n^{(0)} + \psi_n^{(1)}$ in order to calculate matrix elements of the operator in question.

For example, transition probability, $\mathbf{x}$: for electric dipole transitions, the transition probability is $P_{n'\leftarrow n} \propto \left|x_{nn'}\right|^2$

For H-O $n \rightarrow n \pm 1$ only

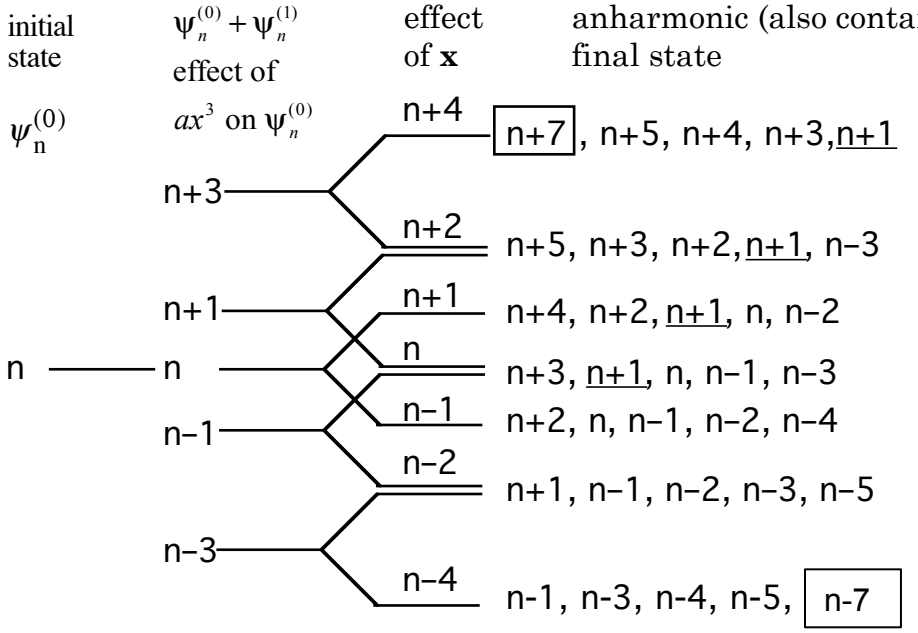
$$\left|\mathrm{x}_{\mathrm{nn+1}}\right|^2 = \left(\frac{\hbar}{2m\omega}\right)(n+1)$$

for perturbed H-O $\mathbf{H}^{(1)} = a\mathbf{x}^3$

Standard result. Now allow for both "mechanical" and "electronic" anharmonicity.

$$\psi_n = \psi_n^{(0)} + \Sigma_k' \frac{H_{nk}^{(1)}}{E_n^{(0)} - E_k^{(0)}}\psi_k^{(0)}$$

$$\psi_n = \psi_n^{(0)} + \frac{H_{nn+3}^{(1)}}{-3\hbar\omega}\psi_{n+3}^{(0)} + \frac{H_{nn+1}^{(1)}}{-\hbar\omega}\psi_{n+1}^{(0)} + + \frac{H_{nn-1}^{(1)}}{\hbar\omega}\psi_{n-1}^{(0)} + \frac{H_{nn-3}^{(1)}}{3\hbar\omega}\psi_{n-3}^{(0)}$$



Many paths from initial to final state, which interfere constructively and destructively in $|x_{nn'}|^2$

$$n' = n+7, n+5, n+4, n+3, n+2, \underline{n+1}, n, \underline{n-1}, n-2, n-3, n-4, n-5, n-7$$

only paths for H-O!

The transition strengths may be divided into 3 classes

1. direct: $n \rightarrow n \pm 1$
2. one anharmonic step $n \rightarrow n + 4, n + 2, n, n - 2, n - 4$
3. 2 anharmonic steps $n \rightarrow n + 7, n + 5, n + 3, n + 1, n - 1, n - 3, n - 5, n - 7$

Work thru the $\Delta n = -7$ path

$$\langle n|x|n+7\rangle = \left(\frac{h}{2m\omega}\right)^{3/2+3/2+1/2}\left[\frac{a^2}{(-3h\omega)^2}\right]\left[\underbrace{(n+1)(n+2)(n+3)}_{x_{n,n+3}}\underbrace{(n+4)}_{x_{n+3,n+4}}\underbrace{(n+5)(n+6)(n+7)}_{x_{n+4,n+7}}\right]^{1/2}$$

$x^3_{n,n+3}$ $x^3_{n+4,n+7}$ $x_{n+3,n+4}$

$$|x_{nn+7}|^2 \propto \frac{\hbar^3 a^4 n^7}{3^4 2^7 m^7 \omega^{11}}$$

* you show that the *single-step* anharmonic terms go as

$$\left|x_{nn+4}\right| \propto \left(\frac{\hbar}{2m\omega}\right)^{3/2+1/2} \frac{a}{\left(-3\hbar\omega\right)}\left[\left(n+1\right)\left(n+2\right)\left(n+3\right)\left(n+4\right)\right]^{1/2}$$

$$\left|x_{nn+4}\right|^2 \propto \frac{\hbar^2 a^2 n^4}{3^2 2^4 m^4 \omega^6}$$

* Direct term

$$\left|x_{nn+1}\right|^2 \propto \frac{\hbar^1}{2m^1\omega^1}\left(n+1\right)$$

Each higher order term gets smaller by a factor $\left(\frac{\hbar n^3 a^2}{3^2 2^3 m^3 \omega^5}\right)$, which is a very small dimensionless factor.
RAPID CONVERGENCE OF PERTURBATION THEORY!

What about Quartic perturbing term b$\mathbf{x}^4$?

Note that $E^{(1)} = \left\langle n \middle| b\mathbf{x}^4 \middle| n \right\rangle \neq 0$

and is directly sensitive to the sign of b!

It is very important to know whether perturbation theory can give us the sign of a perturbation parameter.

- an even power of **x** in $a\mathbf{x}^k$ gives contribution to $E_n^{(1)} = H_{nn}^{(1)}$, which depends on the sign of a.
- an odd power of **x** in ax^k gives a zero contribution to $E_n^{(1)}$ and a non-zero contribution proportional to a^2 to $E_n^{(2)}$, which *does not* depend on the sign of a.
- a cross term, as we will see in $B_v = B_e - \alpha(v + 1/2)$, can give the sign of the coefficient of an odd-k term in $\mathbf{H}^{(1)}$. A bit of a surprise!

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5.73 Quantum Mechanics I
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Perturbation Theory III

Last time

1. $V(\mathbf{x}) = \frac{1}{2}k\mathbf{x}^2 + a\mathbf{x}^3$ cubic anharmonic oscillator

   algebra with $\mathbf{x}^3$ vs. operator algebra with $\mathbf{a}$,$\mathbf{a}^\dagger$
   $a\mathbf{x}^3 \leftrightarrow \omega x$ and Y_{00}

   can't know sign of a from vibrational information alone. [Can know it if rotation-vibration interaction is included.]

2. *M*orse Oscillator $V(\mathbf{x}) = D_e\left[\mathbf{1} - e^{-\alpha\mathbf{x}}\right]^{\mathbf{2}}$

   * $D_e, \alpha \leftrightarrow \omega, \omega x, m$

   * $\frac{d^3V}{dx^3} = 6a = -\frac{3\hbar}{2}\frac{\omega^2\alpha^3}{\omega x} = \left.\frac{d^3V_{morse}}{dx^3}\right|_{x=0}$

   * $\omega x = 2\frac{\alpha^2\hbar}{m^3\omega^4}$ direct from Morse vs. $\frac{15}{4}\frac{a^2\hbar}{m^3\omega^4}$

     from pert. theory on $\frac{1}{2}k\mathbf{x}^2 + a\mathbf{x}^3$

$$\left.\begin{array}{r} \therefore \omega x = 2\frac{\alpha^2\hbar}{m^3\omega^4} \\ \text{from pert. theory on power series expansion of Morse potential (page \#15-4)} \quad \omega x = \frac{15}{4}\frac{a^2\hbar}{m^3\omega^4} \end{array}\right] \text{same functional form}$$

Today:

1. Effect of cubic anharmonicity on transition probability orders of pert. theory, convergence [previous lecture: #15-6,7,8].

2. Use of harmonic oscillator basis sets in wavepacket calculations.

3. What happens when $\mathbf{H}^{(0)}$ has (near) degenerate $E_n^{(0)}$'s? Diagonalize block which contains (near) degeneracies. "Perturbations" — accidental and systematic.

4. 2 coupled non-identical harmonic oscillators: "polyads".

One reason that the result from second-order perturbation theory applied directly to V(x) = k$\mathbf{x}^2$/2 + a$\mathbf{x}^3$ and the term-by-term comparison of the power series expansion of the Morse oscillator are not identical is that contributions to the (n + 1/2)^2 term are neglected from higher derivatives of the Morse potential. In particular

$$E_n^{(1)} = V''''(0)\mathbf{x}^4/4! = \left[7/2\frac{\hbar\omega^2\alpha^4}{\omega\mathbf{x}}\right]\mathbf{x}^4/24$$

$$\left\langle n\left|\mathbf{x}^4\right|n\right\rangle = \left(\frac{\hbar}{2m\omega}\right)^2\left[4\left(n+1/2\right)^2+2\right]$$

contributes in first-order of perturbation theory to the $(n + 1/2)^2$ term in E_n.

$$E_n^{(1)} = \frac{7}{12}\omega x\left(n+1/2\right)^2 + \frac{7}{24}\omega x$$

Example 2 Compute some property other than Energy (repeat of pages 15-6, 7, 8)

need $\psi_n = \psi_n^{(0)} + \psi_n^{(1)}$

transition probability: for electric dipole transitions $P_{n'\leftarrow n} \propto \left|x_{nn'}\right|^2$

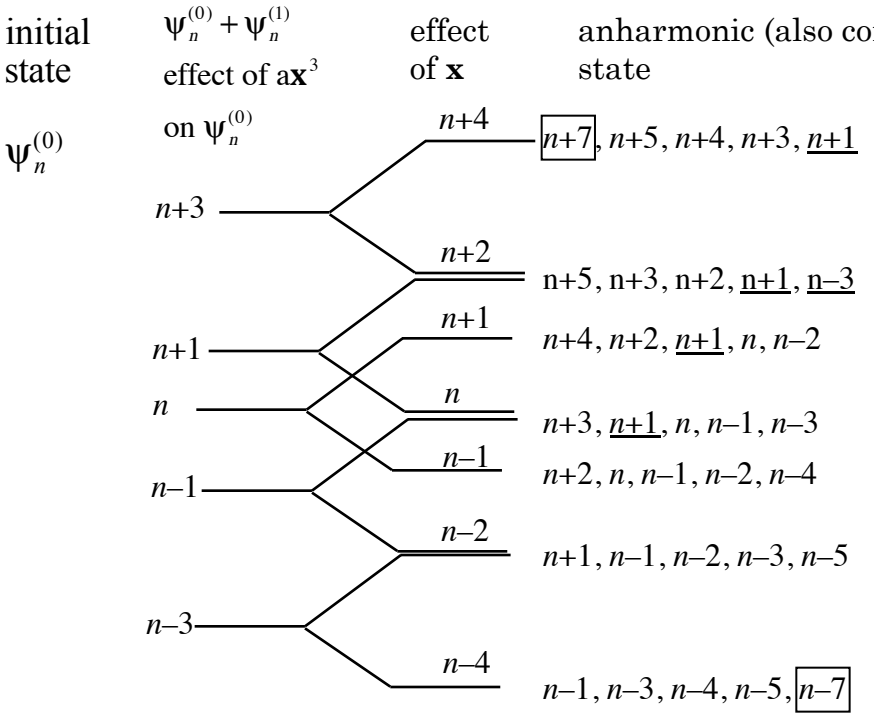
For H-O $n \rightarrow n \pm 1$ only

$$\left|\mathrm{x}_{nn+1}\right|^2 = \left(\frac{\hbar}{2m\omega}\right)(n+1)$$

for perturbed H-O $\mathbf{H}^{(1)} = a\mathbf{x}^3$

$$\psi_n = \psi_n^{(0)} + \sum_k{}' \frac{H_{nk}^{(1)}}{E_n^{(0)} - E_k^{(0)}}\psi_k^{(0)}$$

$$\psi_n = \psi_n^{(0)} + \frac{H_{nn+3}^{(1)}}{-3\hbar\omega}\psi_{n+3}^{(0)} + \frac{H_{nn+1}^{(1)}}{-\hbar\omega}\psi_{n+1}^{(0)} + + \frac{H_{nn-1}^{(1)}}{\hbar\omega}\psi_{n-1}^{(0)} + \frac{H_{nn-3}^{(1)}}{3\hbar\omega}\psi_{n-3}^{(0)}$$



Many paths from initial to final state, which interfere constructively and destructively in $|x_{nn'}|^2$

$n\not\in$ = n + 7, n+5, n+4, n+3, n+2, n+1, n, $n-1$, $n-2$, $n-3$, $n-4$, $n-5$, $n-7$

only paths for H-O!

The transition strengths may be divided into 3 classes

1. direct via $\mathbf{x}$: $n \to n \pm 1$
2. direct plus one anharmonic step $n \to n+4,\ n+2,\ n,\ n-2,\ n-4$
3. direct plus 2 anharmonic steps $n \to n+7,\ n+5,\ n+3,\ n+1,\ n-1,\ n-3,\ n-5,\ n-7$

Work thru the $\Delta n = -7$ path

$$\langle n|x|n+7\rangle = \left(\frac{h}{2m\omega}\right)^{3/2+3/2+1/2}\left[\frac{a^2}{(-3h\omega)^2}\right]\left[\underbrace{(n+1)(n+2)(n+3)}_{x_{n,n+3}}\underbrace{(n+4)}_{x_{n+3,n+4}}\underbrace{(n+5)(n+6)(n+7)}_{x_{n+4,n+7}}\right]^{1/2}$$

$x^3_{n,n+3}$ $x^3_{n+4,n+7}$ $x_{n+3,n+4}$

$$|x_{nn+7}|^2 \propto \frac{\hbar^3 a^4 n^7}{3^4 2^7 m^7 \omega^{11}}$$

* you show that the single-step anharmonic terms go as

$$\left|x_{nn+4}\right| \propto \left(\frac{\hbar}{2m\omega}\right)^{3/2+1/2} \frac{a}{(-3\hbar\omega)}\left[(n+1)(n+2)(n+3)(n+4)\right]^{1/2}$$

$$\left|x_{nn+4}\right|^2 \propto \frac{\hbar^2 a^2 n^4}{3^2 2^4 m^4 \omega^6}$$

* Direct term

$$\left|x_{nn+1}\right|^2 \propto \frac{\hbar^1}{2m^1\omega^1}(n+1)$$

each higher order term gets smaller by a factor $\left(\frac{\hbar n^3 a^2}{3^2 2^3 m^3 \omega^5}\right)$, which is a very small dimensionless factor.
RAPID CONVERGENCE OF PERTURBATION THEORY!

What about the Quartic perturbing term b$\mathbf{x}^4$?

Note that $E^{(1)} = \left\langle n\left|b\mathbf{x}^4\right|n\right\rangle \neq 0$

and is directly sensitive to sign of b!

We get scaling with respect to powers of a, n, ω, and m.
We get magnitudes.
Sometimes we get signs.

2. What about wave-packet calculations?

ψ_n expressed as superposition of $\psi_k^{(0)}$ terms

$\Psi(x,0)$ expanded as superposition of $\psi_k^{(0)}$ terms (usually only one term, called the "bright state"). But we must instead expand $\psi_k^{(0)}$ as a superposition of eigenbasis, ψ_k, terms.

$\Psi(x,t)$ oscillates at $e^{-iE_n t/\hbar}$ ← $E_n = E_n^{(0)} + E_n^{(1)} + E_n^{(2)}$

A time-evolving state, which is initially in a pure zero-order state, $\psi_n^{(0)}$, will dephase, then exhibit partial recurrences at

any integer $\longrightarrow m2\pi \approx \omega t \quad t = \dfrac{m2\pi}{\omega}$

but * the rephasing is not perfect since, due to anharmonicity:

$$E_n - E_m \neq \hbar\omega(n-m).$$

These are not-quite-integer multiples of a common factor ($\hbar\omega$)!

* t*ime* of 1st recurrence will depend on $\langle E \rangle$!

because $\dfrac{E_{n+1} - E_{n-1}}{2}$ decreases as n increases.

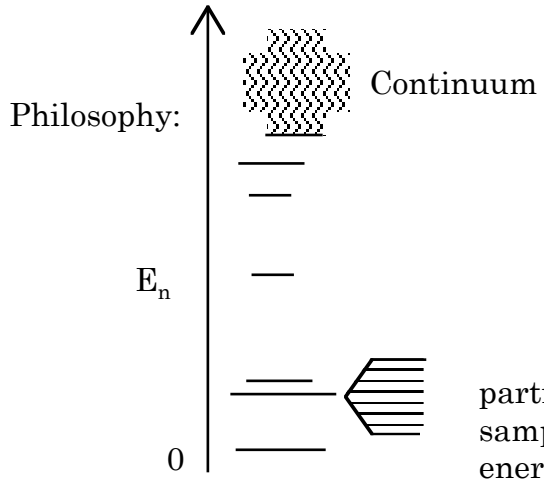
Degenerate and Near Degenerate $E_n^{(0)}$

* Ordinary nondegenerate perturbation theory treats **H** as if it can be "diagonalized" by simple algebra.
* CTDL, pages 1104-1107 → find linear combination of degenerate $\psi_n^{(0)}$ for which $\mathbf{H}^{(1)}$ lifts degeneracy.
* This problem is usually treated in an abstract way by people who have never actually used perturbation theory!

$$\text{Whenever } \left|\frac{H_{nk}^{(1)}}{E_n^{(0)} - E_k^{(0)}}\right| \approx 1$$

must diagonalize the n,k 2×2 block of $\mathbf{H} = \mathbf{H}^{(0)} + \mathbf{H}^{(1)}$

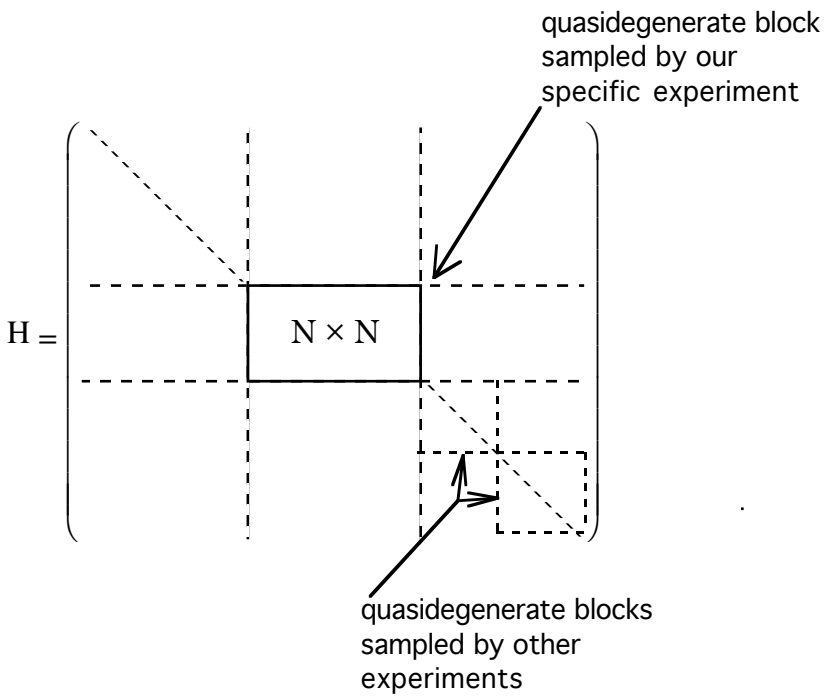
accidental degeneracy — "spectroscopic perturbations"
systematic degeneracy — 2-D isotropic H-O, "polyads"
quasi-degeneracy — special chunk of **H**
effects of remote states — Van Vleck Perturbation Theory - next lecture



particular class of experiment does not sample <u>all</u> E_n's - only samples over a given energy range and only at a given energy resolution!

Want a model that replaces the ∞ dimension **H** by a simpler *finite* one that does really well for the class of states sampled by the particular experiment.

NMR	nuclear spins (hyperfine)	don't care about excited vibrational or electronic states
IR	vibr. and rotation	don't care about Zeeman splittings
UV	electronic	don't care about Zeeman splittings



Each finite block along the diagonal of **H** is described by an $\mathbf{H}^{\text{effective}}$ "fit model". We want these fit models to be as accurate and physically realistic as possible.

* fold important out-of-block effects into N × N block → involves 2 strips of **H**
* diagonalize augmented N × N block - refine parameters that define the block against observed energy levels.

next time review V-V transformation

4. Best to illustrate with an example — 2 coupled harmonic oscillators: "Fermi Resonance" [approx. integer ratios between characteristic frequencies of vibrational subsystems]

$$\mathbf{H} = \left[\frac{\mathbf{p}_1^2}{2m_1} + \frac{1}{2}k_1\mathbf{x}_1^2\right] + \left[\frac{\mathbf{p}_2^2}{2m_2} + \frac{1}{2}k_2\mathbf{x}_2^2\right] + k_{122}\mathbf{x}_1\mathbf{x}_2^2$$ why not $k_{12}\mathbf{x}_1\mathbf{x}_2$?

$$\psi^{(0)}_{n_1 n_2} = \psi^{(1)}_{n_1}(x_1)\psi^{(0)}_{n_2}(x_2)$$

$\mathbf{H}_1^{(0)}$ $\mathbf{H}_2^{(0)}$

$$E^{(0)}_{n_1} = \hbar\omega_1\left(n_1 + 1/2\right)$$
$$E^{(0)}_{n_2} = \hbar\omega_2\left(n_2 + 1/2\right)$$
$$\Rightarrow E^{(0)}_{nm} = \hbar\left[\omega_1\left(n + 1/2\right) + \omega_2\left(m + 1/2\right)\right]$$

let $\omega_1 = 2\omega_2 \quad (m_1 = m_2, k_1 = 4k_2)$

There are systematic degeneracies.

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$$\mathbf{H}^{(1)} = k_{122}\mathbf{x}_1\mathbf{x}_2^2 = k_{122}\left(\frac{\hbar}{2m}\right)^{3/2}\left(\frac{1}{\omega_1\omega_2^2}\right)^{1/2}\left[\left(\mathbf{a}_1+\mathbf{a}_1^\dagger\right)\left(\mathbf{a}_2^2+\mathbf{a}_2^{\dagger 2}+\mathbf{a}_2\mathbf{a}_2^\dagger+\mathbf{a}_2^\dagger\mathbf{a}_2\right)\right]$$

$$\mathbf{a}\mathbf{a}^\dagger+\mathbf{a}^\dagger\mathbf{a}=2\mathbf{a}^\dagger\mathbf{a}+1$$

$\mathbf{H}^{(1)}$ = (constants)

6 types of terms

	$\mathbf{H}^{(1)}_{nm;k\ell}$		
	n – k	m – ℓ	$\mathbf{H}^{(1)}$
$\mathbf{a}_1\mathbf{a}_2^2$	–1	–2	$[(n+1)(m+2)(m+1)]^{1/2}$
$\mathbf{a}_1\mathbf{a}_2^{\dagger 2}$	–1	+2	$[(n+1)(m)(m-1)]^{1/2}$
$\mathbf{a}_1\left(2\mathbf{a}_2^\dagger\mathbf{a}_2+1\right)$	–1	0	$[(n+1)(2m+1)^2]^{1/2}$
$\mathbf{a}_1^\dagger\mathbf{a}_2^2$	+1	–2	$[(n)(m+2)(m+1)]^{1/2}$
$\mathbf{a}_1^\dagger\mathbf{a}_2^{\dagger 2}$	+1	+2	$[(n)(m)(m-1)]^{1/2}$
$\mathbf{a}_1^\dagger\left(2\mathbf{a}_2^\dagger\mathbf{a}_2+1\right)$	+1	0	$[(n)(2m+1)^2]^{1/2}$

Seems complicated – but all we need to do is look for systematic near degeneracies Recall $\omega_1 = 2\omega_2$

List of "Polyads" by Membership		$E^{(0)}/\hbar\omega_2$	"Polyad Number" $P = 2n_1 + n_2$
		$[2(n_1 + 1/2) + (n_2 + 1/2)]$	
(n_1, n_2)	degeneracy		
(0,0)	1	1+ 1/2 = 3/2	0
(0,1)	1	1 + 3/2 = 5/2	1
(1,0), (0,2)	2	3 + 1/2 = 7/2	2
(1,2), (0,3)	2	3 + 3/2 = 9/2; 1 + 7/2 = 9/2	3
(2,0), (1,2), (0,4)	3	11/2	4
(2,1),(1,3),(0,5)	3	13/3	5
?	4	15/2	6
?	4	17/2	7
	etc.	19/2	8

General P block:

$$E_P^{(0)}/\hbar\omega_2 = \frac{3}{2} + (2n_1 + n_2) = P + 3/2$$

of terms in P block depends on whether P is even or odd

$\frac{P+2}{2}$ states even P $\left(n_1 = \frac{P}{2}, n_2 = 0\right), \left(n_1 = \frac{P}{2} - 1, 2\right), \ldots (0, P)$

$\frac{P+1}{2}$ states odd P $\left(n_1 = \frac{P-1}{2}, n_2 = 1\right), \ldots (1, P)$

not 0 because $P = 2n_1 + n_2$ is odd

$$\left(\frac{\mathbf{H}^{(1)}}{\hbar^{3/2} m^{-3/2} \omega_1^{-1/2} \omega_2^{-1} k_{122} 2^{-3/2}}\right) = \mathbf{a}_1 \mathbf{a}_2^{\dagger 2} + \mathbf{a}_1^{\dagger} \mathbf{a}_2^2 + \mathbf{a}_1 \mathbf{a}_2^2 + \mathbf{a}_1^{\dagger} \mathbf{a}_2^{\dagger 2} + \mathbf{a}_1 \left(2\mathbf{a}_2^{\dagger} \mathbf{a}_2 + 1\right) + \mathbf{a}_1^{\dagger} \left(2\mathbf{a}_2^{\dagger} \mathbf{a}_2 + 1\right)$$

$\Delta P =$ 0 (inside polyad) 0 | −4 +4 −2 +2 (between polyad blocks)

POLYAD

$$\frac{\mathbf{H}_P^{(0)}}{\hbar\omega_2} = \begin{pmatrix} P+3/2 & 0 & 0 & 0 \\ 0 & P+3/2 & 0 & 0 \\ 0 & 0 & \ddots & 0 \\ 0 & 0 & 0 & P+3/2 \end{pmatrix}$$

$$\frac{\mathbf{H}_P^{(1)}}{\text{stuff}} = \begin{array}{c} n,m \\ \frac{P}{2},0 \\ \frac{P}{2}-1,2 \\ \vdots \\ 1,P-2 \\ 0,P \end{array} \begin{array}{c} \begin{array}{ccccc} \frac{P}{2},0 & \frac{P}{2}-1,2 & \frac{P}{2}-2,4 & \cdots & \cdots 0,P \end{array} \\ \begin{pmatrix} 0 & \left[\left(\frac{P}{2}\right)(2\bullet 1)\right]^{1/2} & 0 & 0 & 0 \\ sym & 0 & \left[\left(\frac{P}{2}-1\right)(3\bullet 4)\right]^{1/2} & 0 & 0 \\ 0 & sym & 0 & \cdots & 0 \\ 0 & 0 & sym & 0 & \left[(1)(P)(P-1)\right]^{1/2} \\ 0 & 0 & 0 & sym & 0 \end{pmatrix} \end{array} \quad \text{(even } P\text{)}$$

Note that all matrix elements may be written in terms of a general formula — computer decides memberships in each polyad and sets up the matrix.

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So now we have listed ALL of the connections of $P = 6$ to all other blocks!
So we use these results to add some correction terms to the $P = 6$ block according to the formula suggested by Van Vleck.

$$\mathbf{H}^{(2)}_{P_{nm}} = \sum_{P'} \frac{H^{(1)}_{nk} H^{(1)}_{km}}{\frac{E^{(0)}_n + E^{(0)}_m}{2} - E^{(0)}_k}$$

for our example*, the denominator is $\hbar\omega_2\left[P - P'\right]$

* For this particular example there are no cases where there are nonzero elements for n ≠ m (many other problems exist where there are nonzero n ≠ m terms)

$$\underbrace{\frac{\hbar\omega_2 H^{(2)}_6}{\hbar^3 m^{-3}\omega_1^{-1}\omega_2^{-2}k_{122}^2 2^{-3}}}_{\text{dimensionless}} = \begin{matrix} 30 \\ 22 \\ 14 \\ 06 \end{matrix} \begin{pmatrix} \frac{3}{2}-\frac{4}{2}-\frac{8}{4}=-\frac{5}{2} & & & \\ & \frac{50}{2}-\frac{75}{3}+\frac{4}{4}-\frac{36}{4}=-8 & & \\ & & \frac{81}{2}-\frac{162}{2}+\frac{12}{4}-\frac{60}{4}=-\frac{105}{2} & \\ & & & -\frac{169}{2}-\frac{56}{4}=-\frac{197}{2} \end{pmatrix}$$

Computers can easily set these things up.
Could add additional perturbation terms such as diagonal anharmonicities that cause $\omega_1 : \omega_2$ resonance to detune from 2 : 1.

For concreteness, look at $P = 6$ polyad, which includes the (n_1, n_2) states for $2n_1 + n_2 = 6$: (3,0), (2,2), (1,4), (0,6)

$\mathbf{H}_6^{(1)}$ / stuff

	30	22	14	06
30	0	$(3\cdot 2\cdot 1)^{1/2}$	0	0
22	sym	0	$(2\cdot 4\cdot 3)^{1/2}$	0
14	0	sym	0	$(1\cdot 5\cdot 6)^{1/2}$
06	0	0	sym	0

We need to identify what are **all** of the out-of-block elements of the $\mathbf{x}_1\mathbf{x}_2^2$ term that affect the $P = 6$ block?

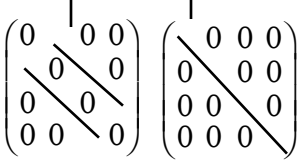
		$P = 6, P'$	$\mathbf{H}^{(1)}$/stuff	$E_P^{(0)} - E_{P-2}^{(0)}$
$P = 6$, $\Delta P = -2$ $P' = P-2 = 4$	$\mathbf{a}_1\left(2\mathbf{a}_2^\dagger\mathbf{a}_2 + 1\right)$	3,0 ~ 2,0	$3^{1/2}$	$+2\hbar\omega_2$
		2,2 ~ 1,2	$2^{1/2}\cdot 5$	$+2\hbar\omega_2$
		1,4 ~ 0,4	$1^{1/2}\cdot 9$	$+2\hbar\omega_2$
		0,6 ~ —	—	—
$\Delta P = +2$ $P' = P+2 = 8$	$\mathbf{a}_1^\dagger\left(2\mathbf{a}_2^\dagger\mathbf{a}_2 + 1\right)$	3,0 ~ 4,0	$4^{1/2}$	$-2\hbar\omega_2$
		2,3 ~ 3,2	$3^{1/2}\cdot 5$	$-2\hbar\omega_2$
		1,4 ~ 2,4	$2^{1/2}\cdot 9$	$-2\hbar\omega_2$
		0,6 ~ 1,6	$1^{1/2}\cdot 13$	$-2\hbar\omega_2$
$\Delta P = -4$ $P' = P-4 = 2$	$\mathbf{a}_1\mathbf{a}_2^2$	3,0 ~ —	—	—
		2,2 ~ 1,0	$2^{1/2}(2\cdot 1)^{1/2}$	$+4\hbar\omega_2$
		1,4 ~ 0,2	$1^{1/2}(4\cdot 3)^{1/2}$	$+4\hbar\omega_2$
		0,6 ~ —	—	—
$\Delta P = +4$ $P' = P+4 = 10$	$\mathbf{a}_1^\dagger\mathbf{a}_2^{\dagger 2}$	3,0 ~ 4,2	$[4\cdot 2\cdot 1]^{1/2}$	$-4\hbar\omega_2$
		2,2 ~ 3,4	$[3\cdot 4\cdot 3]^{1/2}$	$-4\hbar\omega_2$
		1,4 ~ 2,6	$[2\cdot 6\cdot 5]^{1/2}$	$-4\hbar\omega_2$
		0,6 ~ 1,8	$[1\cdot 8\cdot 7]^{1/2}$	$-4\hbar\omega_2$

Within $P = 6$ block elements of $\mathbf{H}^{(1)}$

$P' \neq P = 6$: out-of-$P = 6$ block interactions folded into the $P = 6$ block on-diagonal

$$\mathbf{H}_{P=6}^{\text{eff}} = \mathbf{H}_6^{(0)} + \mathbf{H}_6^{(1)} + \mathbf{H}_6^{(2)}$$

$\hbar\omega_2(6 + 3/2)$



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5.73 Quantum Mechanics I
Fall 2018

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Vibrations of Polyatomic Molecules
H is ∞ dimension
How do we arrange and "read" it?
How do we truncate it? $\mathbf{H} \rightarrow \mathbf{H}^{eff}$
Near Degeneracies – accidental or "intentional"

molecular dynamics is uniquely sensitive to "resonances"

unique sensitivity to inter- and intra-molecular interactions
*IVR – Intramolecular Vibrational Redistribution
LOCAL MODES *x-k relationships
*Polyads
Two frequently occurring polyad cases 2:1 and 2:2 resonances
Van Vleck Transformation for inter-polyad interactions
Isomerization

Not just for small molecules in gas phase

Mark Johnson for H-bonding in H_2O clusters.

molecule with N atoms: 3N-6 normal modes

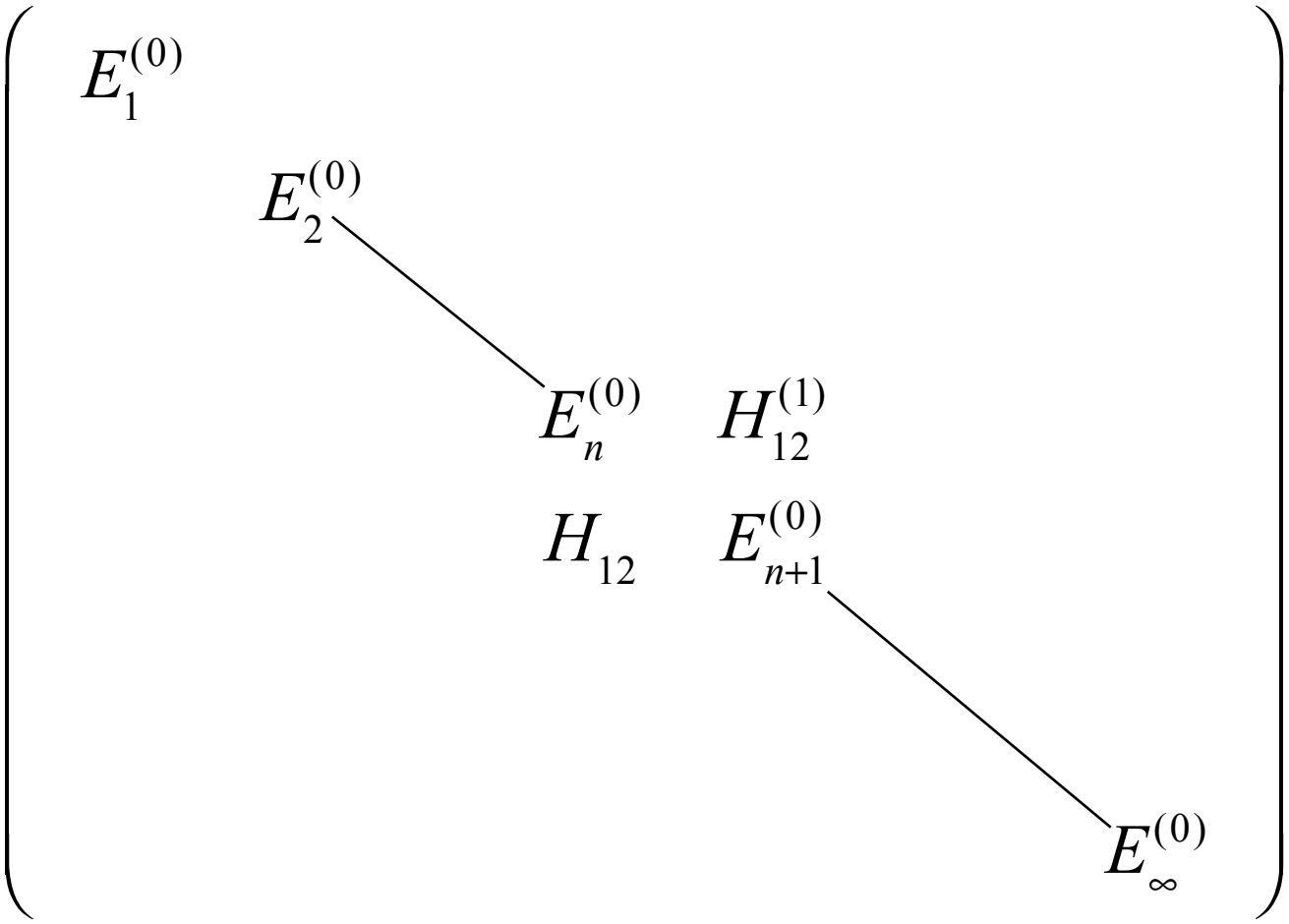
What is a Normal Mode?
synchronized dance of all N atoms
Classical Mechanics
Wilson Decius and Cross "Molecular Vibrations"

QM

$$\omega_e = [k/\mu]^{1/2}$$

↓

μ is generalized



$$\text{usually } \left| \frac{H_{nm}^{(1)}}{E_n^{(0)} - E_m^{(0)}} \right| \ll 1$$

but we will see 2 things:

matrix elements of $\mathbf{x}^k$ grow as constant $\otimes$ $n^{k/2}$ and contribute to energy levels as n^{k-1}

density of states increases so some get very small

We need to deal with anharmonic interactions, which become increasingly important E ↑

N-Atom Polyatomic Molecules

[use **BOLD CAPITAL** letters to symbolize many vibrational modes]

3N–6 normal modes

What is a normal mode?

$$V(\mathbf{Q})^{(0)} = \sum_{i=1}^{3N-6} \frac{1}{2} k_i Q_i^2$$

$$E^{(0)}_{\{n_i\}} = \sum_{i=1}^{3N-6} \hbar\omega_i \left(n_i + 1/2\right)$$

synchronized dance of *all* atoms

↓

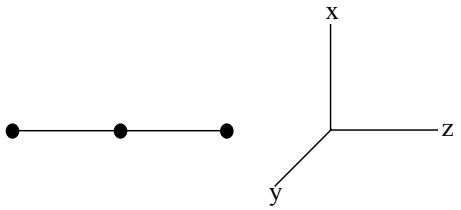
what happens if one dancer has a sprained ankle?

↓

everyone for themself! (i.e., "local modes")

transitions $\mu(\mathbf{Q}) = \mu_0 + \underbrace{\frac{d\mu}{dQ_i}}_{\text{change of } Q_i \text{ causes change of } \mu} \mathbf{Q}_i$

CO_2 is symmetric $\mu_0 = 0$



mode 1	symmetric stretch	$\frac{d\mu}{dQ_1} = 0$	$\Delta\mathbf{V} = 0$
mode 2	band (⊥) type	$\frac{d\mu}{dQ_2} \neq 0$	$\Delta n_2 = \pm 1$
mode 3	anti-symmetric stretch (‖) type	$\frac{d\mu}{dQ_3} \neq 0$	$\Delta n_3 = \pm 1$

Expect simple spectra, exclusively $\Delta n_2 = 1$ or $\Delta n_3 = 1$

NOT BOTH $\Delta n_2 = 1$ and $\Delta n_3 = 1$ simultaneously

Expect energy initially deposited in specific $(n_1, n_2, n_3)^0$ basis state to stay in that state. "Mode Specific Chemistry": an unfulfilled dream

Expectations are violated because of anharmonic interaction terms.

$$V^{(1)}(\mathbf{Q}) = \sum_{i=1} \{\text{cubic terms}\} + \{\text{quartic terms}\}$$

cubic

$$\begin{cases} \frac{1}{6}k_{111}Q_1^3 + \frac{1}{6}k_{222}Q_2^3 + \frac{1}{6}k_{333}Q_3^3 \\ \frac{1}{2}k_{112}Q_1^2Q_2 + \frac{1}{2}k_{113}Q_1^2Q_3 + \frac{1}{2}k_{123}Q_1Q_2Q_3 \end{cases}$$

quartic $\frac{1}{24}k_{1111}Q_1^4$ etc.

Each of these terms has explicitly known quantum number dependent selection rules and magnitude scaling rules.

Seems like a nightmare! But actually it is quite the opposite.

Most of the anharmonic terms connect basis states that are energetically remote from each other. These interactions can be dealt with by 2nd order Perturbation Theory.

A few interaction terms connect near degenerate basis states. These dominate the dynamics and give rise to "polyads"

- must be diagonalized
- can lead to local modes
- can enable isomerization

→ broken picture of level splittings and transition intensities

Let's look at a simple example that illustrates all of the steps.

There are two special cases.

Fermi Resonance	$\omega_1 \approx 2\omega_2$	2:1
Darling-Dennison	$2\omega_1 \approx 2\omega_3$	2:2

Symmetric stretch and anti-symmetric stretch usually have similar frequency, but one is symmetric and the other is anti-symmetric.

Q_1	$\otimes$	Q_1	is symmetric
	Q_3		is anti-symmetric
Q_3	$\otimes$	Q_3	is symmetric

$\longrightarrow$[these 2:2 anharmonic terms lead to "local stretchers"]

wave my arms

Fermi Resonance involves a cubic anharmonic term.

$$k_{133}Q_1Q_3^2$$

put this into $\mathbf{a},\mathbf{a}^\dagger$ form

$$\mathrm{Q}_1 = \left(\frac{\hbar}{2m_1\omega_1}\right)^{1/2}\left(\mathbf{a}_1 + \mathbf{a}_1^\dagger\right)$$

$$\mathrm{Q}_3 = \left(\frac{\hbar}{2m_3\omega_3}\right)^{1/2}\left(\mathbf{a}_3 + \mathbf{a}_3^\dagger\right)$$

Let $m_1 = m_3$ (this is a bit of a fraud because the mass factors are not quite so simple)

$$k_{133}Q_1Q_3^2 = \left[\frac{\hbar}{2m}\right]^{3/2}\left[\omega_1\omega_3^2\right]^{-1/2}\left(\mathbf{a}_1 + \mathbf{a}_1^\dagger\right)\left(\mathbf{a}_3 + \mathbf{a}_3^\dagger\right)^2$$

Operator algebra: intermode $\left[\mathbf{a}_i, \mathbf{a}_j\right] = \left[\mathbf{a}_i, \mathbf{a}_j^\dagger\right] = 0$

$$\left(\mathbf{a}_1 + \mathbf{a}_1^\dagger\right)\left(\mathbf{a}_3 + \mathbf{a}_3^\dagger\right) = \mathbf{a}_1\left[\mathbf{a}_3^2 + \mathbf{a}_3^{\dagger 2} + \left(2N_3 + 1\right)\right] + \mathbf{a}_1^\dagger\left[\mathbf{a}_3^2 + \mathbf{a}_3^{\dagger 2} + \left(2N_3 + 1\right)\right]$$

6 terms $\qquad H_{n_1,n_3;n_1',n_3'} \equiv \left\langle n_1, n_3 \middle| \mathbf{H}^{(1)} \middle| n_1', n_3' \right\rangle$

operator	$n_1 - n_1'$	$n_3 - n_3'$	$\mathbf{H}^{(1)}_{n_1,n_3;n_1',n_3'}$	$\Delta E = E_{n_1,n_3} - E_{n_1',n_3'} = \hbar\left[\left(n_1 - n_1'\right)\omega_1 + \left(n_3 - n_3'\right)\omega_3\right]$
$\mathbf{a}_1, \mathbf{a}_3^2$	–1	–2	$(n_1+1)^{1/2}[(n_3+2)(n_3+1)^{1/2}$	$\hbar[-\omega_1-2\omega_3]$
$\mathbf{a}_1, \mathbf{a}_3^{\dagger 2}$	–1	+2	$(n_1+1)^{1/2}[n(n-1)]^{1/2}$	$\hbar[-\omega_1+2\omega_3]$
$\mathbf{a}_1(2\mathbf{N}_3+1)$	–1	0	$(n_1+1)^{1/2}[(2n_3+1)]^{1/2}$	$\hbar[-\omega_1]$
$\mathbf{a}_1^\dagger, \mathbf{a}_3^2$	+1	–2	$(n_1)^{1/2}[(n_3+2)(n_3+1)]^{1/2}$	$\hbar[\omega_1-2\omega_3]$
$\mathbf{a}_1^\dagger, \mathbf{a}_3^{\dagger 2}$	+1	+2	$(n_1)^{1/2}[n(n-1)]^{1/2}$	$\hbar[\omega_1+2\omega_3]$
$\mathbf{a}_1^\dagger\left(2\mathbf{N}_3+1\right)$	+1	0	$(n_1)^{1/2}[2n_3+1]^{1/2}$	$\hbar[\omega_1]$

The two circled terms involve near degeneracy.

They require diagonalization of a near-degenerate block.

All of the other terms can be dealt with by 2nd order non-degenerate Perturbation Theory.

OK, we are interested in all of the terms where $\omega_1 - 2\omega_3 \approx 0$.

$$P = 2n_1 + n_2$$

(n_1: 2 points for $\omega_1 = 2\omega$; n_2: 1 point for $\omega_3 = \omega$)

Polyad #	total membership of states	
P = 0	(0, 0)	1
P = 1	(0, 1)	1
P = 2	(1, 0), (0, 2)	2
P = 3	(1, 1), (0, 3)	2
P = 4	(2, 0), (1, 2), (0, 4)	3
P = 5	(2, 1), (1, 3), (0, 5)	3
P = 6	(3, 0), (2, 2), (1, 4), (0, 6)	4

We have degenerate groups of levels where all within-group matrix elements scale larger with n_1, n_3 and within-group membership increases. Worse and worse!!!!! Or is it?

This amounts to increasingly strong interactions among larger groups of states. But it is consistent with a mode. **NOT ERGODIC!!!**

This is a dominant feature of both spectrum and dynamics.

$\mathbf{k}_{133}$ can either make the bend softer or stiffer as the stretch increases.

We can rewrite everything so far in terms of P.

$$\frac{E_p^{(0)}}{\hbar\omega} = 2\left(n_1 + 1/2\right) + \left(n_3 + 1/2\right)$$
$$= 2n_1 + n_3 + 3/2 = P + 3/2$$

even-P: states are $\left(n_1 = \frac{P}{2}, n_3 = 0\right), \left(n_1 = \frac{P}{2} - 1, n_3 = 2\right), \ldots (0, P)$

there are $\frac{P+2}{2}$ states in polyad

odd-P: states are $\left(n_1 = \frac{P-1}{2} - 1, n_3 = 1\right), \left(n_1 = \frac{P-3}{2}, n_3 = 3\right), \ldots (0, P)$

there are $\frac{P+1}{2}$ states in polyad

$$\left(\frac{\mathbf{H}^{(1)}}{\hbar^{3/2}m^{-3/2}\omega_1^{-1/2}\omega_3^{-1}k_{122}2^{-3/2}}\right)=\underbrace{\mathbf{a}_1\mathbf{a}_3^{\dagger 2}+\mathbf{a}_1^{\dagger}\mathbf{a}_3^{2}}_{\Delta P=0}+\underbrace{\underbrace{\mathbf{a}_1\mathbf{a}_3^{2}+\mathbf{a}_1^{\dagger}\mathbf{a}_3^{\dagger 2}}_{-4\quad -4}\underbrace{\mathbf{a}_1^{\dagger}\left(N_2+1\right)+\mathbf{a}\left(N_2+1\right)}_{-2\quad -2}}_{\text{between polyads}}$$

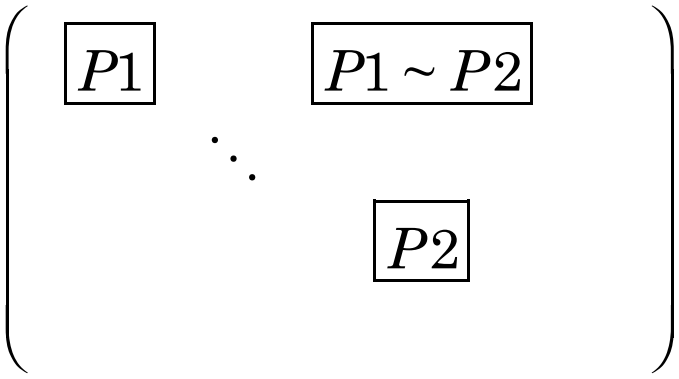
This is a recipe for adding <u>interpolyad interactions</u> via 2nd order Perturbation Theory.

The 133 Polyad itself:

$$\frac{\mathbf{H}_P^{(0)}}{\hbar\omega}=\begin{pmatrix} P+3/2 & 0 & 0 & 0 \\ 0 & P+3/2 & 0 & 0 \\ 0 & 0 & \ddots & 0 \\ 0 & 0 & 0 & P+3/2 \end{pmatrix}$$

	$\frac{\mathbf{H}_P^{(1)}}{\text{stuff}}=\frac{P}{2},0$	$\frac{P}{2}-1,2$	$\frac{P}{2}-2,4$	…	$0,P$
even P	$\frac{P}{2},0$	0	$\left[\left(\frac{P}{2}\right)(2-1)\right]$		
	$\frac{P}{2}-1,2$	sym	0	$\left[\left(\frac{P}{2}-1\right)(3\cdot 4)\right]^{1/2}$	
	$\frac{P}{2}-2,4$		sym	0	
	$\vdots$				$\left[(1)(P)(P-1)\right]^{1/2}$
	$0,P$			sym	0

Inter-Block Interactions



Van Vleck Transformation [HLB-RWF, pages 237-243]

Second-Order corrections along diagonal

$$E_1^{(0)} + \frac{\mathbf{H}_{ij}^{(1)2}}{E_i^{(0)} - E_1^{(0)}}$$

Off diagonal

$$\mathbf{H}_{ab}^{(1)} + \sum_{\alpha} \frac{\mathbf{H}_{a\alpha}^{(1)}\mathbf{H}_{\alpha b}^{(1)}}{\frac{E_a^{(0)} + E_b^{(0)}}{2} - E_{\alpha}^{(0)}}$$

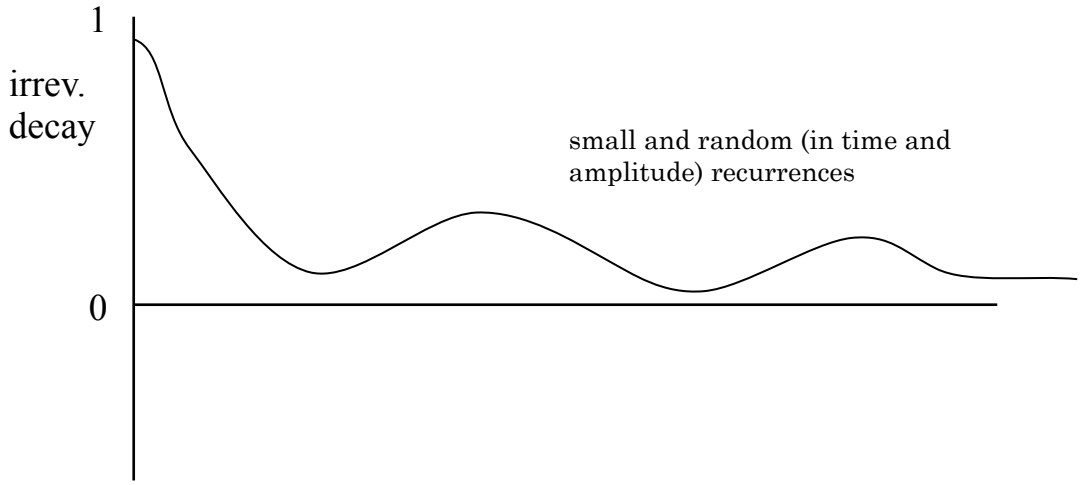
Intramolecular Vibrational Redistribution (IVR)

The Pluck: $\Psi(\mathbf{Q},0) = \phi_{\{\mathbf{V}\}}^{(0)}$

$\phi_{\{\mathbf{V}\}}^{(0)}$ expressed as a sum of $a_{\{\mathbf{V}\}}\psi_{\{\mathbf{V}\}}$

$$\Psi\left(\mathbf{Q},t\right) = \sum_{\{\mathbf{V}\}} a_{\{\mathbf{V}\}} e^{-iE_{\{\mathbf{V}\}}t/\hbar}$$

"Survival Probability": $P = \left|\int \Psi^*(\mathbf{Q},0)^* \Psi(\mathbf{Q},t) d\mathbf{Q}\right|^2$



how fast?
minor recurrences?

"x-k" relationships

$$E_{\{\mathbf{v}\}} = \hbar\sum_{\{\mathbf{v}\}} \omega_i \left(n_i + 1/2\right) - \sum_{\{\mathbf{v}\},\{\mathbf{v}^1\}} x_{ij}\left(n_i + 1/2\right)\left(n_j + 1/2\right)$$

$$\left\{x_{ij}\right\} \leftrightarrow k_{iij}, k_{iiij}, k_{iiiij}$$

from Perturbation Theory

Polyads – a model of what is supposed to happen far above where normal modes are well known to be a bad approximation.

a "broken pattern" that extrapolates well → a "pattern of broken patterns"

But at high E, some normal modes are replaced by local modes.

⇩ a frequent path to isomerization

trans-cis in S_1 acetylene [*Science* **350**, 1338 (2015)]
acetylene-vinylidene in S_0 acetylene [*Science* **358**, 336 (2017)]

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Variational Method

(See CTDL 1148-1155, [Variational Method]
252-263, 295-307[Density Matrices])

Last time:

Quasi-Degeneracy → Diagonalize a part of infinite **H**
* sub-matrix : $\mathbf{H}^{(0)} + \mathbf{H}^{(1)}$
* corrections for effects of out-of-block elements: $\mathbf{H}^{(2)}$
(the Van Vleck transformation)
*diagonalize $\mathbf{H}^{eff} = \mathbf{H}^{(0)} + \mathbf{H}^{(1)} + \mathbf{H}^{(2)}$

coupled H-O's example: the 2 : 1 ($\omega_1 \approx 2\omega_2$) Fermi resonance polyads

1. Perturbation Theory vs. Variational Method: non-orthognal → **S** (overlap matrix)
2. Variational Theorem
3. Stupid nonlinear variation
4. Linear Variation → new kind of secular Equation
5. Linear combined with nonlinear variation
6. Strategies for criteria of goodness — various kinds of variational calculations

1. Perturbation Theory vs. Variational Method

Perturbation Theory effectively uses ∞ basis set
non-degenerate: diagonalize $\mathbf{H}^{eff}$
quasi-degenerate: non-diagonal $\mathbf{H}^{eff}$ (model with quantum number scaling)
goals: parametrically parsimonious fit model, $\mathbf{H}^{eff}$
fit parameters (molecular constants) ↔ parameters that define V(x)

order-sorting $\dfrac{H_{nk}^{(1)}}{E_n^{(0)} - E_k^{(0)}} < 1$ — errors smaller than this "mixing angle" times the previous order non–zero correction term

(n is in-block, k is out-of block) because diagonalization is to ∞ order (within block).

Variational Method



best possible estimate for lowest few E_n, ψ_n (and properties derivable from these) using a finite (often huge) basis set and the exact form of **H**.

Vast majority of computer time in Chemistry is spent in variational calculations
Goal is numbers. Insight is secondary.
"*Ab Initio*" vs. "semi-empirical" or "fitting"

[intentionally bad basis set: Hückel, tight binding –
qualitative behavior obtained by a fit to a few microscopic–<u>like</u>
control parameters]

2. Variational Theorem

not necessarily normalized

any observable

If ϕ is approximation to eigenfunction of $\hat{\mathbf{A}}$
that belongs to the lowest eigenvalue, a_0, then

$$\alpha \equiv \frac{\langle \phi | \mathbf{A} | \phi \rangle}{\langle \phi | \phi \rangle} \geq a_0$$

the Variational Theorem

PROOF: eigenbasis (which we do not know – but know that it must exist)

$$\mathbf{A}|n\rangle = a_n |n\rangle$$

expand $|\phi\rangle$ in eigenbasis of **A**, exploiting completeness

$$|\phi\rangle = \sum_n |n\rangle\langle n|\phi\rangle$$

completeness

$$\langle \phi | \mathbf{A} | \phi \rangle = \sum_{n,n'} \langle \phi | n \rangle \underbrace{\langle n | \mathbf{A} | n' \rangle}_{a_n \, \delta_{nn'}} \langle n' | \phi \rangle = \sum_n |\langle \phi | n \rangle|^2 a_n$$

eigenbasis for **A**

$$\langle \phi | \phi \rangle = \sum_n \langle \phi | n \rangle \langle n | \phi \rangle = \sum_n |\langle \phi | n \rangle|^2$$

$$\alpha \equiv \frac{\langle \phi | \mathbf{A} | \phi \rangle}{\langle \phi | \phi \rangle} = \frac{\sum_n a_n |\langle n | \phi \rangle|^2}{\sum_{n'} |\langle n' | \phi \rangle|^2}$$

all terms in both sums are ≥ 0

subtract a_0 from both sides

$$\alpha - a_0 = \frac{\sum_n (a_n - a_0) |\langle n | \phi \rangle|^2}{\sum_{n'} |\langle n' | \phi \rangle|^2} \geq 0$$

again, all terms in both sums are ≥ 0

because $\sum_n |\langle n | \phi \rangle|^2 = \sum_{n'} |\langle n' | \phi \rangle|^2$

We are done because, by definition of a_0, $a_n \geq a_0$ for all n and all terms in sum are $\therefore \geq 0$.

$\therefore \quad \alpha \geq a_0$. QED (but useless because we do not know a_n or $\langle n|\phi\rangle$ in advance)

It is possible to perform a variational calculation for any **A**, not limited to **H**.

---

3. Stupid Nonlinear Variation
   Use the wrong functional form or the wrong variational criterion to get poor results — illustrates that the variational function must have sufficient flexibility and the variational criterion must be as it is specified in the variational theorem, as opposed to a clever shortcut.

The H atom Schrödinger Equation ($\ell = 0$)

$$\mathbf{H} = \underbrace{-\frac{1}{2}\frac{1}{r^2}\frac{\partial}{\partial r}r^2\frac{\partial}{\partial r}}_{T} \quad \underbrace{-\frac{1}{r}}_{V}$$

and we know $\begin{cases} \psi_{1s}(r) = \langle r|1s\rangle = \pi^{-1/2}e^{-r} \\ E_{1s} = -1/2 \; au \end{cases}$ $\left[1\; au = 219475\; cm^{-1}\right]$

but try $\langle r|\phi\rangle = \left[\xi^3/2\pi\right]^{1/2}(\xi r)e^{-\xi r}$ normalized for all ξ

ξ is a scale factor that controls overall size of ϕ(r)

[actually this is the form of ψ_{2p}(r)] which at $\xi = 1$ is necessarily orthogonal to ψ_{1s}!
STUPID!

$\left(\phi(0) = 0 \qquad but \qquad \psi_{1s}(0) = \pi^{-1/2}\right)$

$$\varepsilon = \frac{\langle\phi|\mathbf{H}|\phi\rangle}{\langle\phi|\phi\rangle} = \frac{4}{3}\left(\frac{\xi^2 - 3\xi}{8}\right)$$ skipped a lot of algebra

minimize ε: $\quad \frac{d\varepsilon}{d\xi} = 0 \quad \xi_{min} = 3/2 \rightarrow \varepsilon_{min} = -3/8\; au$

FAILURE! $\left[c.f.\text{ the true values: } E_{1s} = -1/2\; au, E_{2s} = -\frac{1}{8}au\right]$

[insufficiently flexible variational function.

exact result (usually not known)

Try something clever (but lazy):
What is the value of ξ that maximizes $\langle\phi|1s\rangle$?

for the best variational $\xi = 3/2, C_{1s} \equiv \left\langle \phi\left(\xi = 3/2\right) \middle| 1s \right\rangle = 0.9775$

but if instead we maximize C_{1s} *wrt.* $\xi : \xi = 5/3 \rightarrow C_{1s} = 0.9826$ Is this better?

No. The value of ε = –0.370 results from maximizing C_{1s}, which is a poorer bound to ε than obtained by minimizing ε.
$\xi = 3/2 \rightarrow \ \varepsilon \ = -0.375$

* need flexibility in ϕ

* can't improve on $\frac{d\varepsilon}{d\xi} = 0$ by employing an alternative variational strategy

This was stupid anyway because we would never use the variational method when we already know the answer!

---

4. Linear Variation → Secular Equation

KEY TOPIC for this lecture

$$\phi = \sum_{n=1}^{N} c_n \chi_n$$

distributed Gaussians

$$\left\langle \chi_n \middle| \mathbf{H} \middle| \chi_{n'} \right\rangle = H_{nn'}$$

$$\left\langle \chi_n \middle| \chi_{n'} \right\rangle = S_{nn'}$$

overlap integrals
(non-orthogonal basis sets are often convenient)
also not necessarily normalized

$$\varepsilon = \frac{\left\langle \phi \middle| \mathbf{H} \middle| \phi \right\rangle}{\left\langle \phi \middle| \phi \right\rangle} = \frac{\sum_{n,n'} c_n c_{n'} H_{nn'}}{\sum_{m,m'} c_m c_{m'} S_{mm'}}$$

Rearrange this equation.

To find the minimum value of ε,

$$\varepsilon \sum_{m,m'} c_m c_{m'} S_{mm'} = \sum_{n,n'} c_n c_{n'} H_{nn'}$$

take $\frac{\partial}{\partial c_j}$ for each j and require that

$$\frac{\partial \varepsilon}{\partial c_j} = 0 \text{ for each j}$$

linear variation!

We are seeking to minimize ε with respect to each c_j. Find the global minimum of the e(c_1,c_2,…,c_N) hypersurface.

The only terms in the sums that survive $\frac{\partial}{\partial c_j}$ are those that include c_j.

$$\varepsilon\sum_m c_m\left(S_{mj}+S_{jm}\right)=\sum_n c_n\left(H_{jn}+H_{nj}\right)$$

if $\{\chi_n\}$ are real $\quad S_{ij}=S_{ji}, H_{ij}=H_{ji}$

$$0=\sum_{n=1}^{N} c_n\left(H_{jn}-\varepsilon S_{jn}\right)$$

These are all of the surviving terms (i.e. those that include j). Each j term appears twice in both sums, once as a bra and once as a ket.

We get one such equation for each j (same set of unknown $\{c_n\}$).

N linear homogeneous equations in N unknown c_n's
Non trivial $\{c_n\}$ only if $|\mathbf{H}-\varepsilon\mathbf{S}|=0$
(Not the same simple form as $|\mathbf{H}-\mathbf{1}\mathrm{E}|=0$, but we can deal with this extra computational complexity.)

The result is N special values of ε that satisfy this equation.

CTDL show: all N ε-values are upper bounds to the lowest-N E_n's and all obtained $\{\phi_n\}$'s are othogonal! (orthogonal provided that they belong to different values of E_n)

How to solve $|\mathbf{H}-\varepsilon\mathbf{S}|=0$:
series of transformations on both **S** and **H**

1. Diagonalize **S**

$$\mathbf{U}^\dagger\mathbf{S}\mathbf{U}=\tilde{\mathbf{S}} \qquad \tilde{S}_{ij}=s_i\delta_{ij}$$

(orthogonalize $\{\chi\}$ basis)

2. Normalize $\tilde{\mathbf{S}}$

$$\underbrace{\left(\tilde{\mathbf{S}}\right)^{-1/2}\tilde{\mathbf{S}}\left(\tilde{\mathbf{S}}\right)^{-1/2}}_{\text{product of 3 diagonal matrices}}=\mathbf{1}=\tilde{\tilde{\mathbf{S}}}=\mathbf{T}^\dagger\mathbf{S}\mathbf{T} \qquad \text{where } \mathbf{T}=\mathbf{U}\tilde{\mathbf{S}}^{-1/2}$$

$$\left(\tilde{\mathbf{S}}^{-1/2}\right)^\dagger=\tilde{\mathbf{S}}^{-1/2}=\begin{pmatrix} s_1^{-1/2} & 0 & 0 \\ 0 & s_2^{-1/2} & 0 \\ 0 & 0 & \ddots \end{pmatrix}$$

unitary

This is not an orthogonal transformation of $\tilde{\mathbf{S}}$, but it does not destroy orthogonality because each eigenfunction of $\tilde{\mathbf{S}}$ is only being multiplied by a constant.

3. Transform **H** to orthonormalized basis set

$$\tilde{\tilde{\mathbf{H}}} = \underbrace{\tilde{\mathbf{S}}^{-1/2}\left(\mathbf{U}^{\dagger}\right)}_{\mathbf{T}^{\dagger}}\mathbf{H}\underbrace{\mathbf{U}\tilde{\mathbf{S}}^{-1/2}}_{\mathbf{T}}$$

U diagonalizes **S** not **H**

Obtain a new secular equation:

$$\left|\tilde{\tilde{\mathbf{H}}} - \varepsilon\tilde{\tilde{\mathbf{S}}}\right| = 0 \quad but \quad \tilde{\tilde{\mathbf{S}}} = \mathbf{1}$$

thus $\left|\tilde{\tilde{\mathbf{H}}} - \varepsilon\mathbf{1}\right| = 0$ by which $\tilde{\tilde{\mathbf{H}}}$ is diagonalized by the usual procedure

---

5. <u>Combine Linear and Nonlinear Variation</u>

typically done in *ab initio* electronic structure calculations

Basis set: $\chi_n(\xi_n r)$ ↓ linear variation in $\{\chi_n\}$, but where ξ_n is a radial scale factor, one for each χ_n

$\psi = \sum_n c_n \chi_n(\xi_n r)$ ↑ nonlinear variation

get $S_{nn'}(\xi_n, \xi_{n'}), H_{nn'}(\xi_n, \xi_{n'})$

0. pick arbitrary set of $\{\xi_i\}$
1. calculate all $H_{ij}(\xi_i, \xi_j)$ and $S_{ij}(\xi_i, \xi_j)$
2. *Solve* $|\mathbf{H}-\varepsilon\mathbf{S}| = 0$
   - *a*. $\mathbf{S} \rightarrow \tilde{\mathbf{S}}$ diagonalize **S** (orthogonalize)
   - b. $\left(\tilde{\mathbf{S}}\right)^{-1/2}$ (normalize)
   - *c*. $\mathbf{H} \rightarrow \tilde{\tilde{\mathbf{H}}}$
   - *d*. diagonalize $\tilde{\tilde{\mathbf{H}}}$

and now the nonlinear variation begins — find global minimum of $\varepsilon_{\text{lowest}}$ with respect to ξ_i.

3. change ξ_1 *from* $\xi_1^{(0)} \rightarrow \xi_1^{(1)} = \xi_1^{(0)} + \delta$
4. recalculate all integrals in **H** and **S** involving χ_1
5. *Solve* $|\mathbf{H}\text{-}\varepsilon\mathbf{S}| = 0$ to obtain a new set of $\{\varepsilon_i\}$.
   Pick lowest ε_i.
6. *calculate* $\dfrac{\partial \varepsilon_{lowest}}{\partial \xi_1} = \dfrac{\varepsilon_{lowest}^{old} - \varepsilon_{lowest}^{new}}{\xi_1^{(0)} - \xi_1^{(1)}}$

7. repeat #3 – 6 for each ξ_i (always looking only at lowest ε_i)
   This defines a gradient on a multidimensional lowest-$\varepsilon(\xi_1,\ldots\xi_N)$ surface. We seek the minimum of this hypersurface. Take a step in direction of steepest descent by an amount determined by $|\partial \varepsilon / \partial \xi_{steepest}|$ (small slope, small step; large slope, large step).

   This completes 1st iteration. All values of $\{\xi_i\}$are improved.

8. Return to #3, iterate #3-7 until convergence is obtained.

Nonlinear variations are much slower than linear variations.
Typically use ENORMOUS LINEAR $\{\chi\}$ basis set.
Contract this basis set by optimizing the nonlinear parameters (exponential scale factors) in a SMALL BASIS SET to match the lowest $\{\phi\}$'s that had initially been expressed in large basis set. Least squares fitting of wavefunctions.

* Large linear variation to get the linear combination of ϕ's that have the lowest energies;

* define a set of functions $\{\psi_i(\xi_i x)\}$ that contain non-linear scale parameters;

* perform a least squares fit of the $\{\xi\}$ to match the lowest few energy states from the linear variation;

* ortho-normalize the small set of $\psi_i(\xi_i x)$ functions and use them in a linear variation, thereby replacing the many-component functions from the massive linear variation by few-component functions (contracted basis set) from the hybrid linear variation.

6. Alternative Strategies

* rigorous variational minimization of E_{lowest}: "*ab initio*"
* constrain variational functions to be orthogonal to specific subset of variationally optimized functions

e.g. orthogonal to ground state – to get variational convergence. Applied to higher energy members of specific symmetry class
or orthogonal to core: frozen-core approximation. "Pseudopotentials" (use some observed energy levels to determine $Z^{eff}(r)$ of frozen core)

* least squares fitting to truncated **H** i.e. $\mathbf{H}^{eff}$

minimize differences between a set of measured energy levels (or other properties) and a set of computed variational eigen-energies (or other properties computed from variational wavefunctions).

$$\{\text{observed } E_n\} \leftrightarrow \{\text{parameters in } \mathbf{H}^{eff}\}$$

molecular constants
⇓
experimental Ψ's in finite variational basis set

* semi-empirical model

replace exact $\hat{\mathbf{H}}$ by a grossly simplified form and restrict basis set to a simple form too. Then adjust parameters in **H** to match some observed pattern of energy splittings. Confirm by using parameters to predict unobserved properties. Use values of fit parameters to build insight.

Never-ending battle between accuracy and insight!

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Density Matrices I

(See CTDL pp. 252-263, 295-307**, 153-163, 199-202, 290-294)

Last time: Variational Method

Linear variation: $0 = \left|\mathbf{H} - \varepsilon\mathbf{S}\right| \Rightarrow 0 = \left|\tilde{\tilde{H}} - \varepsilon\mathbf{1}\right|$

$$\psi = \sum_n c_n \phi_n \quad \frac{d\varepsilon}{dc_n} = 0$$

[Variational method vs. perturbation theory]

TODAY

ψ phase ambiguity – but for every observable each state always appears as a bra and a ket.

What is needed to encode motion in the probability density? A superposition of eigenstates belonging to several different values of E.

Coherent superposition vs. statistical mixture: think about polarized light.

ρ no phase ambiguity in density matrix, $|\psi\rangle\langle\psi|$, an N × N matrix
* "**coherences**" in off-diagonal position
* "**populations**" along diagonal

$$\langle\mathbf{A}\rangle = \mathrm{Tr}(\rho\mathbf{A}) = \mathrm{Tr}(\mathbf{A}\rho)$$

Quantum Beats

prepared state → ρ

detection → **D** (detect or destroy coherences)

$\rho(t)$

$\langle\mathbf{A}\rangle_t$ equations of motion

$$\frac{d}{dt}\langle\mathbf{A}\rangle = \frac{i}{\hbar}\langle[\mathbf{H},\mathbf{A}]\rangle + \left\langle\frac{\partial\mathbf{A}}{\partial t}\right\rangle$$

expectation value

each element of ρ encodes important information

$$i\hbar\frac{d\rho}{dt} = [\mathbf{H}(t),\rho]$$

* state: ρ
* evolution: **H**
* detection: **D**

modified 8/13/20 1:33 PM

Let us define a quantity called "Density Matrix"

$$\rho \equiv |\psi\rangle\langle\psi|$$

ψ can be any sort of QM wavefunction
* eigenstate of **H**
* coherent superposition of several eigenstates of **H**

but ψ <u>cannot</u> represent a <u>statistical</u> (i.e. incoherent) mixture of several different ψ's

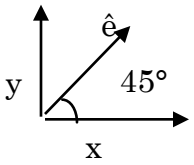
However, ρ <u>can</u> represent a statistical (i.e. equilibrium) mixture of states!

$$\rho \equiv \sum_k p_k |\psi_k\rangle\langle\psi_k| = \sum_k p_k \rho_k$$

$\sum p_k = 1$ ← probability of each contributing term to ρ

<u>Example</u>

* *one* beam of linearly polarized light, with its polarization axis at 45° (ε -field)



$$\hat{e} = 2^{-1/2}\left(\hat{e}_x + \hat{e}_y\right)$$

* *two* superimposed beams of linearly polarized light, 50% along $\hat{e}_x$, 50% along $\hat{e}_y$. Call this a statistical mixture state.

These 2 cases seem to be identical if you make 2 measurements with analyzer polarizers along $\hat{e}_x$ then $\hat{e}_y$. But the 2 cases are different with respect to 2 measurements with analyzer polarizers along $2^{-1/2}\left(\hat{e}_x + \hat{e}_y\right)$ and then along $2^{-1/2}\left(\hat{e}_x - \hat{e}_y\right)$.

In the statistical mixture, it does not matter how the analyzer is oriented.

What are the properties of ρ?

1. ρ for a pure state is Hermitian with positive elements along the diagonal and other elements off-diagonal. $\rho \equiv |\psi\rangle\langle\psi|$ so evaluate matrix elements of ρ

$$\rho_{nm} = \langle n|\psi\rangle\langle\psi|m\rangle$$

c_n ← $|\psi\rangle$ → c_m^*

$$\rho_{nm} = c_n c_m^*$$

$|\psi\rangle = \sum c_n |n\rangle$ can expand $|\psi\rangle$ in any basis set, but **H** eigenbasis is most useful.

$$\text{but } \left(\rho^\dagger\right)_{nm} = \rho^*_{mn} = \left[\langle m|\psi\rangle\langle\psi|n\rangle\right]^*$$

$$= \langle\psi|m\rangle\langle n|\psi\rangle = \langle n|\psi\rangle\langle\psi|m\rangle = \rho_{nm}$$

$\therefore \rho^\dagger = \rho$

$\therefore$ ρ passes the Hermiticity test that all observable quantities must pass!
So if ρ is observable, what does it tell us?

$$\rho_{nn} = \langle n|\psi\rangle\langle\psi|n\rangle = c_n c_n^* = |c_n|^2 \geq 0$$

positive along diagonal

---

2. 2 × 2 Example Coherent Superposition vs. Statistical Mixture

$$|\psi\rangle = 2^{-1/2}\begin{pmatrix} 1 \\ \pm 1 \end{pmatrix}$$ a coherent superposition state

$$\rho_{cs} = \frac{1}{2}\begin{pmatrix} 1 \\ \pm 1 \end{pmatrix}\begin{pmatrix} 1 & \pm 1 \end{pmatrix} = \frac{1}{2}\begin{pmatrix} 1 & \pm 1 \\ \pm 1 & 1 \end{pmatrix}$$

Trace ρ = 1

$$\rho^2 = \frac{1}{4}\begin{pmatrix} 2 & \pm 2 \\ \pm 2 & 2 \end{pmatrix} = \frac{1}{2}\begin{pmatrix} 1 & \pm 1 \\ \pm 1 & 1 \end{pmatrix}$$

$$\rho^2 = \rho$$

Now consider a statistical mixture state.

$$\rho_{sm} = \frac{1}{2}\begin{pmatrix}1\\0\end{pmatrix}(1\ 0) + \frac{1}{2}\begin{pmatrix}0\\1\end{pmatrix}(0\ 1)$$

$$= \frac{1}{2}\begin{pmatrix}1 & 0\\0 & 1\end{pmatrix} \qquad \text{trace } \rho = 1$$

The difference is in the off-diagonal positions of ρ

diagonal elements → "populations"
off-diagonal elements → "coherences"

(statistical mixture states have strictly diagonal ρ)

Expectation values of $\hat{\mathbf{A}}$ in terms of ρ

completeness

$$\langle \mathbf{A} \rangle = \langle \psi | \mathbf{A} | \psi \rangle = \sum_{j,k} \langle \psi | k \rangle \langle k | \mathbf{A} | j \rangle \langle j | \psi \rangle$$

these are three simple numbers and can be rearranged in any order.

$$= \sum_{j,k} \langle j | \psi \rangle \langle \psi | k \rangle \mathbf{A}_{kj}$$

$$\rho_{jk}$$

$$= \sum_{j} (\rho \mathbf{A})_{jj} \equiv Trace(\rho \mathbf{A})$$

a fantastic labor saving and insight generating result!

$\langle \mathbf{A} \rangle = $ Trace ($\boldsymbol{\rho}\mathbf{A}$)!

[Could have arranged the factors $\sum_{j,k} \mathbf{A}_{kj} \langle j | \psi \rangle \langle \psi | k \rangle = \sum_{k} (\mathbf{A}\rho)_{kk} = \text{Trace}(\mathbf{A}\rho)$]

$$\langle \mathbf{A} \rangle = \text{Trace}(\mathbf{A}\rho) = \text{Trace}(\rho \mathbf{A})$$

So ρ describes the state of system, **A** describes a measurement to be made on the system

simple prescription for calculating ⟨**A**⟩

The separation between *initial preparation, evolution*, and *measurement* of a specific observable becomes very convenient and instructive.

Example: Quantum Beats

Preparation, evolution, detection

magically prepare some coherent superposition state $\Psi(t)$

$$\Psi(t) = N\sum_n \underbrace{a_n \psi_n e^{-iE_n t/\hbar}}_{\text{Several eigenstates of H. Evolve freely without any time-dependent intervention}}$$

$$\boxed{N = \left[\sum_n |a_n|^2\right]^{-1/2}}$$

normalization

$$\rho(t) = |\Psi(t)\rangle\langle\Psi(t)|$$

Case (1): Detection: only one of the eigenstates, ψ_1, in the superposition is capable of giving fluorescence that our detector can "see". (Build a detector matrix out of the same form as the selected bright state in **ρ**.)

Thus $\mathbf{D} = |\psi_1\rangle\langle\psi_1| = \begin{pmatrix} 1 & 0 & \cdots \\ 0 & 0 & 0 \\ \vdots & 0 & 0 \end{pmatrix}$ a projection operator

(designed to project out only the $|\psi_1\rangle$ part of the state vector or the ρ_{11} part of **ρ**.

$$\rho = N^2 \begin{pmatrix} |a_1|^2 & a_1 a_2^* e^{-i(E_1-E_2)t/\hbar} & \cdots \\ & |a_2|^2 & \\ & & |a_3|^2 \\ & & \ddots \end{pmatrix}$$

This particular **D** picks out only 1st row of **ρ**.

$$\rho_{12} = \langle 1|\Psi\rangle\langle\Psi|2\rangle$$

$$\rho_{12} = N^2 a_1 e^{-iE_1 t/\hbar} a_2^* e^{+iE_2 t/\hbar}$$

$$\langle \mathbf{D} \rangle_t = \text{Trace}(\mathbf{D}\rho) = N^2 \,\text{Trace} \begin{pmatrix} |a_1|^2 & a_1 a_2^* e^{-i\omega_{12} t} & \text{stuff} & \cdots \\ 0 & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots \end{pmatrix}$$

$$= N^2 |a_1|^2$$

no time dependence!

You do not need to work out the full **Dρ** matrix product!

case (2): a particular linear combination of eigenstates is bright: the initial (i.e. at t = 0) state $2^{-1/2}(\psi_1 + \psi_2)$ has $\langle \mathbf{D} \rangle = 1$.

$$\mathbf{D} = \frac{1}{2}\left(|\psi_1\rangle + |\psi_2\rangle\right)\left(\langle\psi_1| + \langle\psi_2|\right)$$

$$= \frac{1}{2}\left[|\psi_1\rangle\langle\psi_1| + |\psi_2\rangle\langle\psi_2| + |\psi_1\rangle\langle\psi_2| + |\psi_2\rangle\langle\psi_1|\right]$$

$$= \frac{1}{2}\left[\begin{pmatrix} 1 & 0 & 0 & \cdots \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ \vdots & 0 & 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 & 0 & \cdots \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ \vdots & 0 & 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 1 & 0 & \cdots \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ \vdots & 0 & 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 & 0 & \cdots \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ \vdots & 0 & 0 & 0 \end{pmatrix}\right]$$

$$\mathbf{D} = \frac{1}{2}\begin{pmatrix} 1 & 1 & 0 & \cdots \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ \vdots & 0 & 0 & 0 \end{pmatrix}$$

a projection operator. How much of the original state is present in the evolved state?

[if the bright state had been $2^{-1/2}(\psi_1 - \psi_2)$, then $\mathbf{D} = \frac{1}{2}\begin{pmatrix} 1 & -1 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$]

$$\text{Trace}(\mathbf{D}\boldsymbol{\rho}) = \frac{1}{2}N^2 \quad \text{Trace}\left(\quad\right)$$

why do we need to look at only the 1,2 block of $\boldsymbol{\rho}$

$$\left(\mathbf{D}\boldsymbol{\rho}\right)_{11} = \frac{1}{2}N^2\left[|a_1|^2 + a_1^* a_2 e^{+i(E_1 - E_2)t/\hbar}\right]$$

$$\left(\mathbf{D}\boldsymbol{\rho}\right)_{22} = \frac{1}{2}N^2\left[|a_2|^2 + a_1 a_2^* e^{-i(E_1 - E_2)t/\hbar}\right]$$

The 1,2 block is the only part of $\mathbf{D}$ that picks out something that can appear along the diagonal of $\mathbf{D}\boldsymbol{\rho}$

$$\text{Trace}\left(\mathbf{D}\boldsymbol{\rho}\right) = \frac{1}{2}N^2\left[|a_1|^2 + |a_2|^2 + 2\,\text{Re}\left[a_1^* a_2 e^{+i\omega_{12}t}\right]\right]$$

beat note at ω_{12}

[if the bright state had been $2^{-1/2}(\psi_1 - \psi_2)$, then Tr($\mathbf{D}\,\boldsymbol{\rho}$) would be the same except for –2Re[ ]]

If $|a_1|^2 = |a_2|^2$ (and a_1, a_2 real), $\text{Trace}\left(\mathbf{D}\boldsymbol{\rho}\right) = N^2|a_1|^2\left[1 \pm \cos\omega_{12}t\right]$ ($N^2 = 1/2$)

QUANTUM BEAT! 100% modulation! Either $2N^2|a_1|^2$ at t = 0 or 0 at t = 0.

So we see that the same $\Psi(x,t)$ or $\boldsymbol{\rho}(t)$ can look simple or complicated depending on the nature of the measurement operator! The measurement operator is designed to be sensitive (can detect or "destroy" a particular coherence) only to specific coherences (i.e. locations in $\boldsymbol{\rho}$) which oscillate at ω_{ij}. THIS IS THE REASON WHY WE CAN SEPARATE PREPARATION AND OBSERVATION SO CLEANLY.

Time evolution of ρ_{nm} and $\langle \mathbf{A} \rangle$

Start with the time-dependent Schrödinger equation:

$$\mathbf{H}\Psi = i\hbar\frac{\partial\Psi}{\partial t} \quad \begin{cases} \mathbf{H}|\Psi\rangle = i\hbar\dfrac{\partial}{\partial t}|\Psi\rangle \\ \langle\Psi|\mathbf{H} = -i\hbar\dfrac{\partial}{\partial t}\langle\Psi| \end{cases} \qquad \text{Hermitian conjugate.}$$

for time-independent $\mathbf{H}$ we know $\Psi(t) = \sum_n a_n \psi_n e^{-iE_n t/\hbar}$

1. $\boldsymbol{\rho}(t)$

$$\rho(t) = |\Psi(t)\rangle\langle\Psi(t)|$$

$$\rho_{nn}(t) = \langle n|\Psi(t)\rangle\langle\Psi(t)|n\rangle = |a_n|^2$$ a time independent "**population**" in state n.

$$\rho_{nm}(t) = a_n a_m^* e^{-i(E_n - E_m)t/\hbar} = a_n a_m^* e^{-i\omega_{nm}t}$$ a "**coherence**" which oscillates at ω_{nm} (eigenstate energy differences /$\hbar$)

2. $\langle \mathbf{A} \rangle_t$

Recall $i\hbar\dfrac{\partial\Psi}{\partial t} = \mathbf{H}\Psi$

$$\frac{\partial}{\partial t}\langle\mathbf{A}\rangle = \left[\frac{\partial}{\partial t}\langle\Psi|\right]\mathbf{A}|\Psi\rangle + \left\langle\Psi\left|\frac{\partial\mathbf{A}}{\partial t}\right|\Psi\right\rangle + \langle\Psi|\mathbf{A}\left[\frac{\partial}{\partial t}|\Psi\rangle\right]$$

$$= \left[\frac{-1}{i\hbar}\langle\Psi|\mathbf{H}\right]\mathbf{A}|\Psi\rangle + \left\langle\frac{\partial\mathbf{A}}{\partial t}\right\rangle + \langle\Psi|\mathbf{A}\left[\frac{1}{i\hbar}\mathbf{H}|\Psi\rangle\right]$$

$$= \frac{i}{\hbar}\langle[\mathbf{H},\mathbf{A}]\rangle + \left\langle\frac{\partial\mathbf{A}}{\partial t}\right\rangle$$

Heisenberg Equation of Motion

Note that nothing has been assumed here about the time-dependence of $\mathbf{H}$. This is a simple prescription for calculating the motion of $\langle A \rangle$. One observable quantity.

If **A** commutes with **H** (regardless of whether **H** is time-dependent), there is no dynamics as far as observable **A** is concerned. However, if **A** does not commute with **H**, there can be dynamics of $\langle \mathbf{A} \rangle$ even if both **A** and **H** are time-independent.

Similarly, can derive $i\hbar \frac{\partial \rho}{\partial t} = [\mathbf{H}(t), \rho]$, which describes evolution of ρ under **H**(t).

If **H** is time dependent

This is a matrix equation. It specifies the time dependence of each element of ρ. Usually has the form of many coupled first-order differential equations.

Summarize

$$\langle \mathbf{A} \rangle = \mathrm{Tr}(\rho \mathbf{A}) = \mathrm{Tr}(\mathbf{A} \rho)$$

info about quantity being measured

info about state on which measurement is to be made

$$i\hbar \frac{\partial \rho}{\partial t} = [\mathbf{H}, \rho]$$

time evolution

state

initial state : ρ

time evolution of ρ: **H**

observable quantity : **A**

each expressed independently in the form of matrices which can be easily read (or designed!).

NMR pulse gymnastics

---

statistical mixture states - use the same machinery BUT add the independent ρ_k matrices with weights p_k that correspond to their fractional populations [populations have no phase].

ρ is Hermitian so it can be diagonalized by $\mathbf{T}^\dagger \rho \mathbf{T} = \tilde{\rho}$. However, if ρ is time-dependent, **T** would have to be time-dependent. This transformation gives a representation without any coherences in $\tilde{\rho}$, even if we started with a coherent superposition state. No problem, because this transformation will undiagonalize **H**, thereby reintroducing time dependences.

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5.73 Quantum Mechanics I
Fall 2018

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Density Matrices II

Read CTDL, pages 643-652.

Last time: ψ, $|\ \rangle$, $\rho = |\ \rangle\langle\ |$
* coherent superposition vs. statistical mixture
* $\boldsymbol{\rho}$ can have non-zero off-diagonal elements if it is a statistical mixture, that includes one coherent superposition state.
* populations along diagonal, coherences off-diagonal
$\langle \mathbf{A} \rangle = \text{Trace}(\boldsymbol{\rho}\,\mathbf{A}) = \text{Trace}(\mathbf{A}\,\boldsymbol{\rho})$

Today: Quantum Beats
prepared state $\boldsymbol{\rho}$
detection as projection operator **D**

What part of **D** samples a specific off-diagonal element of $\boldsymbol{\rho}$?
Optimize the magnitude of quantum beats

"[partial traces]"

* system consisting of 2 parts — e.g. coupled oscillators
* motion in state-space vs. motion in coordinate space.
* "entanglement"

The material on pages 20–2, –3, –5, and –7 is an exact duplication of pages 19–5, –6, –7 and –8.

revised August 13, 2020

Example: Quantum Beats

Preparation, evolution, detection

magically prepare some coherent superposition state $\Psi(t)$

$$\Psi(t) = N\sum_n \underbrace{a_n \psi_n e^{-iE_n t/\hbar}}_{\text{Several eigenstates of } \mathbf{H}. \text{ Evolve freely without any time-dependent intervention}}$$

$$\boxed{N = \left[\sum_n |a_n|^2\right]^{-1/2}}$$

normalization

$$\rho(t) = |\Psi(t)\rangle\langle\Psi(t)|$$

Case (1): Detection: only one of the eigenstates, ψ_1, in the superposition is capable of giving fluorescence that our detector can "see".

Thus $\mathbf{D} = |\psi_1\rangle\langle\psi_1| = \begin{pmatrix} 1 & 0 & \cdots \\ 0 & 0 & 0 \\ \vdots & 0 & 0 \end{pmatrix}$

a projection operator
(designed to project out only $|\psi_1\rangle$ part of state vector or ρ_{11} part of $\boldsymbol{\rho}$.

$$\rho = N^2 \begin{pmatrix} |a_1|^2 & a_1 a_2^* e^{-i(E_1 - E_2)t/\hbar} & \cdots \\ & |a_2|^2 & \\ & & |a_3|^2 \\ & & \ddots \end{pmatrix}$$

$$\rho_{12} = \langle 1|\Psi\rangle\langle\Psi|2\rangle$$

$$\rho_{12} = N^2 a_1 e^{-iE_1 t/\hbar} a_2^* e^{+iE_2 t/\hbar}$$

D picks out only 1st row of $\boldsymbol{\rho}$.

$$\langle \mathbf{D} \rangle_t = \text{Trace}(\mathbf{D}\rho) = N^2 \,\text{Trace}\begin{pmatrix} |a_1|^2 & a_1 a_2^* e^{-i\omega_{12} t} & \text{stuff} & \cdots \\ 0 & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots \end{pmatrix}$$

$$= N^2 |a_1|^2$$

no time dependence!

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case (2): a particular linear combination of eigenstates is bright: the initial state $2^{-1/2}(\psi_1 + \psi_2)$ has $\langle \mathbf{D} \rangle = 1$.

$$\mathbf{D} = \frac{1}{2}\left(|\psi_1\rangle + |\psi_2\rangle\right)\left(\langle\psi_1| + \langle\psi_2|\right)$$

$$= \frac{1}{2}\left[|\psi_1\rangle\langle\psi_1| + |\psi_2\rangle\langle\psi_2| + |\psi_1\rangle\langle\psi_2| + |\psi_2\rangle\langle\psi_1|\right]$$

$$= \frac{1}{2}\left[\begin{pmatrix} 1 & 0 & 0 & \cdots \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ \vdots & 0 & 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 & 0 & \cdots \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ \vdots & 0 & 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 1 & 0 & \cdots \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ \vdots & 0 & 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 & 0 & \cdots \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ \vdots & 0 & 0 & 0 \end{pmatrix}\right]$$

a projection operator. How much of the original state is present in the evolved state?

$$\mathbf{D} = \frac{1}{2}\begin{pmatrix} 1 & 1 & 0 & \cdots \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ \vdots & 0 & 0 & 0 \end{pmatrix}$$

[if the bright state had been $2^{-1/2}(\psi_1 - \psi_2)$, then $\mathbf{D} = \frac{1}{2}\begin{pmatrix} 1 & -1 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$]

$$\text{Trace}(\mathbf{D}\rho) = \frac{1}{2}N^2 \quad \text{Trace}\left(\quad\right)$$

why do we need to look at only the 1,2 block of $\boldsymbol{\rho}$?

$$\left(\mathbf{D}\rho\right)_{11} = \frac{1}{2}N^2\left[|a_1|^2 + a_1^* a_2 e^{+i(E_1 - E_2)t/\hbar}\right]$$

$$\left(\mathbf{D}\rho\right)_{22} = \frac{1}{2}N^2\left[|a_2|^2 + a_1 a_2^* e^{-i(E_1 - E_2)t/\hbar}\right]$$

$$Trace\left(\mathbf{D}\rho\right) = \frac{1}{2}N^2\left[|a_1|^2 + |a_2|^2 + 2\text{Re}\left[a_1^* a_2 e^{+i\omega_{12}t}\right]\right]$$

beat note at ω_{12}

[if the bright state had been $2^{-1/2}(\psi_1 - \psi_2)$, then Tr($\mathbf{D}\,\rho$) would be the same except for –2Re[ ]]

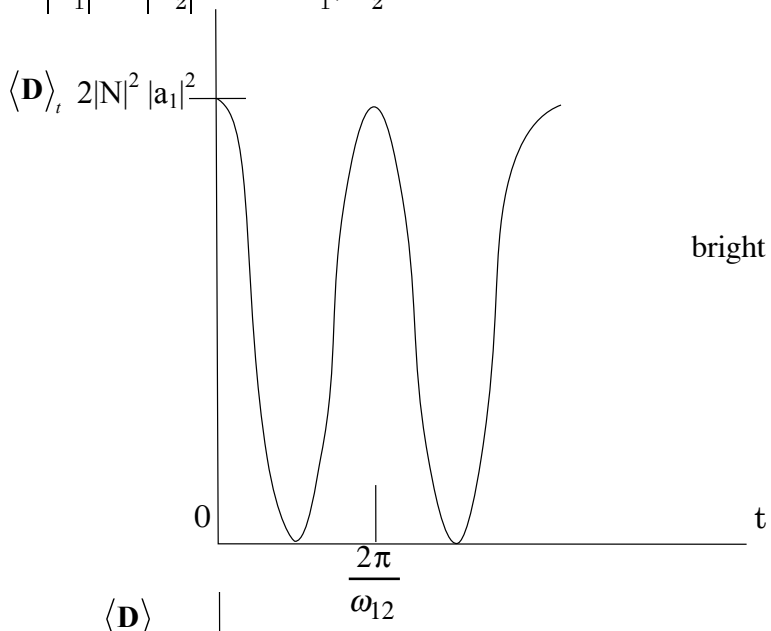
If $|a_1|^2 = |a_2|^2$ (and a_1, a_2 real), $\text{Trace}(\mathbf{D}\rho) = N^2|a_1|^2\left[1 \pm \cos\omega_{12}t\right]$ ($N^2 = 1/2$)

QUANTUM BEAT! 100% modulation!

Either $2N^2|a_1|^2$ at t = 0 (+ sign) or 0 at t = 0 (– sign)

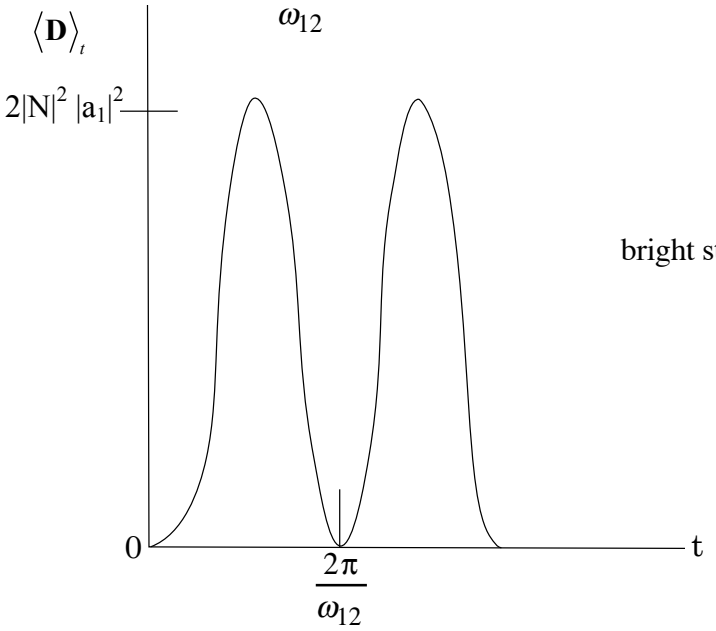
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if $|a_1|^2 = |a_2|^2$ and a_1, a_2 *real*



bright state $2^{-1/2}(\psi_1 + \psi_2)$

$$\mathbf{D} = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$



bright state $2^{-1/2}(\psi_1 - \psi_2)$

$$\mathbf{D} = \begin{pmatrix} 1 & -1 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Usually $|a_1|_t^2 = e^{-t/\tau}$ — exponential decay: we have sinusoidal beats superimposed on exponential decay

---

what happens if $|a_1|^2 \neq |a_2|^2$?

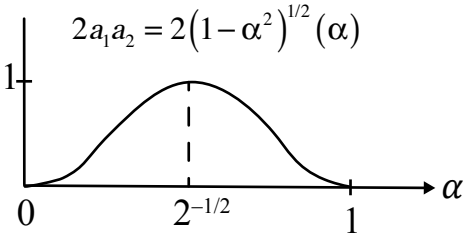
try $1-\alpha^2, \alpha^2$ as mixing fractions

$|a_1|^2 + |a_2|^2 = 1$

$2a_1a_2 = 2(1-\alpha^2)^{1/2}(\alpha)$

Terms in Trace (**Dρ**)

$2a_1a_2$



maximum beat amplitude

occurs when $|a_1|^2 = |a_2|^2$

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So we see that the same $\Psi(x,t)$ or $\boldsymbol{\rho}(t)$ can look simple or complicated depending on the nature of the measurement operator! The measurement operator is designed to be sensitive only to specific coherences (i.e. locations in $\boldsymbol{\rho}$) which oscillate at ω_{ij}. THIS IS THE REASON WHY WE CAN SEPARATE PREPARATION AND OBSERVATION SO CLEANLY.

Time evolution of ρ_{nm} and $\langle \mathbf{A} \rangle$

Start with the time-dependent Schrödinger equation:

$$\mathbf{H}\Psi = i\hbar\frac{\partial\Psi}{\partial t} \quad \begin{cases} \mathbf{H}\left|\Psi\right\rangle = i\hbar\dfrac{\partial}{\partial t}\left|\Psi\right\rangle \\ \left\langle\mathbf{H}\right| H = -i\hbar\dfrac{\partial}{\partial t}\left\langle\Psi\right| \end{cases}$$

for time-independent $\mathbf{H}$ we know $\Psi\left(t\right) = \sum_n a_n \psi_n e^{-iE_n t/\hbar}$

1. $\boldsymbol{\rho}(t)$

$$\boldsymbol{\rho}(t) = \left|\Psi(t)\right\rangle\left\langle\Psi(t)\right|$$

$$\rho_{nn}(t) = \left\langle n\middle|\Psi(t)\right\rangle\left\langle\Psi(t)\middle|n\right\rangle = \left|a_n\right|^2$$

a time independent "**population**" in state n.

$$\rho_{nm}(t) = a_n a_m^* e^{-i\left(E_n - E_m\right)t/\hbar} = a_n a_m^* e^{-i\omega_{nm}t}$$

a "**coherence**" which oscillates at ω_{nm} (eigenstate energy differences /ħ)

2. $\langle \mathbf{A} \rangle_t$

Recall $i\hbar\dfrac{\partial\Psi}{\partial t} = \mathbf{H}\Psi$

$$\frac{\partial}{\partial t}\left\langle\mathbf{A}\right\rangle = \left[\frac{\partial}{\partial t}\left\langle\Psi\right|\right]\mathbf{A}\left|\Psi\right\rangle + \left\langle\Psi\middle|\frac{\partial\mathbf{A}}{\partial t}\middle|\Psi\right\rangle + \left\langle\Psi\middle|\mathbf{A}\left[\frac{\partial}{\partial t}\middle|\Psi\right\rangle\right]$$

$$= \frac{-1}{i\hbar}\left[\left\langle\Psi\middle|\mathbf{H}\right]\mathbf{A}\left|\Psi\right\rangle + \left\langle\frac{\partial\mathbf{A}}{\partial t}\right\rangle + \left\langle\Psi\middle|\mathbf{A}\left[\frac{1}{i\hbar}\mathbf{H}\middle|\Psi\right\rangle\right]$$

$$= \frac{i}{\hbar}\left\langle\left[\mathbf{H},\mathbf{A}\right]\right\rangle + \left\langle\frac{\partial\mathbf{A}}{\partial t}\right\rangle$$

Heisenberg Equation of Motion

This is a scalar equation, not a matrix equation. It tells us about the motion of the "center" of a wavepacket. Note that nothing has been assumed about the time-dependence of **H**. Motion of **A**. Example of one observable quantity.

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Nonlecture

$$\frac{\partial \boldsymbol{\rho}}{\partial t} = \frac{\partial}{\partial t}\left[|\Psi\rangle\langle\Psi|\right] = \left[\frac{\partial}{\partial t}|\Psi\rangle\right]\langle\Psi| + |\Psi\rangle\left[\frac{\partial}{\partial t}\langle\Psi|\right]$$

$$= \left[\frac{1}{i\hbar}\mathbf{H}|\Psi\rangle\right]\langle\Psi| + |\Psi\rangle\left[\frac{-1}{i\hbar}\langle\Psi|\mathbf{H}\right]$$

$$= \frac{1}{i\hbar}\left[\mathbf{H}\boldsymbol{\rho} - \boldsymbol{\rho}\mathbf{H}\right]$$

$$i\hbar\frac{\partial \boldsymbol{\rho}}{\partial t} = \left[\mathbf{H}, \boldsymbol{\rho}\right]$$

no requirement that **H** be independent of t.

But if **H** is independent of t, then take matrix elements of both sides of equation.

$$i\hbar\dot{\rho}_{jk} = \left\langle j\left|\mathbf{H}\boldsymbol{\rho} - \boldsymbol{\rho}\mathbf{H}\right|k\right\rangle$$

$$= E_j\rho_{jk} - \rho_{jk}E_k = \left(E_j - E_k\right)\rho_{jk}$$

$$\dot{\rho}_{jk} = -\frac{i}{\hbar}\left(E_j - E_k\right)\rho_{jk}$$

You already knew this, but not so elegantly.

$$\rho_{jk}(t) = e^{-\frac{i}{\hbar}\left(E_j - E_k\right)t}\rho_{jk}(0)$$

Time evolution of all coherences in the absence of external manipulation!

External manipulation can cause coupling between differential equations.

If **A** commutes with **H** (regardless of whether **H** is time-dependent), there is no dynamics as far as observable **A** is concerned. However, if **A** does not commute with **H**, there can be dynamics of $\langle \mathbf{A} \rangle$ even if both **A** and **H** are time-independent.

Similarly can derive $i\hbar \frac{\partial \rho}{\partial t} = [\mathbf{H}(t), \rho]$ evolution of ρ under **H**(t).

If **H** is time dependent

This is a matrix equation. It specifies the time dependence of each element of ρ.

Often we have coupled differential equations where ρ_{ij} is related to ρ_{ii}, ρ_{jj} and perhaps other things too.

Summarize

$$\langle \mathbf{A} \rangle = \mathrm{Tr}(\rho \mathbf{A}) = \mathrm{Tr}(\mathbf{A} \rho)$$

info about quantity being measured

info about state on which measurement is to be made

$$i\hbar \frac{\partial \rho}{\partial t} = [\mathbf{H}, \rho]$$

time evolution

state

initial state : ρ

time evolution of ρ: **H**

observable quantity : **A**

each expressed independently in the form of matrices which can be easily read (or designed!).

NMR pulse gymnastics

---

statistical mixture states – use the same machinery BUT add the independent ρ_k matrices with weights p_k that correspond to their fractional populations. [Populations have no phase.]

ρ is Hermitian so ρ can be diagonalized by $\mathbf{T}^\dagger \rho \mathbf{T} = \tilde{\rho}$. However, if ρ is time-dependent, **T** would have to be time-dependent. This transformation gives a representation without any coherences in $\tilde{\rho}$ even if we started with a coherent superposition state. No problem because this transformation will undiagonalize **H**, thereby reintroducing time dependences.

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Systems consisting of 2 parts: method of **partial traces**

e.g. coupled harmonic oscillators (recall anharmonically coupled oscillators, $k_{122}q_1q_2^2$, $\psi(q_1,q_2)=\psi_{v_1}(q_1)\psi_{v_2}(q_2)$)

direct product representation

$$\psi(x_1,x_2)=\psi_{1,n_1}(x_1)\psi_{2,n_2}(x_2) \qquad |n_1,n_2\rangle$$

$$\rho=\rho^{(1)}\otimes\rho^{(2)}$$

ρ has 4 indices

$$\rho_{n_1n_2;n_1'n_2'}=\langle n_1|\langle n_2|\psi\rangle\langle\psi|n_2'\rangle|n_1'\rangle$$

It is still a square matrix with $[(n_{1max}+1)(n_{2max}+1)]^2$ elements.

We might want to measure the expectation value of an operator that operates on both systems 1 and 2: **A**(1,2)

$$\langle\mathbf{A}\rangle=\text{Trace}(\rho\mathbf{A})$$

$$=\sum_{n_1,n_2}(\rho\mathbf{A})_{n_1n_2;n_1n_2}$$

Alternatively, we might want to measure the expectation value of an operator that operates only on system 1: call it **B**(1).
To use the Trace(ρ **B**) method, need the concept of **partial traces** and need to formally extend **B** so that it acts as a dummy operator on system 2.

$$\tilde{B}(1)=B(1)\otimes\mathbf{1}(2)$$

(**1** is diagonal with respect to n_2)

Several types of initial preparation are possible:

1. pure state of 1 ⊗2 (a "tensor product" state)
2. statistical mixture in 1, pure state in 2.
3. statistical mixture in both.

Entanglement! Handout from 10/11/02. *Science* **298**, p369 (2002).

Several types of observation are possible:

1. separate observation of subsystem 1 or 2
2. simultaneous measurement of both systems

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$$\mathbf{H} = \begin{pmatrix} 0 & 5 & 2 \\ 5 & 2 & 0.5 \\ 2 & 0.5 & 20 \end{pmatrix}$$

This **H** has a 2×2 quasi-degenerate block and both members of this block interact weakly with a non-quasi-degenerate remote state.

$$\mathbf{H}^{(0)} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 20 \end{pmatrix}$$

$$\mathbf{H}^{(1)} = \begin{pmatrix} 0 & 5 & 2 \\ 5 & 0 & 0.5 \\ 2 & 0.5 & 0 \end{pmatrix}$$

$$\mathbf{H}^{(2)} = \begin{pmatrix} \dfrac{2^2}{-20} & \dfrac{(2)(0.5)}{\dfrac{0+2}{2} - 20} & 0 \\ -\dfrac{1}{19} & \dfrac{0.5^2}{-18} & 0 \\ 0 & 0 & \left(\dfrac{2^2}{20} + \dfrac{0.5^2}{18}\right) \end{pmatrix}$$

Van Vleck Transformation

$$H^{(2)}_{nn'} \underset{\text{in-block}}{=} \sum_{\substack{k \\ \text{out-of-block}}} \frac{H^{(1)}_{nk} H^{(1)}_{kn'}}{\dfrac{E^{(0)}_n + E^{(0)}_{n'}}{2} - E^{(0)}_k}$$

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CTDL use this definition of $\tilde{\mathbf{B}}(1)$ (page 306) to prove that

$\left\langle \tilde{\mathbf{B}}(1) \right\rangle = \mathrm{Tr}(\boldsymbol{\rho}(1)\mathbf{B}(1))$ calculated as if system 1 were isolated from system 2

---

for coupled H–O system

operator of type (1,2) $\mathbf{a}_1^\dagger \mathbf{a}_1 \mathbf{a}_2^\dagger \mathbf{a}_2$ (a correlated property of two parts of the system)

type (1) $\mathbf{a}_1^\dagger \mathbf{a}_1$

or type (2) $\mathbf{a}_2^\dagger \mathbf{a}_2$

or type (1 + 2) $\left(\mathbf{a}_1^\dagger \mathbf{a}_1 + \mathbf{a}_2^\dagger \mathbf{a}_2\right)$

See *Chem. Phys. Lett.* **320**, 553 (2000).

Suppose $t = 0$ wavepacket is located at turning point of $v_2 = 5$ in oscillator #2 and at $x_1 = 0$ for oscillator #1

$$\Psi(x_1, x_2, t=0) = \sum_{n_2=0}^{\infty} a_{n_2} \left|0, n_2\right\rangle^{(0)}$$

Discuss inital preparation that gives dynamics within a polyad and between polyads. Diagnostics in state and in configuration space.

suppose we have $\omega_1 = 2\omega_2$ $\quad P = 2n_1 + n_2$ polyads.

and only the $\left|0, P\right\rangle^{(0)}$ state is "bright" (i.e. excitation is initially in oscillator #2)

eigenvectors of $\mathbf{H}(1,2)$ expressed in H–O Basis set

need to write $\left|0, P\right\rangle^{(0)}$ as $\sum_{n=0}^{P/2} b_n \left|n, P-2n\right\rangle$

Basis state — Eigen state

$\mathbf{T}^\dagger \mathbf{H}^P \mathbf{T}$

columns of $\mathbf{T}^\dagger$

($\mathbf{H}^P$ is polyad Hamiltonian)

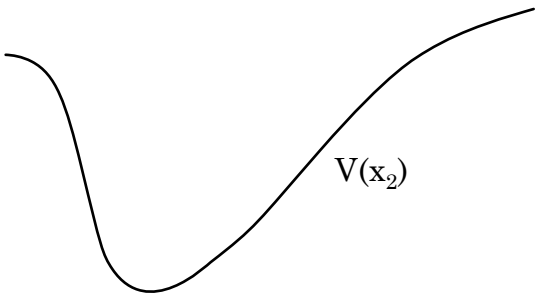
The initial state is a coherent superposition of several polyads. Motion occurs in *both* coordinate space and in state space. Each kind of motion is sampled by a different class of diagnostic.

revised August 13, 2020

so that we can use $E_{P,n}$ in $e^{-iE_{P,n}t/\hbar}$

to express $\Psi\left(x_1, x_2, t\right)$

get motion of w.p. on



get motion of pieces of state vector within each Polyad P.

Could want expectation values of quantities like $N_1, N_2, P, x_1, x_1 x_2^2$:

$$\left.\begin{aligned}\mathbf{N}_1 &= \mathbf{a}_1^\dagger \mathbf{a}_1 \\ \mathbf{N}_2 &= \mathbf{a}_2^\dagger \mathbf{a}_2\end{aligned}\right\} \text{ state space}$$

$$2\mathbf{N}_1(\mathrm{t}) + \mathbf{N}_2(\mathrm{t}) = \mathbf{P}$$

$$\text{coordinate space} \begin{cases} \mathbf{x}_1 = 2^{-1/2}\left(\mathbf{a}_1 + \mathbf{a}_1^\dagger\right) \\ \mathbf{x}_1 \mathbf{x}_2^2 == 2^{-3/2}\left(\mathbf{a}_1 + \mathbf{a}_1^\dagger\right)\left(\mathbf{a}_2 + \mathbf{a}_2^\dagger\right)^2 \end{cases}$$

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Fall 2018

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